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Department of CSE/CS&IT/AI&DS

Cloud Infrastructure and Services

25CS2206E

List of Lab Experiments and Lab Manual

2026-2027 Even Trimester(T5)

List of Lab Experiments

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2	a) Creating First Linux Instance b) S3 Bucket Creation and Upload Objects I. Create an S3 Bucket for file storage. II. Upload 5 objects with different file extensions.	CO1
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Tutorial Session no	Topics	CO-Mapping
10	Cloud Watch Alarm Creation: a) Create a CloudWatch billing alarm that goes off when the estimated charges go above $500. b) Create a CloudWatch alarm which goes off to an Alarm state when the CPU utilization of an EC2 instance goes above 65%. Also, add an SNS topic so that it notifies the person when the threshold is crossed.	CO5
11	a) Create 4 IAM users named "Dev1", "Dev2", "Test1", and "Test2". b) Create 2 groups named "Dev Team" and "Ops Team". c) Add Dev1 and Dev2 to the Dev Team. d) Add Dev1, Test1 and Test2 to the Ops Team	CO6
12	a) Create a DynamoDB table with a partition key as ID. b) Add 5 items to the DynamoDB table. c) Take a backup and delete the table.	CO6

Lab Experiments Solutions

Experiment 1: Amazon Free Tier Account Creation and Enabling Multi-Factor Authentication

1	a) Amazon Free Tier Account Creation b) Enabling Multi-Factor Authentication to Secure Your Access	CO1

Aim

To create an Amazon Web Services (AWS) Free Tier account and to secure the account access by enabling Multi-Factor Authentication (MFA) for an IAM user.

Description

The AWS Free Tier lets a new user try AWS services free of charge within fixed monthly limits for the first 12 months, for example 750 hours per month of a t2.micro EC2 instance and limited S3 and RDS usage. Account creation needs a valid email address, a credit or debit card for identity verification and a reachable phone number for the verification call.

After sign-up, the root user has full access to every resource in the account. AWS recommends that the root user is used only for account level tasks and that day-to-day work is done with IAM users. Multi-Factor Authentication adds a second factor, a time-based one-time code generated by a virtual MFA device such as Google Authenticator, on top of the user name and password. Even if the password is leaked, the account cannot be accessed without the MFA code.

Solution

1.a) Amazon Free Tier Account Creation

Read these conditions before creating a free tier account.

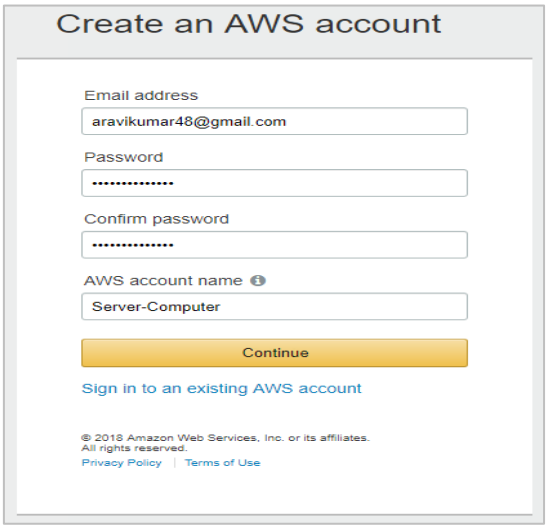
- Amazon Elastic Cloud computer EC2 Linux t2.micro 750Hours per month
- 750 Hours t2.micro windows instance per month
- 2000 Put requests of Amazon S3 (single PUT Request max 5GB)
- 20000 Get requests of Amazon S3 (Each request Get request)
- Amazon RDS MySQL DB instance with t2.micro 5GB storage
- MSSQL Express version t2.micro with 20GB GP-SSD Free tier https://aws.amazon.com/free/

Prerequisites:

Credit card with minimum 1$ available balance

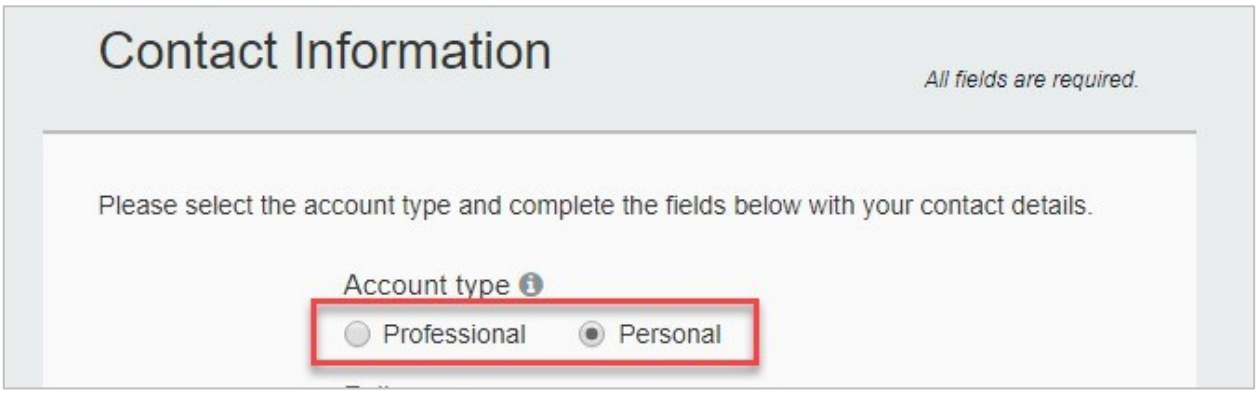
- Reachable mobile number for verification https://aws.amazon.com/console/

Click on 



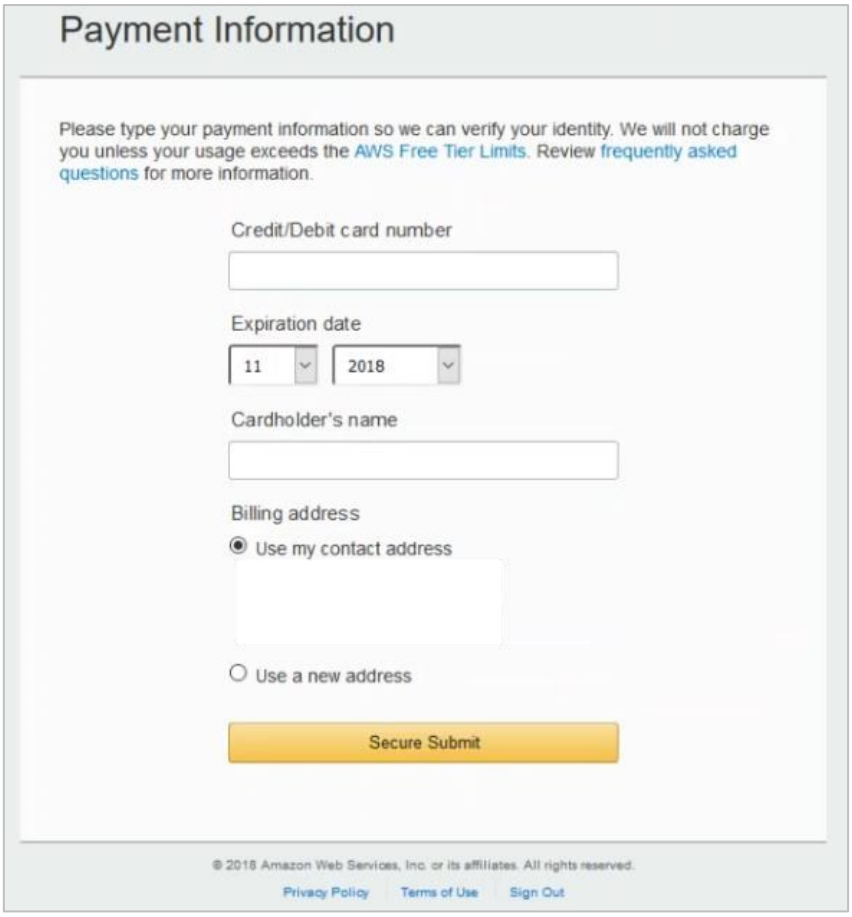
cll

Fill the details example is shown above and **click continue**

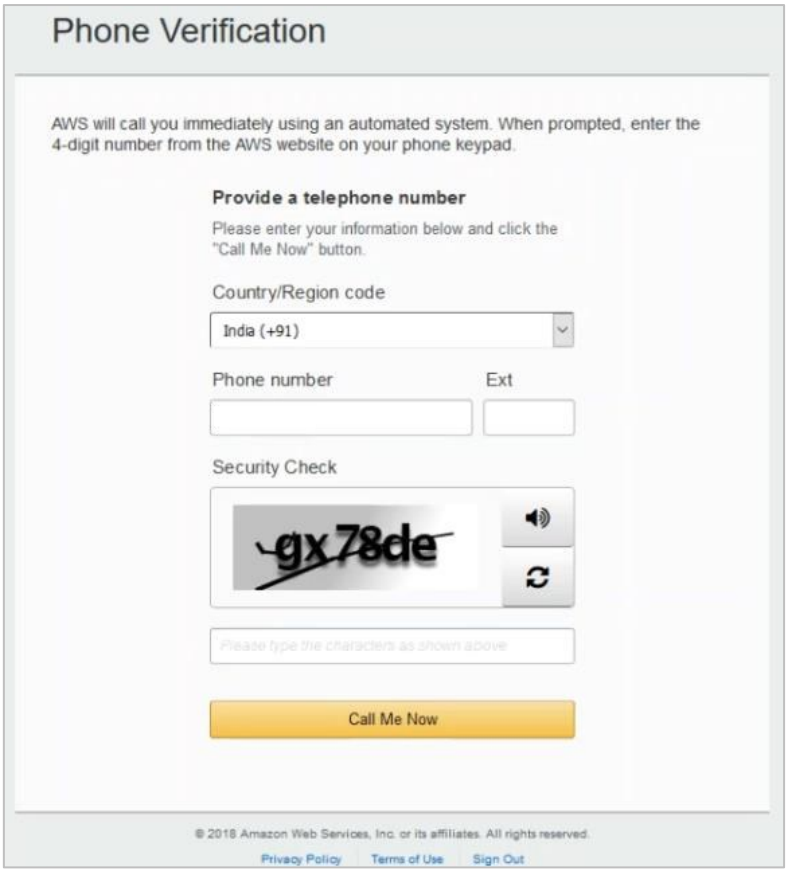


Click on radio button

- Professional is for company
- Personal is for single person



Provide your credit card details correctly, Card Number, Expiry Date and Card Holder Name
Click on **Secure Submit**



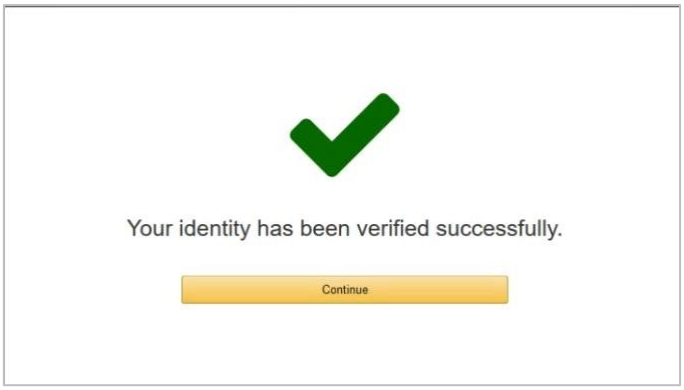
It will ask you to enter phone number, Security check then click on **Call Me Now**



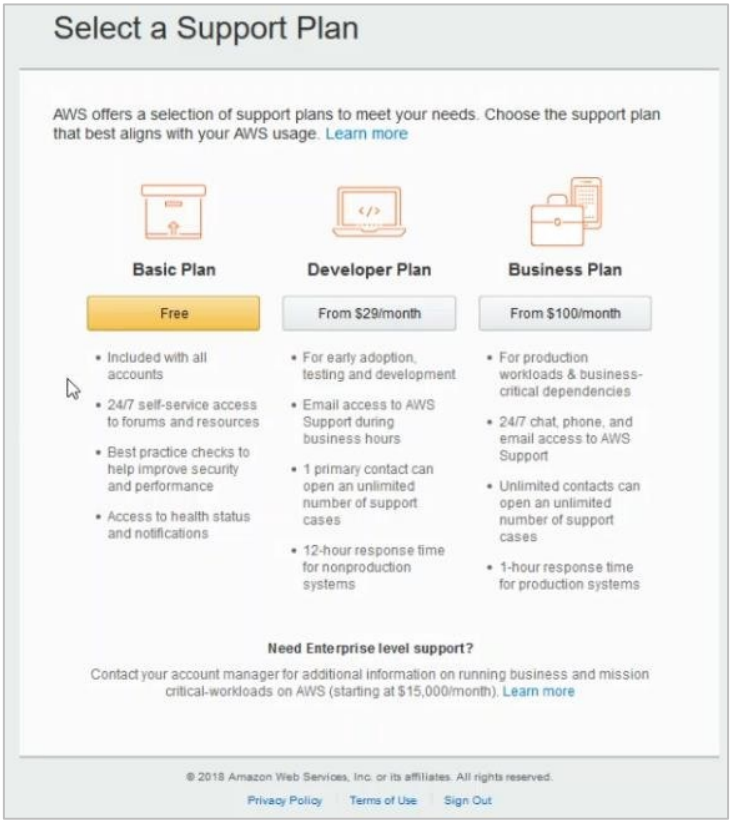
You will receive a call from AWS tele communication and ask you to enter the code displayed on screen.

Note: Listen All the Details carefully and proceed by entering code displayed on screen. After successful verification





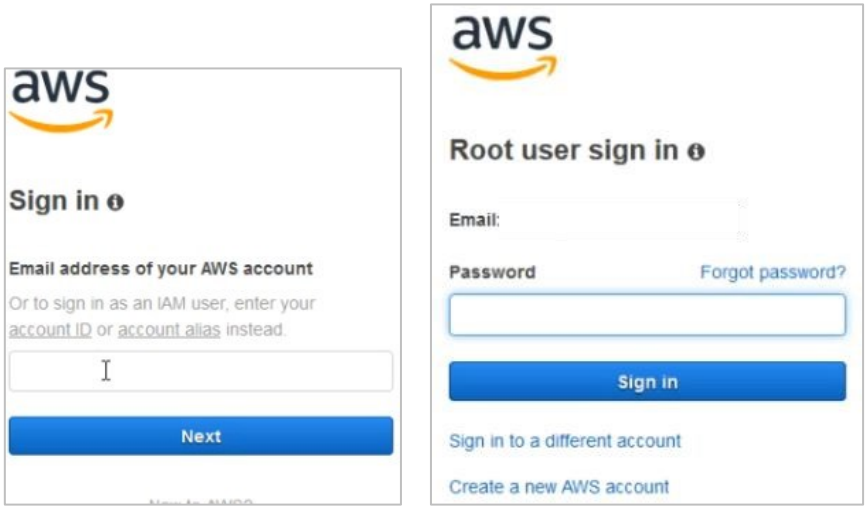
Continue



Select Support plan in this case select **Free**

You successfully completed Free Tier Account Creation. Login and Enjoy AWS Free Tier.

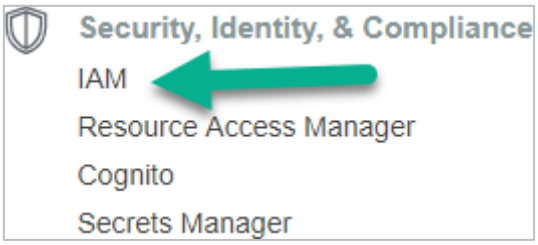
AWS Console



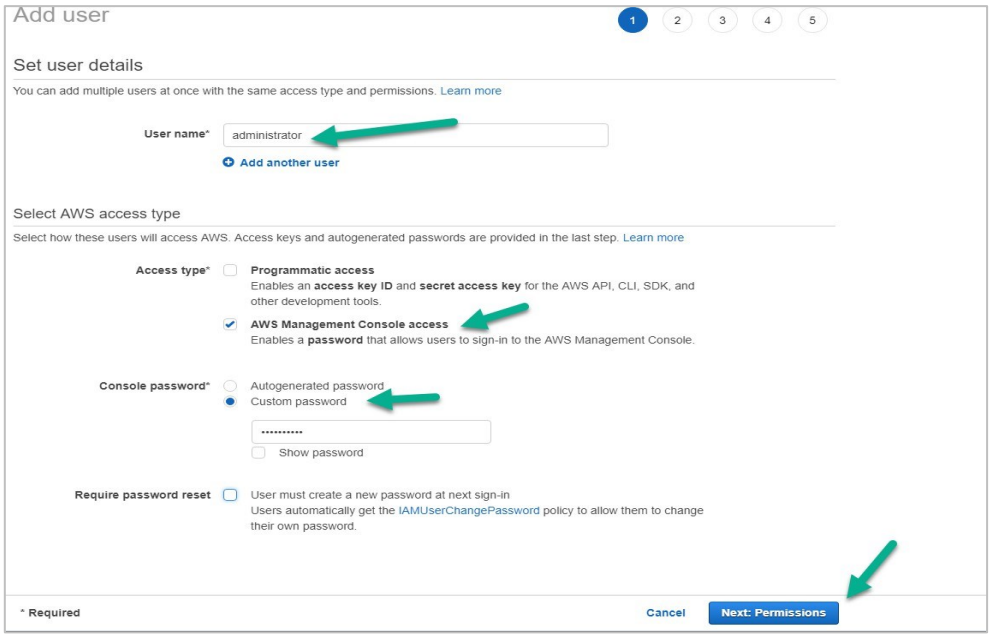
Provide your email address and password to **Sign In**

1.b Enabling Multi-Factor Authentication to Secure Your Access

Go To IAM **Services → Security, Identify & Compliance → IAM**

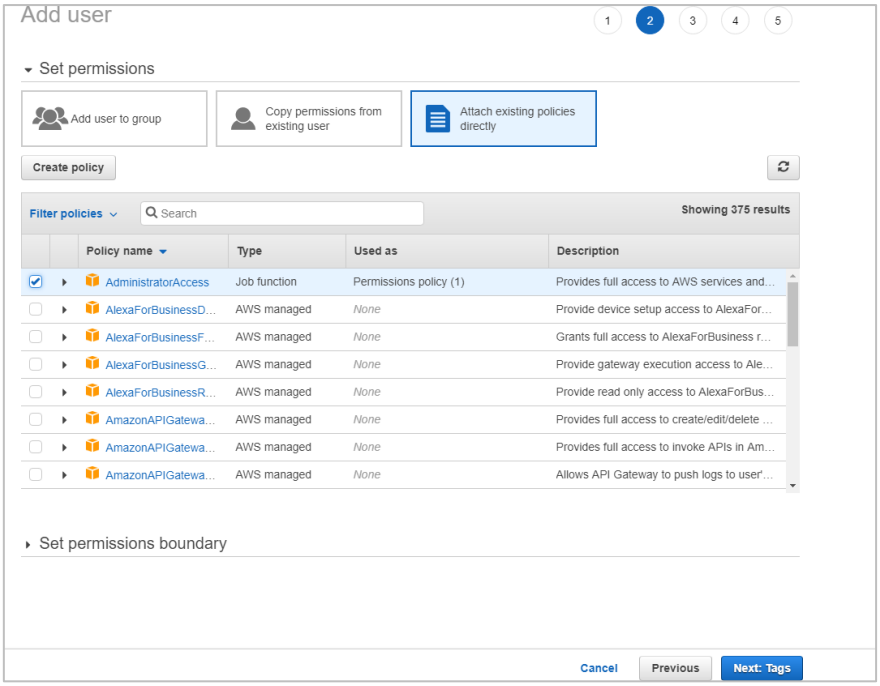


Click on Users → Add User



Provide user name, select access type

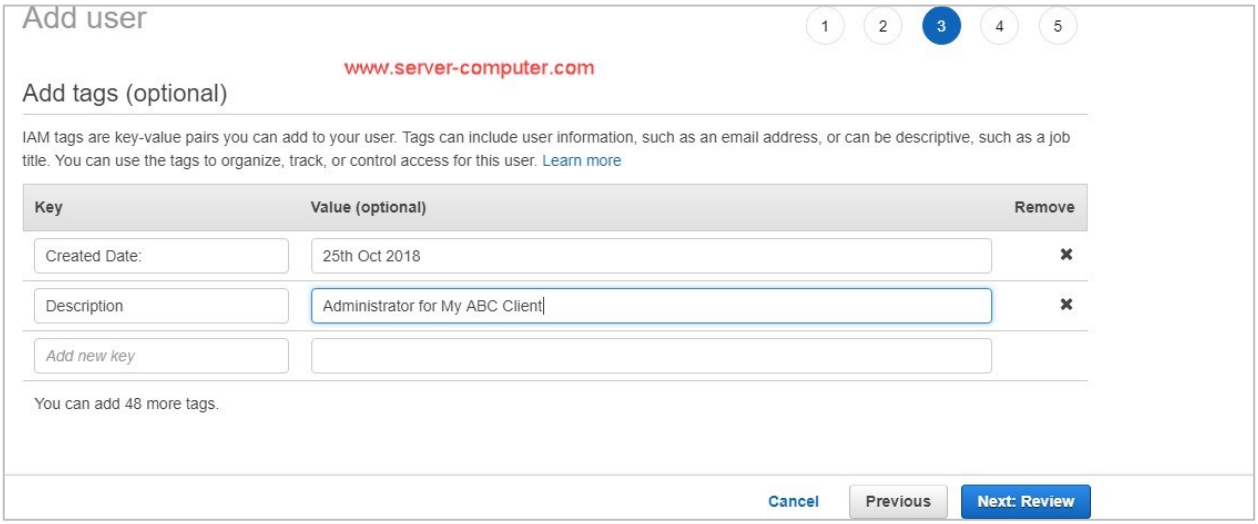
- Programmatic Access – Required for automation, run any operation using programs



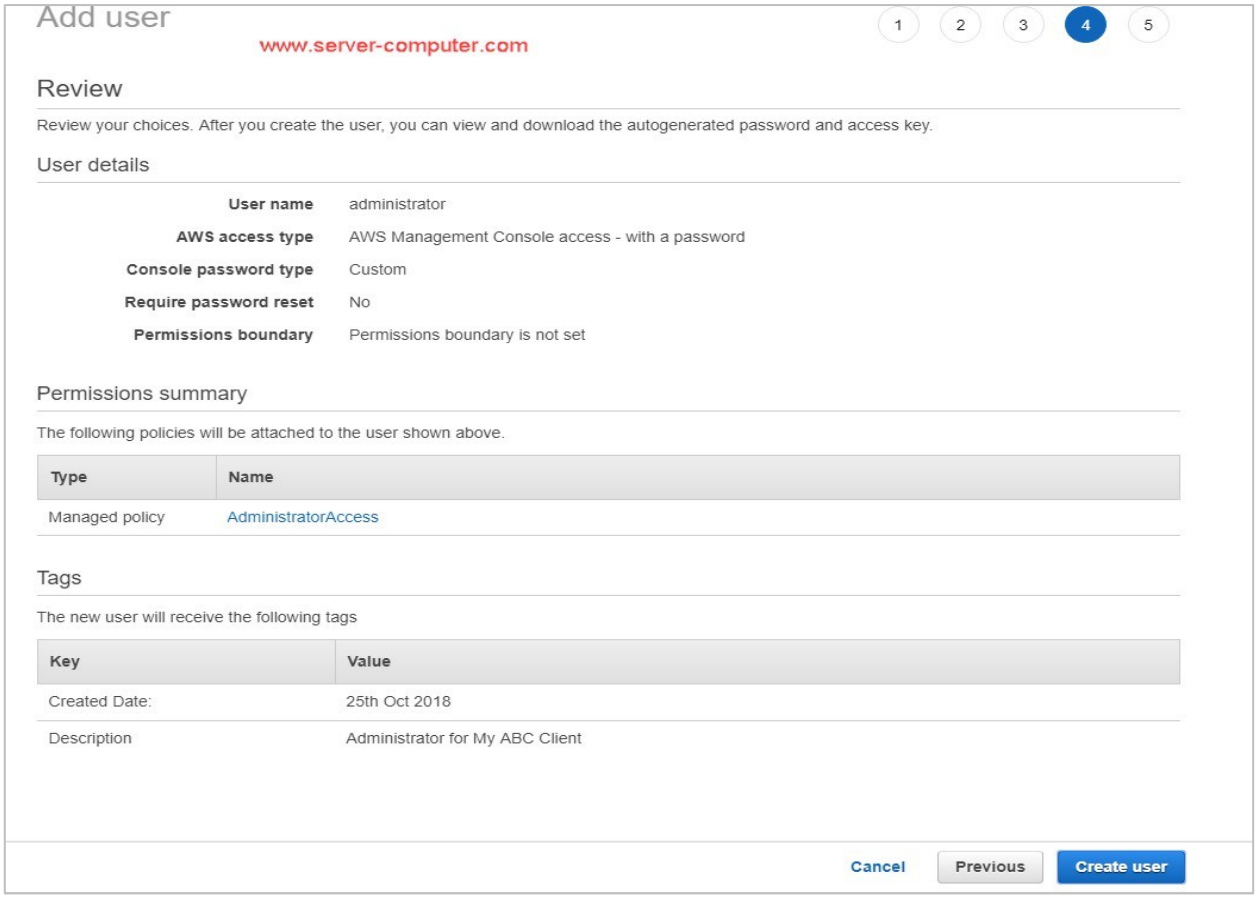
- AWS Management Console Access – User will have web console access Click **Next Permissions**

Click **Next: Tags**

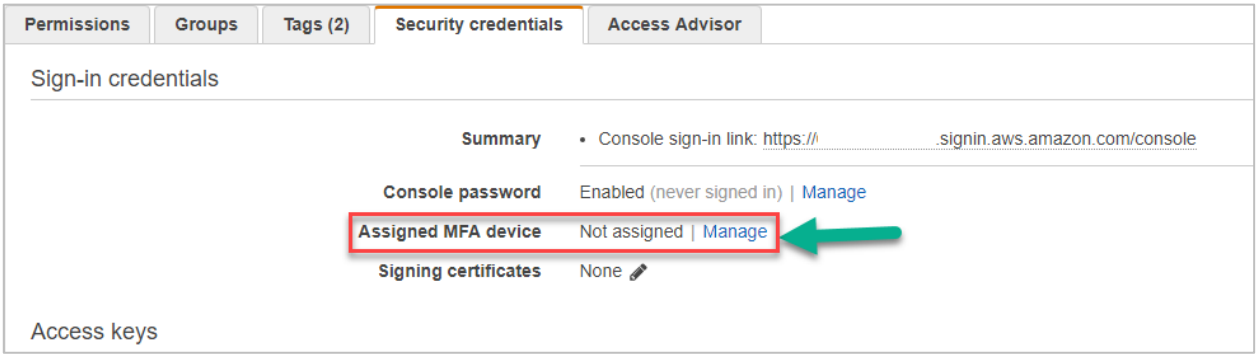
Add tags whatever required to identify user



Click **Next: Review**

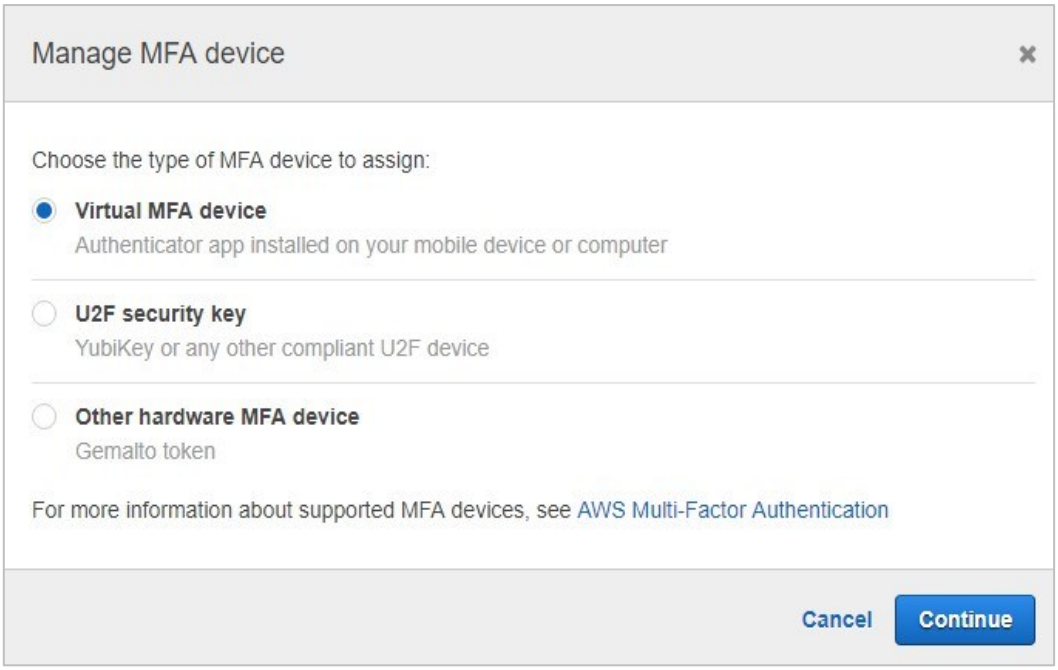


Click **Create User**



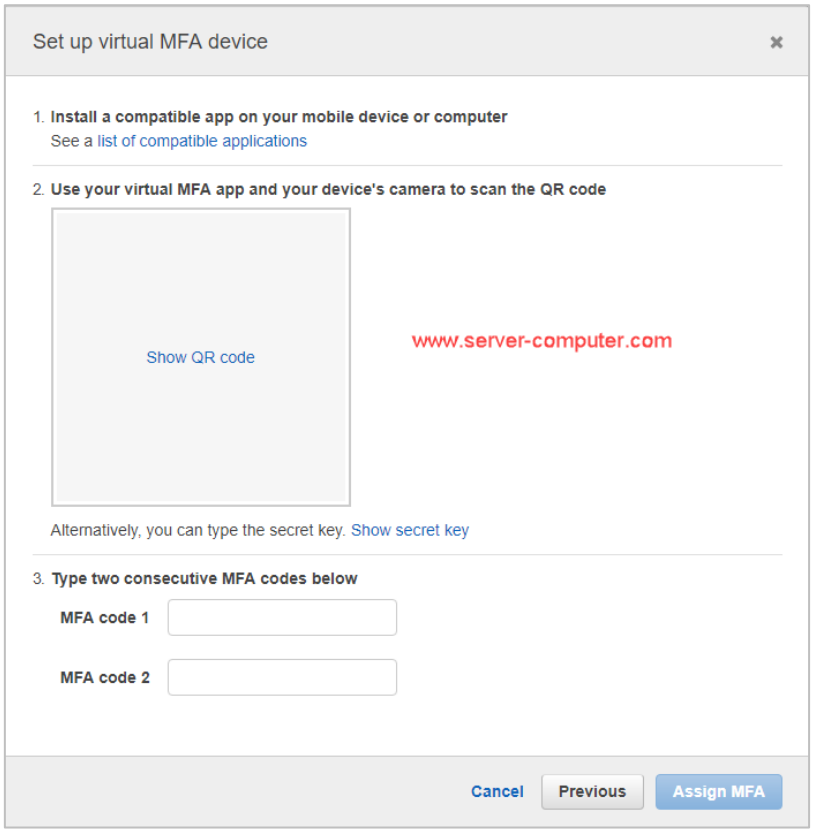
User creation has been completed successfully now you will get on access URL with your account number. Note the URL. Now Click on User name → Security credentials (TAB)

Click on Assigned MFA Device – Manage



Use any method based on your requirement. Here I am showing Virtual MFA Device method Install Google Authenticator in your smart phone and ready to pair

Click **Continue**



Click in **Show QR Code** and scan the same code from your Google authenticator App. It will generate six digit codes enter one code in first MFA code 1 wait 1 minute and second code in MFA Code 2 Click on **Assign MFA**



That's it, now you successfully enabled MFA (Multi-Factor Authentication).

Here after if you want to login, you have to enter credentials and MFA code to Login.

Result

The AWS Free Tier account was created and the sign-in to the AWS Management Console was verified. An IAM user was created and a virtual MFA device was assigned to it, so every console login now requires the password and the MFA code.

Viva Questions

1. What is the AWS Free Tier?

Ans. It is a set of AWS services offered free of charge within fixed monthly usage limits so that new users can try the platform, for example 750 hours per month of a t2.micro EC2 instance for the first 12 months.

2. Why does AWS ask for a credit card during Free Tier sign-up?

Ans. The card is used for identity verification and to bill any usage that goes beyond the free limits.

3. What is the difference between the root user and an IAM user?

Ans. The root user is the identity created along with the account and has unrestricted access. IAM users are identities created inside the account and get only the permissions granted to them through policies.

4. What is Multi-Factor Authentication and why is it needed?

Ans. MFA requires a second factor, such as a one-time code from an authenticator app, in addition to the password. A stolen password alone is then not enough to sign in.

5. What is a virtual MFA device?

Ans. A software application such as Google Authenticator that generates six-digit time-based codes after it is paired with the AWS account by scanning a QR code.

Assignment Questions

1. Create an IAM user with programmatic access only, download the access key and secret key, and document the steps with screenshots. Then set up a billing alert for the account so that an email is received when the Free Tier limit is about to be crossed.

2. Enable MFA on the root user of your account and prepare a short note comparing virtual MFA devices, hardware MFA devices and FIDO security keys.

Experiment 2: Creating First Linux Instance and S3 Bucket Creation

week-2	a) Creating First Linux Instance b) S3 Bucket Creation and Upload Objects I. Create an S3 Bucket for file storage. II. Upload 5 objects with different file extensions.	CO1

Aim

To launch a Linux EC2 instance, connect to it using PuTTY and host a web server on it, and to create an Amazon S3 bucket and upload five objects with different file extensions.

Description

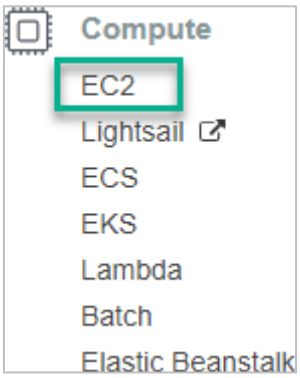
Amazon EC2 provides virtual servers, called instances, that are launched from an Amazon Machine Image. The launch wizard asks for the AMI, the instance type, the network settings, the storage, tags, a security group and a key pair. The security group works as a firewall that controls the ports open on the instance. The key pair is used for SSH authentication. The .pem private key downloaded from AWS is converted to .ppk format with PuTTYgen so that PuTTY can use it.

Amazon S3 is object storage. Data is stored as objects inside buckets. A bucket name must be unique across all AWS accounts and the bucket is created in one region. The Block Public Access settings keep the bucket private by default. Objects of any file type can be uploaded through the console, the CLI or the SDK.

Solution

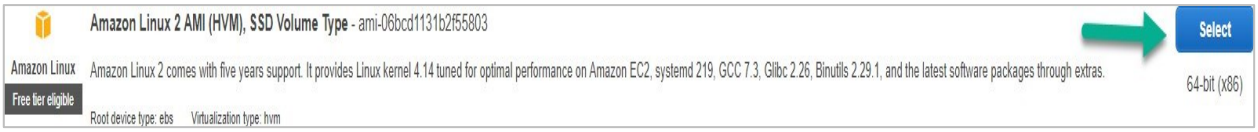
2.a Creating First Linux Instance

Login to AWS console, services drop down click on EC2

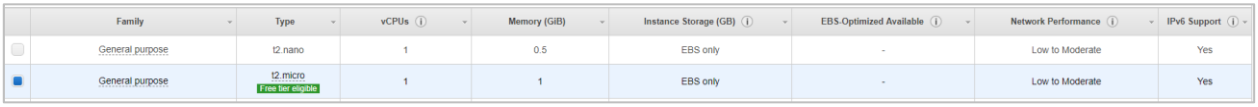


Click on **Launch instance**



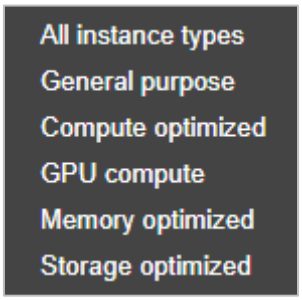


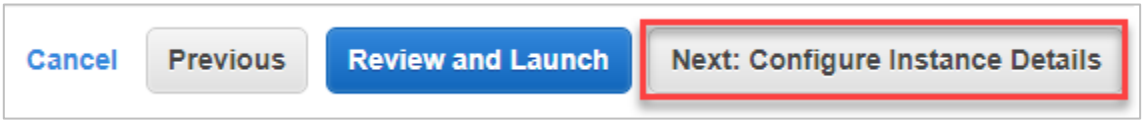
I am selecting Free Tier instance Amazon Linux

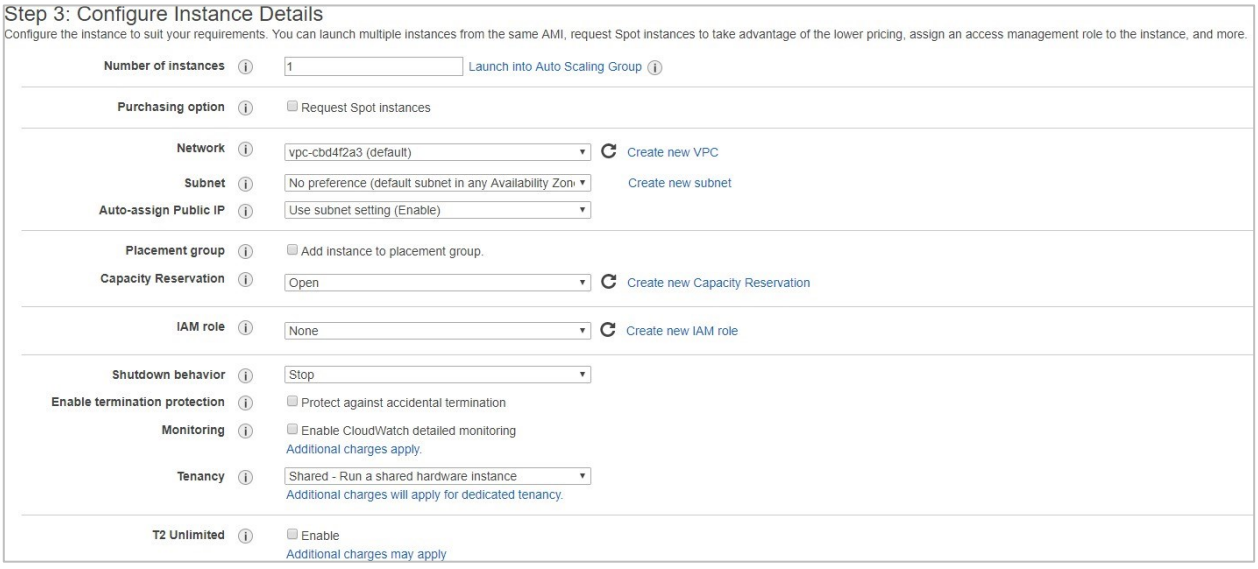


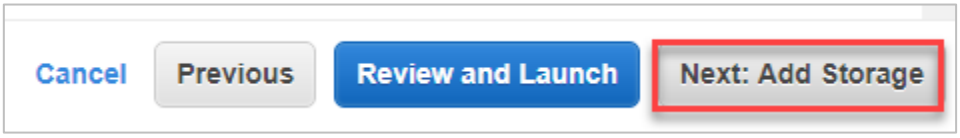
	Family	Type	vCPUs	Memory (GiB)	Instance Storage (GB)	EBS-Optimized Available	Network Performance	IPv6 Support
☐	General purpose	t2.nano	1	0.5	EBS only	-	Low to Moderate	Yes
☑	General purpose	t2.micro Free tier eligible	1	1	EBS only	-	Low to Moderate	Yes

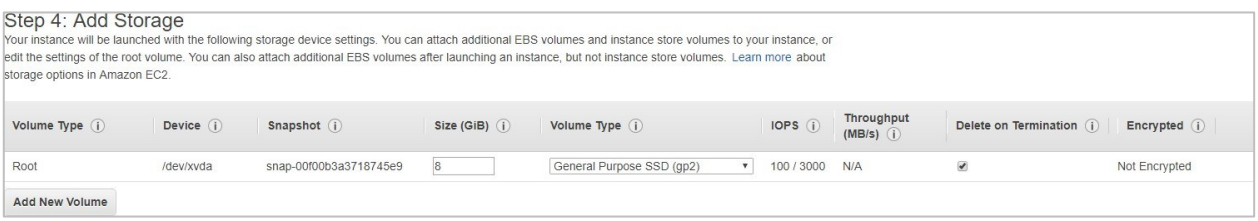
We have below types of instances



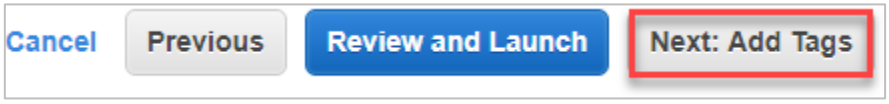




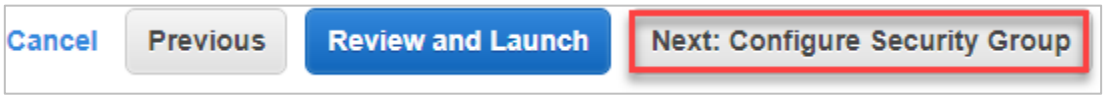




Add storage – EBS Elastic Block Storage volume will attached to your instance

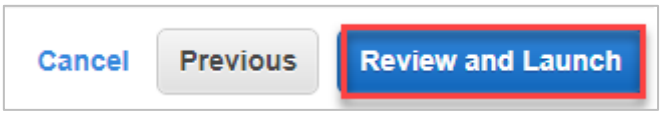


Tags to identify the details about instance (Production/Test/Dev/Client Name)

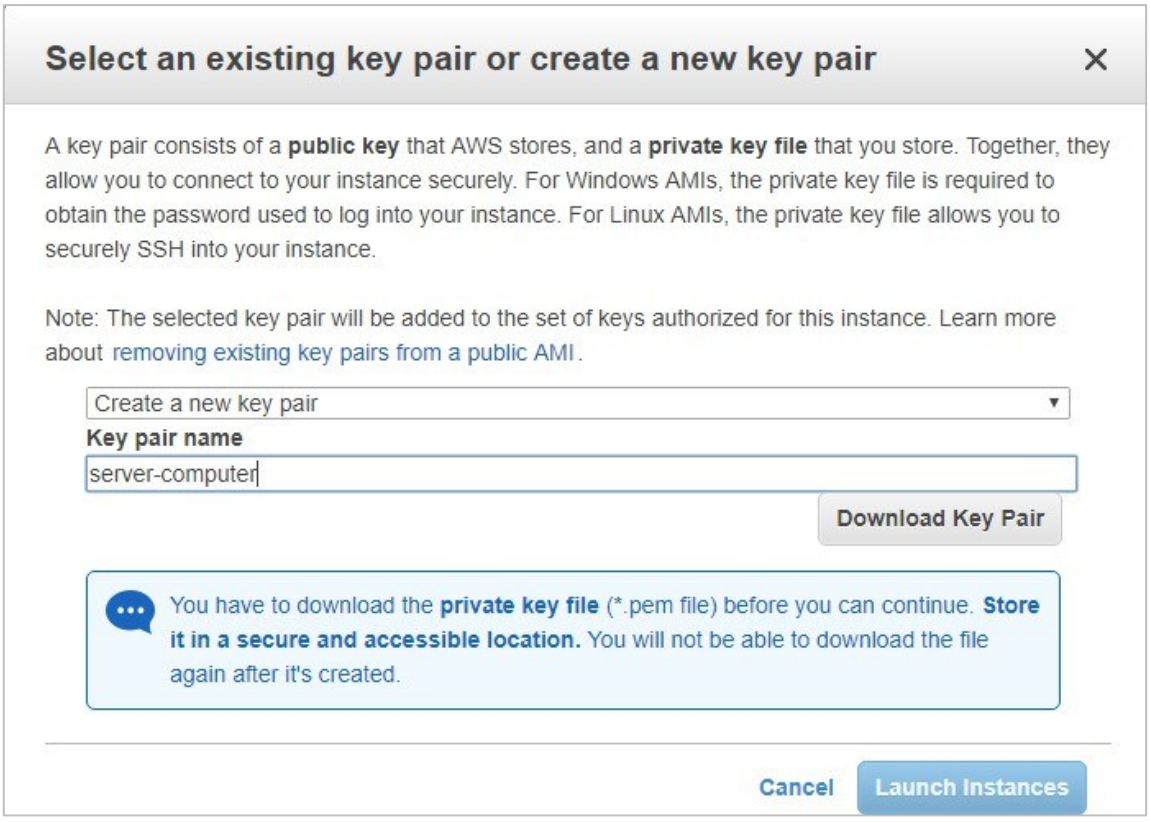




Using security group we can allow/deny any ports



Verify the details and click on **Launch**



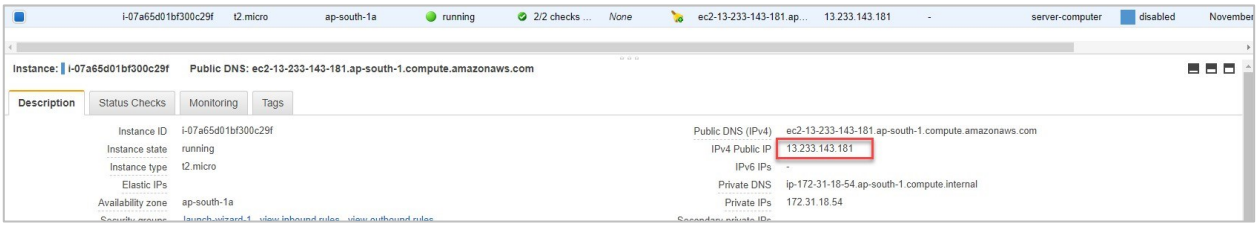
For the first time you **create a new key pair** and **Download Key Pair**

Server-computer.pem file will downloaded, **keep it safe Launch Instances**

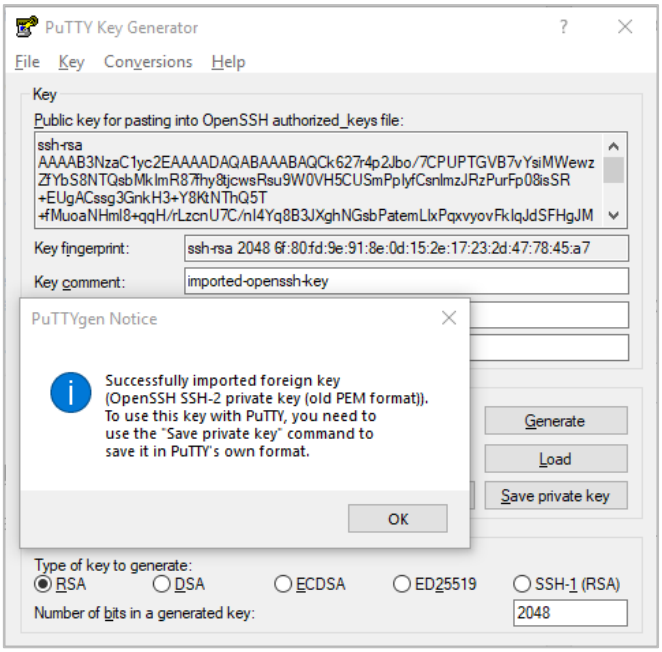
Go to EC2 → See the instances



Click on instance and copy the Public IP Address



Install putty msi installer you will get PuttyGen and Putty for accessing Linux machine Open puttyGen and load server-computer.pem file

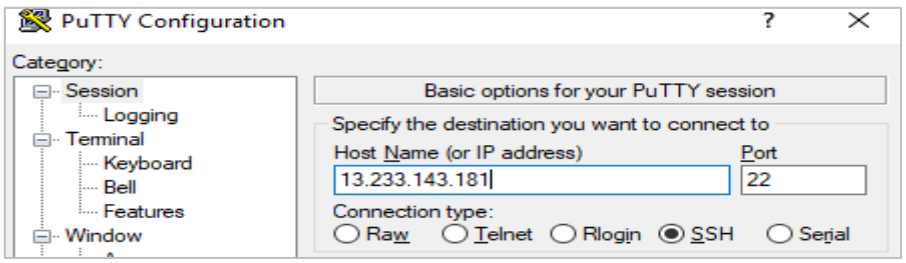


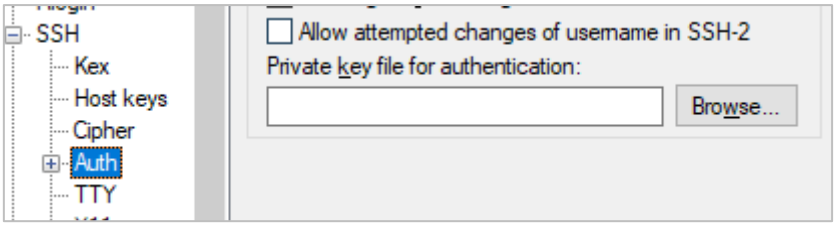
Click Ok.

Save Private Key

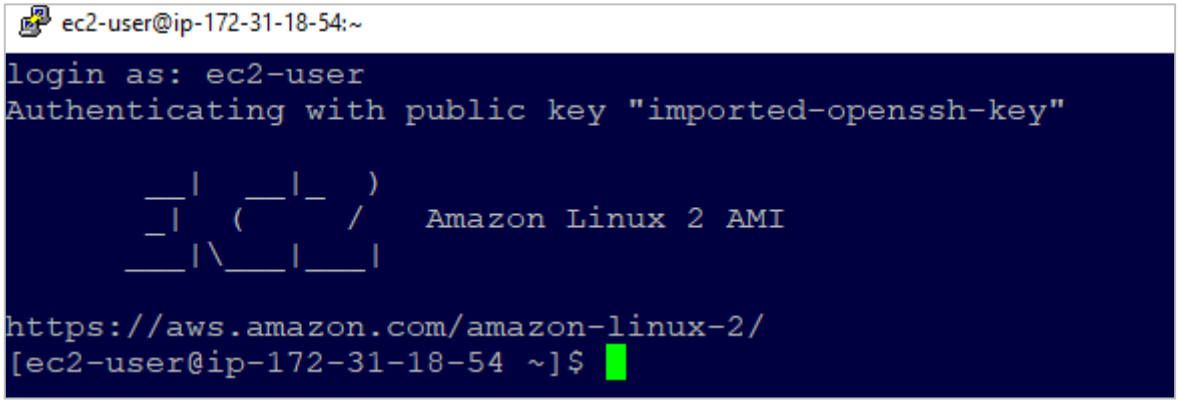
In this case, I have used server-computer1.ppk

Open putty application and type IP address as shown below





Expand SSH → Click on Auth → Browse and attach .ppk file Click on **Open**



You successfully logged into your Amazon Linux instance

As example, we are going to install web server in Linux server and access using web browser https://github.com/techtutorials/aws-lab-guide/blob/aws/webserver.sh

You can also use above shell script to automatically build webserver for you

sudo yum update –y;

sudo yum install httpd –y; sudo service httpd start; sudo service httpd status; sudo chkconfig httpd on;

Now go back to your EC2 → Security Groups and Add 80 port

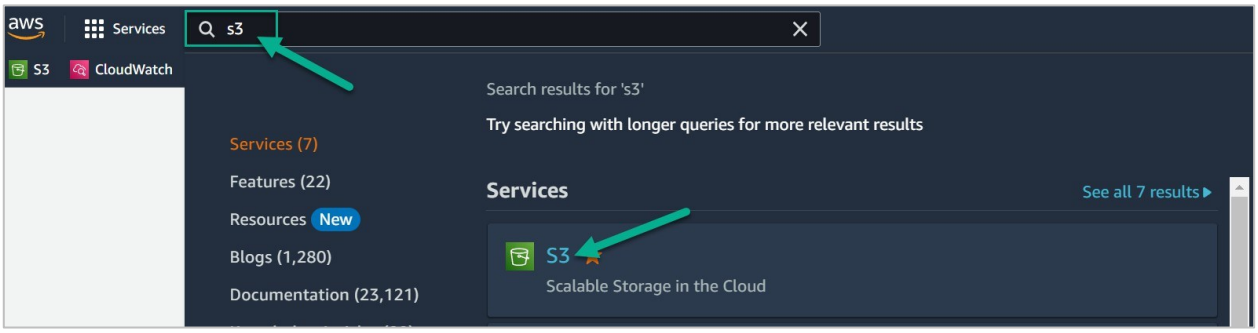


Open browser and type your instance public IP address you can access web-server test page.

2.b S3 Bucket Creation and Upload Objects

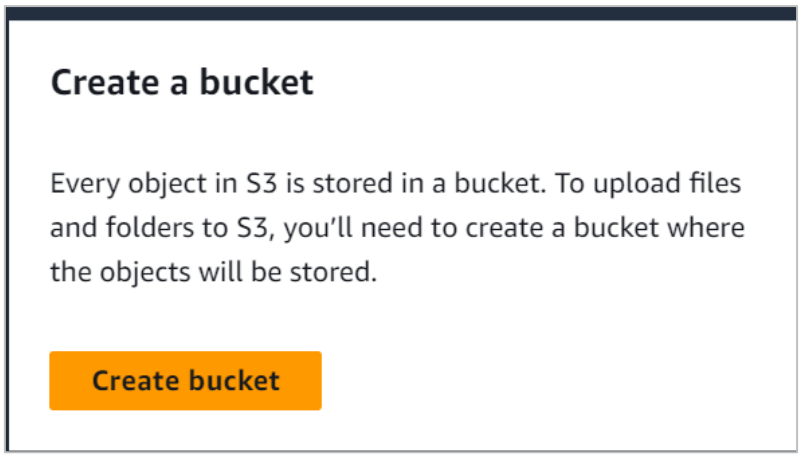
1. Create an S3 Bucket for file storage.
2. Upload 5 objects with different file extensions.

Answer:

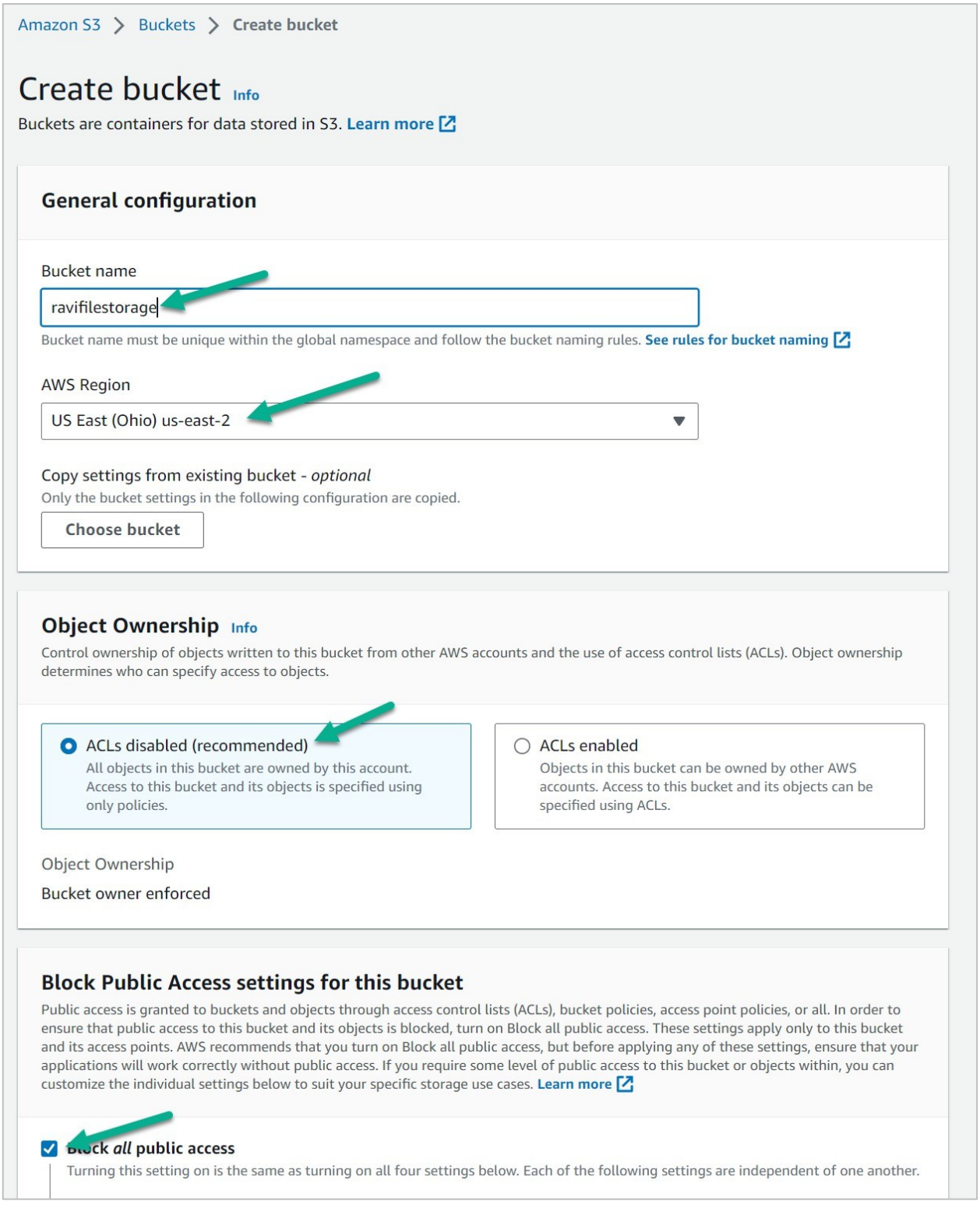


Login to the AWS Console providing your credentials

In Search bar search for S3 then select S3 in results



In S3 Console Click on “Create Bucket”

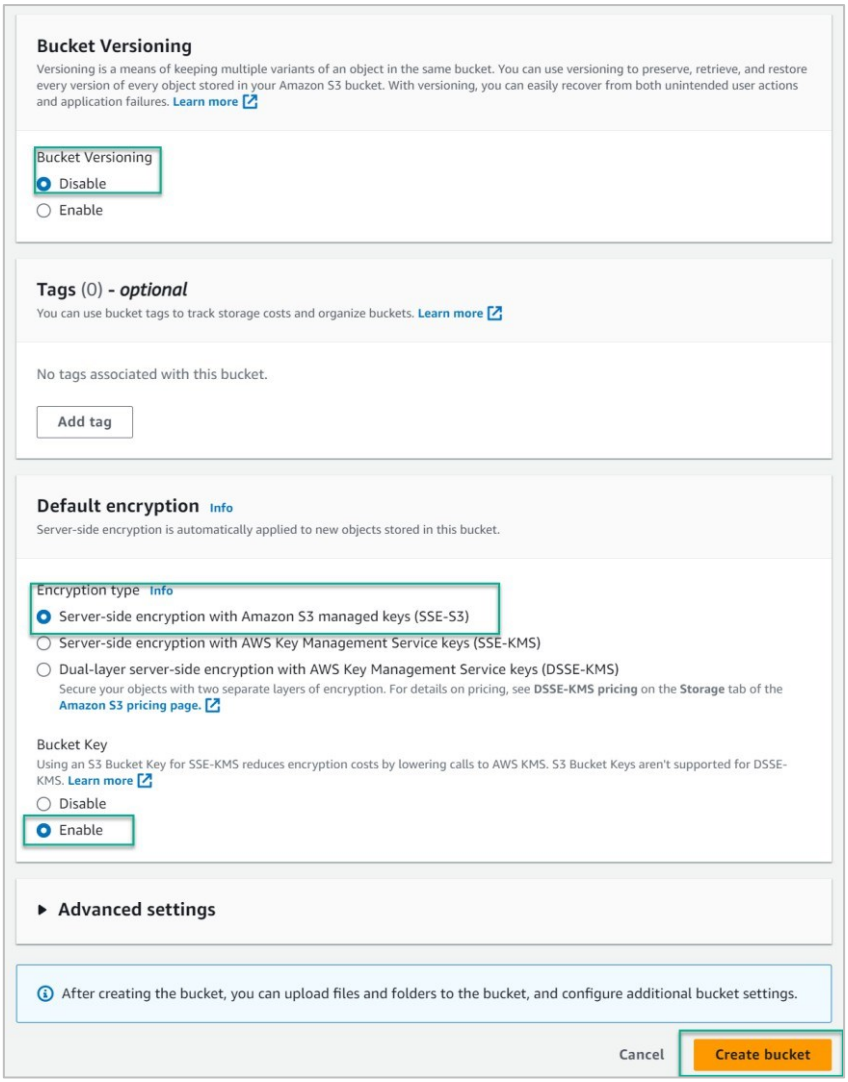


Provide Bucket Name should be unique

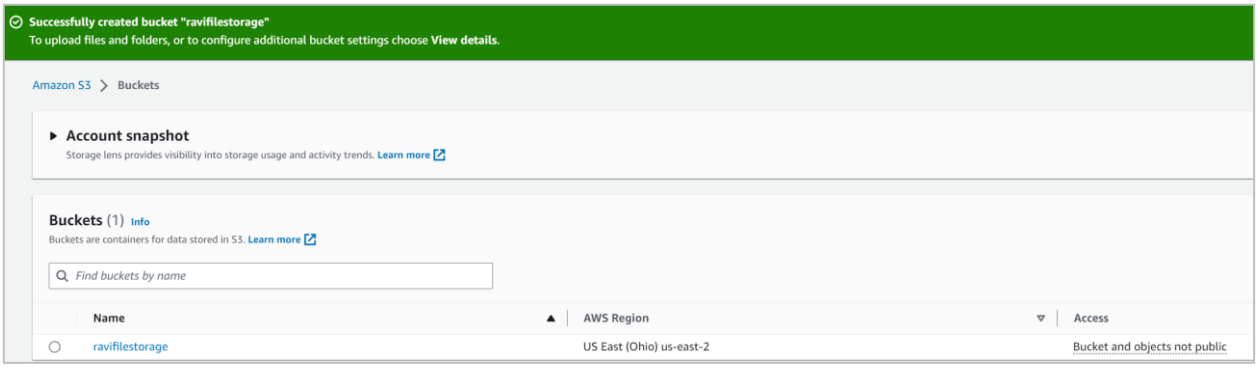
Select the region in which region you wanted to create that bucket

Object Ownership ACLs Disabled

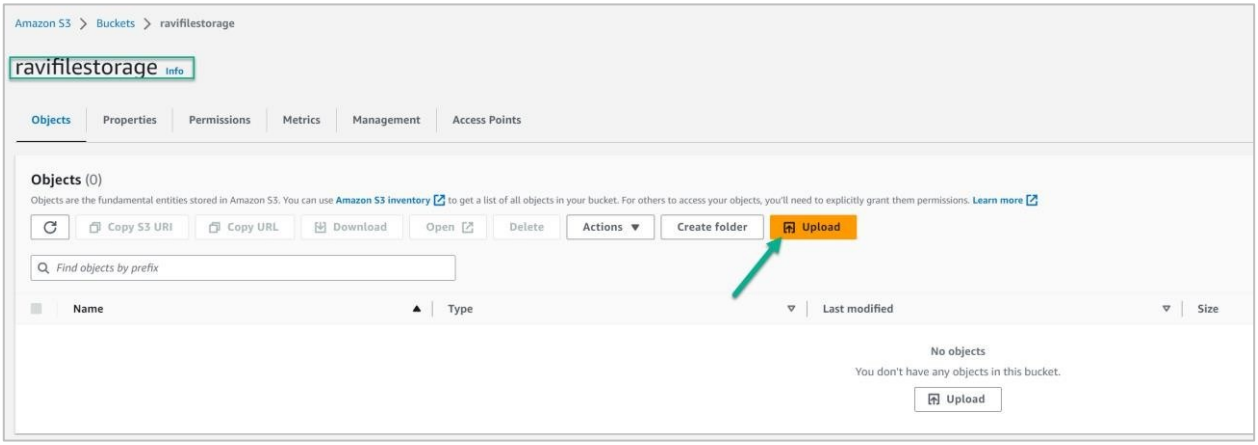
Select “Block All Public Access” to avoid publishing your bucket to public



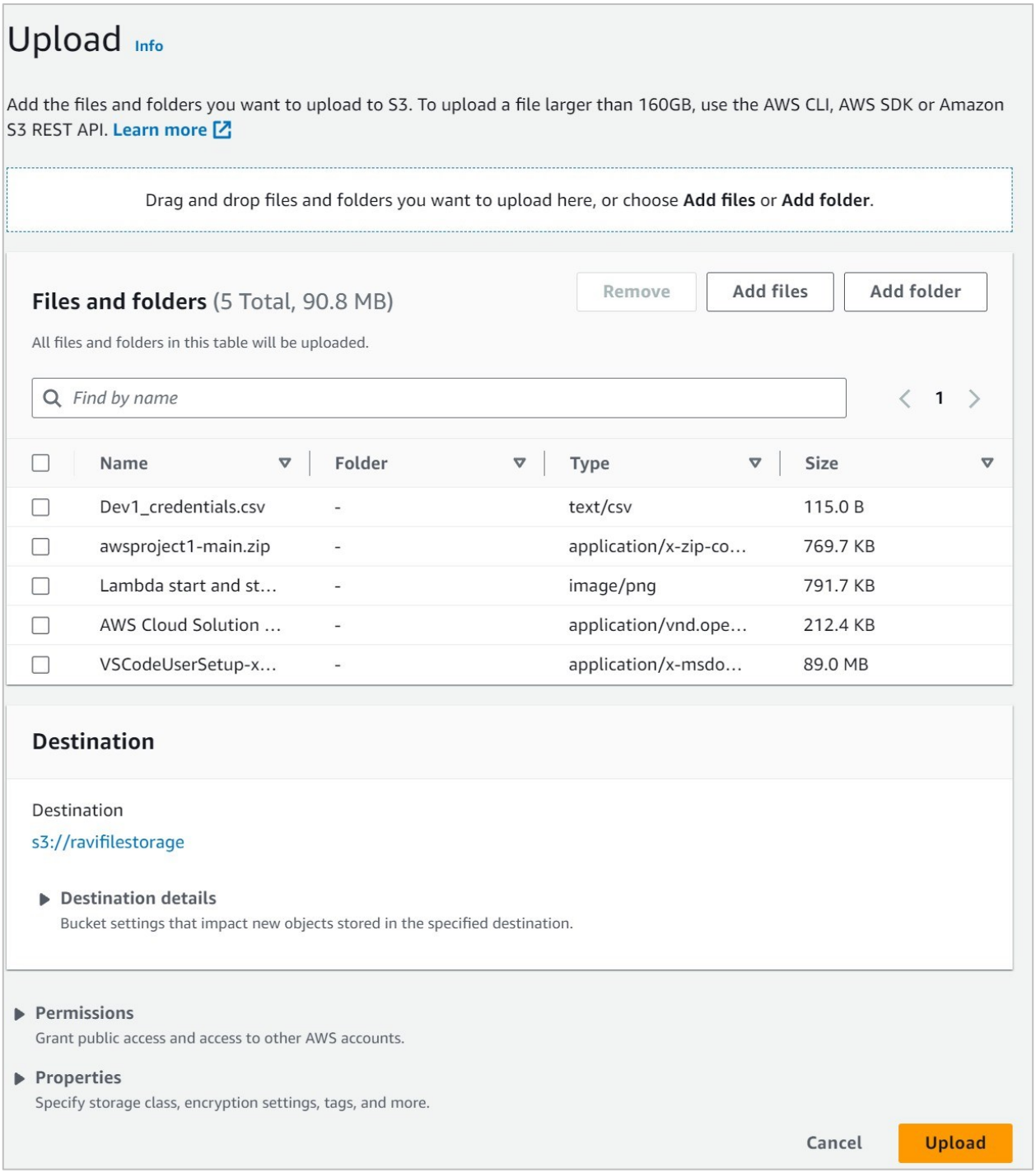
Keep all the Default options Click on “Create Bucket”



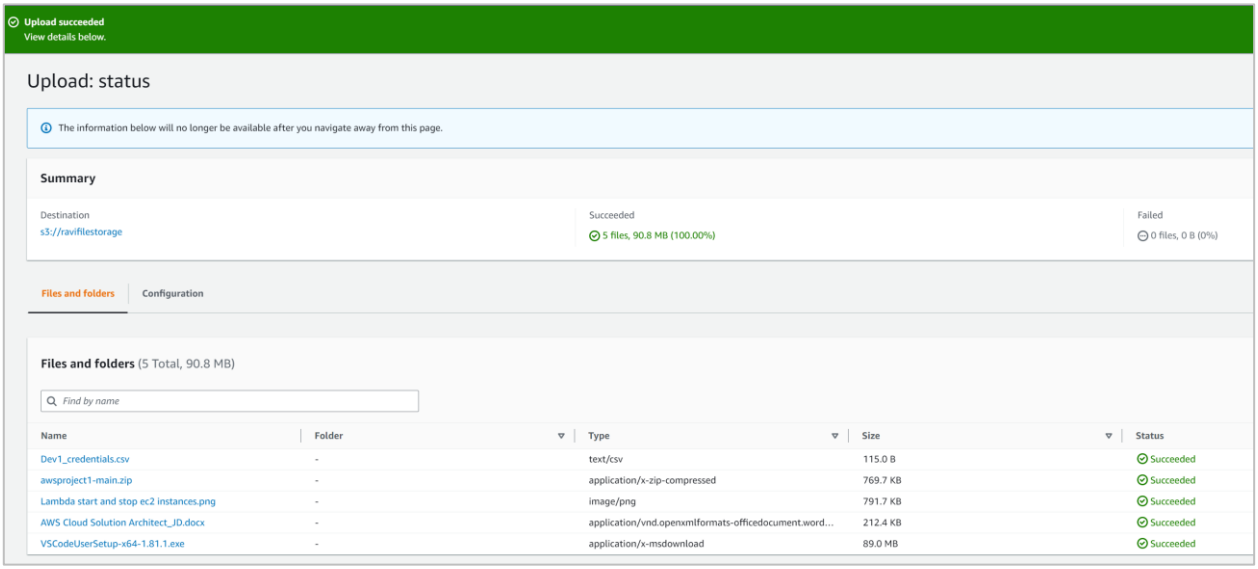
Bucket is creation is successful Now Upload the Objects



Click on the Bucket Name → Click Upload


Upload Info
Add the files and folders you want to upload to S3. To upload a file larger than 160GB, use the AWS CLI, AWS SDK or Amazon S3 REST API. Learn more
Drag and drop files and folders you want to upload here, or choose Add files or Add folder.
Files and folders (5 Total, 90.8 MB)
Remove
Add files
Add folder
All files and folders in this table will be uploaded.
Find by name
1
Name
Folder
Type
Size
Dev1_credentials.csv - text/csv 115.0 B
awsproject1-main.zip - application/x-zip-co... 769.7 KB
Lambda start and st... - image/png 791.7 KB
AWS Cloud Solution ... - application/vnd.ope... 212.4 KB
VSCodeUserSetup-x... - application/x-msdo... 89.0 MB
Destination
Destination
s3://ravifilestorage
Destination details
Bucket settings that impact new objects stored in the specified destination.
Permissions
Grant public access and access to other AWS accounts.
Properties
Specify storage class, encryption settings, tags, and more.
Cancel
Upload

Select 5 types of objects Click on Upload



Objects are uploaded successfully

Result

A t2.micro Amazon Linux instance was launched and accessed through PuTTY with the key pair. The Apache web server was installed and its test page opened in the browser after port 80 was allowed in the security group. An S3 bucket was created and five objects with different file extensions were uploaded into it.

Viva Questions

1. What is an AMI?

Ans. An Amazon Machine Image is a template that contains the operating system, application server and applications used to launch an instance.

2. What is the purpose of a security group?

Ans. It is a virtual firewall attached to the instance that controls which inbound and outbound ports and source addresses are allowed.

3. Why is the .pem file converted to .ppk?

Ans. PuTTY uses its own private key format, so PuTTYgen converts the .pem key downloaded from AWS into a .ppk file.

4. What is an S3 bucket and what rule applies to its name?

Ans. A bucket is a container for objects in S3. Its name must be unique across all AWS accounts and must follow DNS naming rules.

5. What does the Block all public access setting do?

Ans. It prevents the bucket and its objects from being made public through ACLs or bucket policies, so the data stays private.

Assignment Questions

1. Launch an Ubuntu EC2 instance, install the Nginx web server through a user data script at launch time and open the default page in a browser. Record the differences you observed compared to the Amazon Linux and Apache steps.

2. Enable versioning on your S3 bucket, upload the same file twice with changes and show how the earlier version can be restored. Also configure a lifecycle rule that moves objects to S3 Glacier after 30 days.

Experiment 3: Creating Amazon Machine Image (AMI) and Creating EC2 Instance using AWS CLI

week- 3	a) Creating Amazon Machine Image (AMI) b) Creating EC2 Instance using AWS CLI	CO2

Aim

To create a custom Amazon Machine Image from a configured EC2 instance and launch instances from it, and to install and configure the AWS Command Line Interface and create an EC2 instance from the command line.

Description

An AMI captures the root volume, the launch permissions and the block device mapping of an instance. Creating an image from a configured instance lets the same configuration be launched any number of times. Stored AMIs and their snapshots are charged, so unused images should be deregistered. A Windows instance launched from a Windows AMI is accessed through Remote Desktop with the administrator password decrypted using the key pair.

The AWS CLI is a single tool that manages AWS services from the command line and from scripts. After installation it is configured with the aws configure command using an access key, a secret key, a default region and an output format. S3 buckets and objects are managed with the aws s3 and aws s3api commands, and an EC2 instance is launched with aws ec2 run-instances by supplying the AMI ID, instance type, key name, security group ID and subnet ID.

Solution

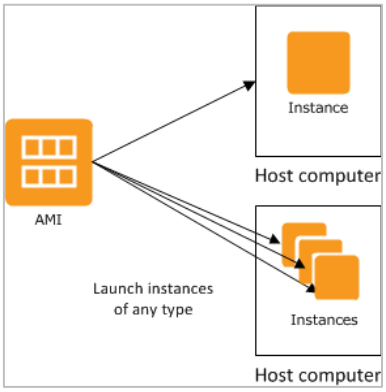
a) Creating Amazon Machine Image (AMI)

An Amazon Machine Image (AMI) provides the information required to launch an instance, which is a virtual server in the cloud. You must specify a source AMI when you launch an instance. You can launch multiple instances from a single AMI when you need multiple instances with the same configuration. You can use different AMIs to launch instances when you need instances with different configurations.

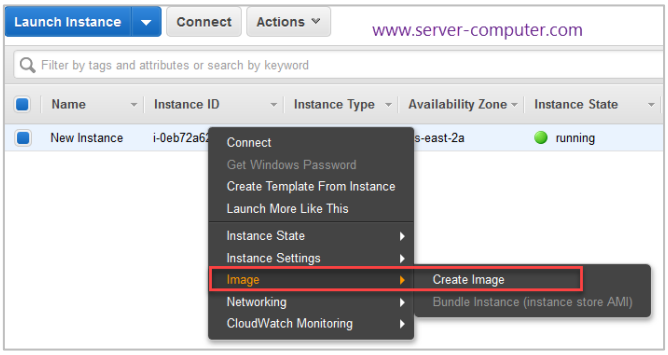
An AMI includes the following:

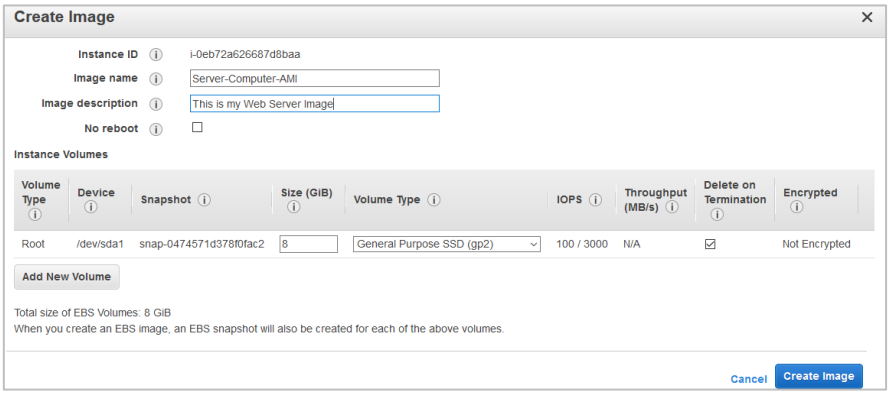
- A template for the root volume for the instance (for example, an operating system, an application server, and applications)
- Launch permissions that control which AWS accounts can use the AMI to launch instances

- A block device mapping that specifies the volumes to attach to the instance when it's launched



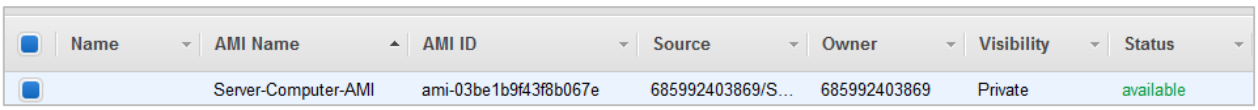
First, follow above steps to create EC2 instance, modify all the required settings, and install required applications. Right click on instance **Image** →**Create Image**



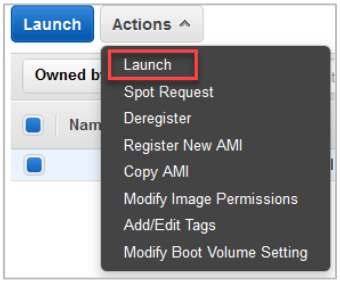


Provide Image name (Easy to Identify), Image Description and Click **Create Image**

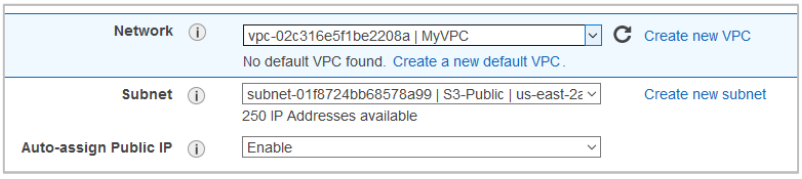
It will take few minutes depends on your EC2 instance size. Go to → EC2 → AMIs



Select AMI → Actions → Launch

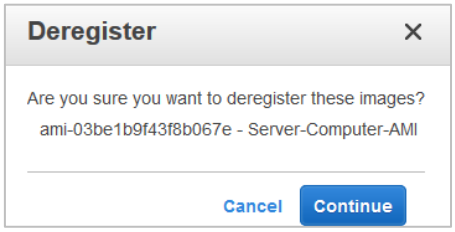


Choose Instance Type → Click Next: Configure Instance Details



Select appropriate details Click **Next: Add Storage** → **Next: Add Tags** → **Next: Configure Security Group** → **Review and Launch** → **Launch**

That is it your application is ready to use.



Note: Storing AMI will be charged based on your EC2 instance size. To delete the AMI select AMI → Actions → Deregister

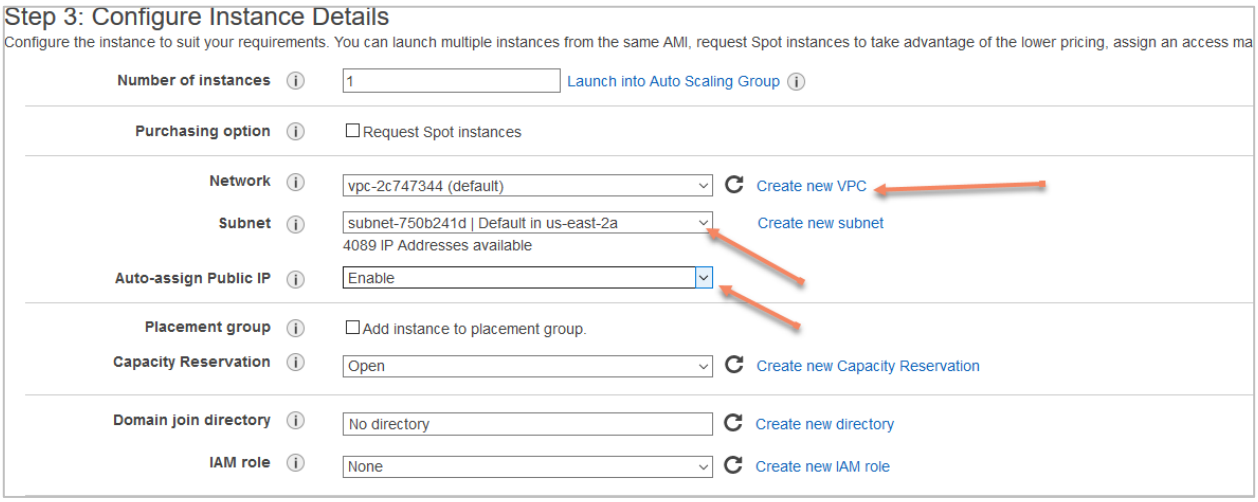
B Create your First EC2 windows instance

Expand services EC2 → Launch Instance



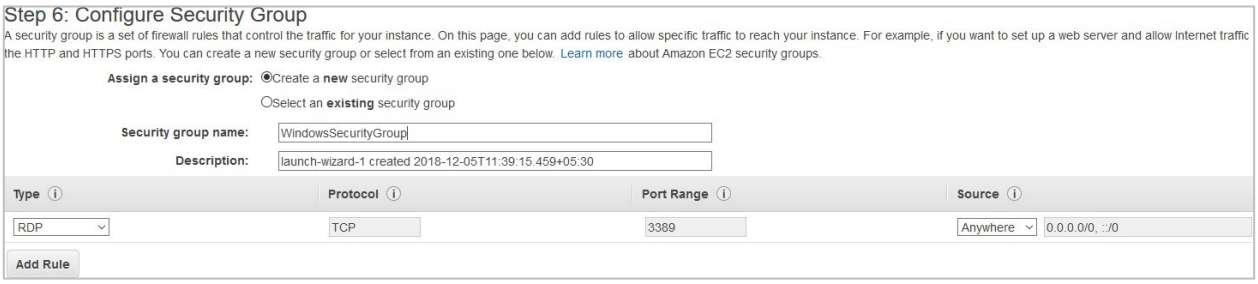
Select Windows Image

Choose an Instance Type → General Purpose (t2.micro) → Click **Next: Configure Instance Details** →



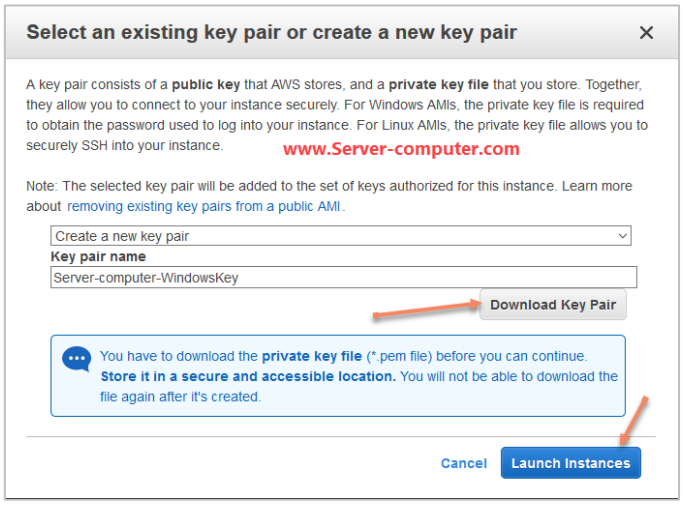
Select VPC, subnet and enable Public IP address. Click **Next: Add Storage**

Click **Next: Add Tags**



Add Tags to identify instance details Like Name, Purpose, Account and so and so Click **Next: Configure Security Group**

Click **Review and Launch**

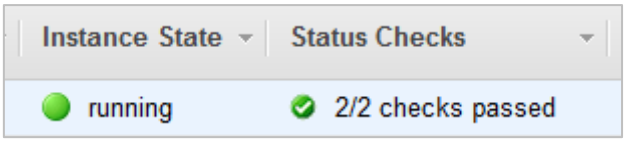


Download Key Pair* and *Launch Instance

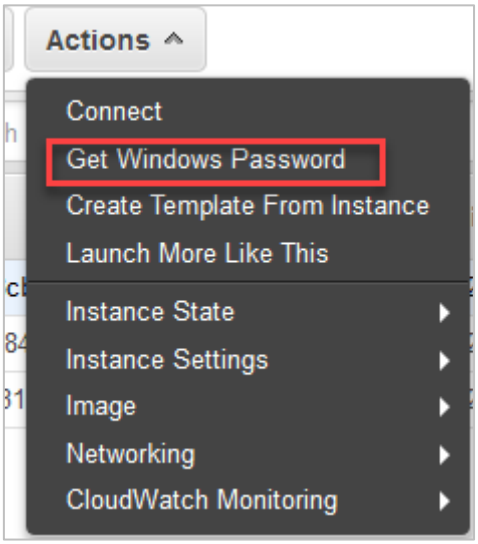
Note: Wait 4 Minutes instance to launch

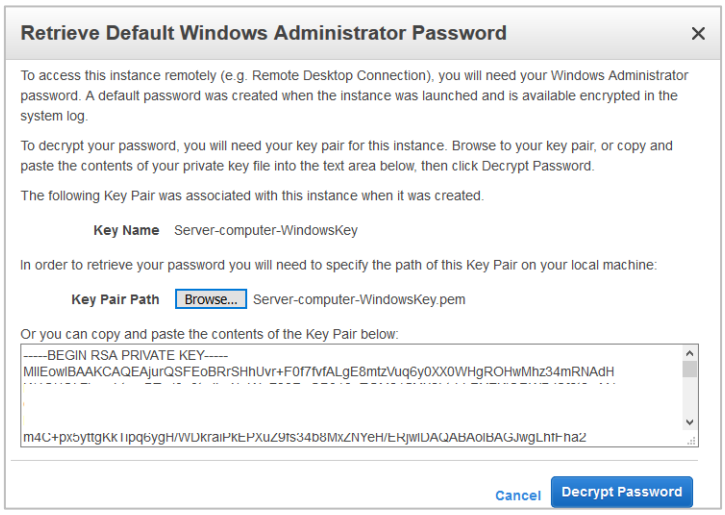
It should display the following:

- Instance State: running
- Status Checks: 2/2 checks passed

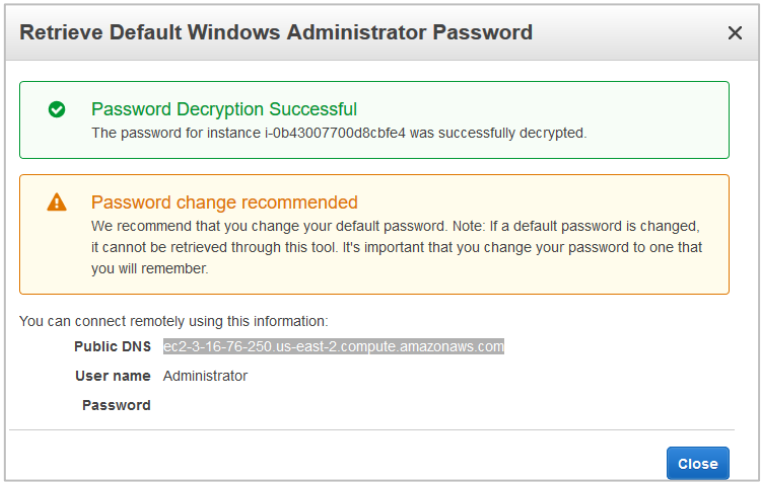


Select instance you have launched → Actions



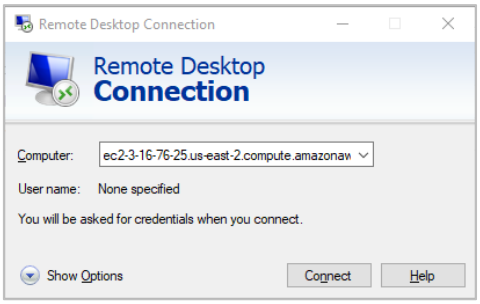


Browse server-computer-WindowsKey.pem file to decrypt and get password



Now you got password successfully. Click **Close**.

Go to your windows machine Start → Run → mstsc → Ok



Click **connect** and type user name and password you are connected to your EC2 windows instance.

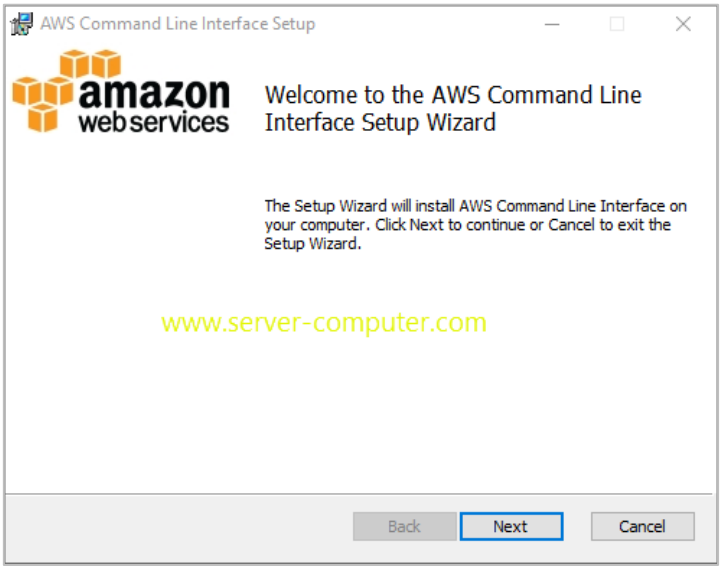
3. AWS CLI

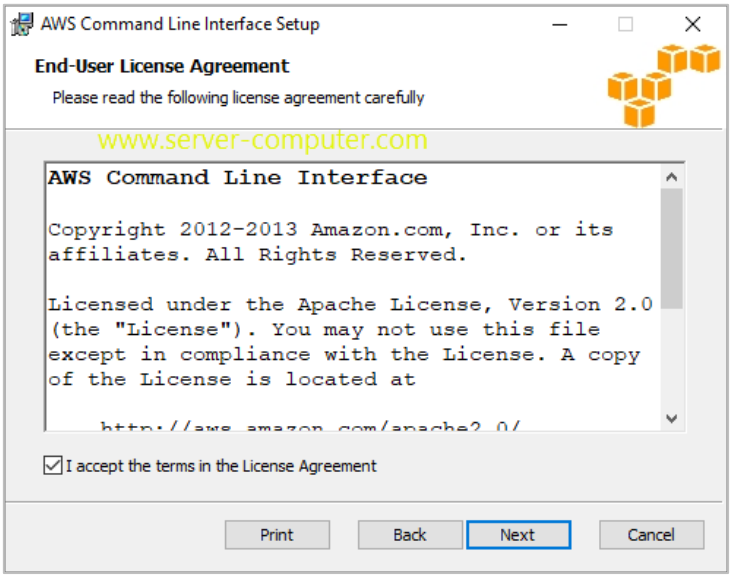
The AWS Command Line Interface (CLI) is a unified tool to manage your AWS services. With just one tool to download and configure, you can control multiple AWS services from the command line and automate them through scripts. The AWS CLI introduces a new set of simple file commands for efficient file transfers to and from Amazon S3.

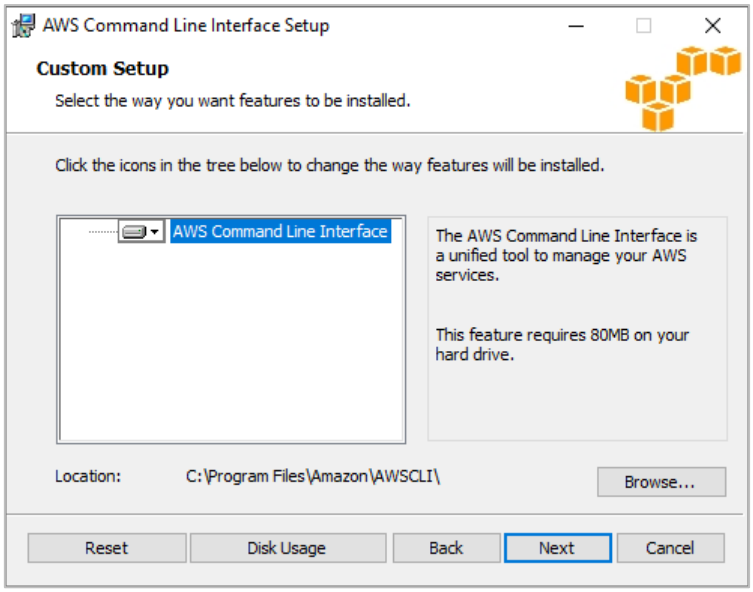
Aws cli configuration for Linux

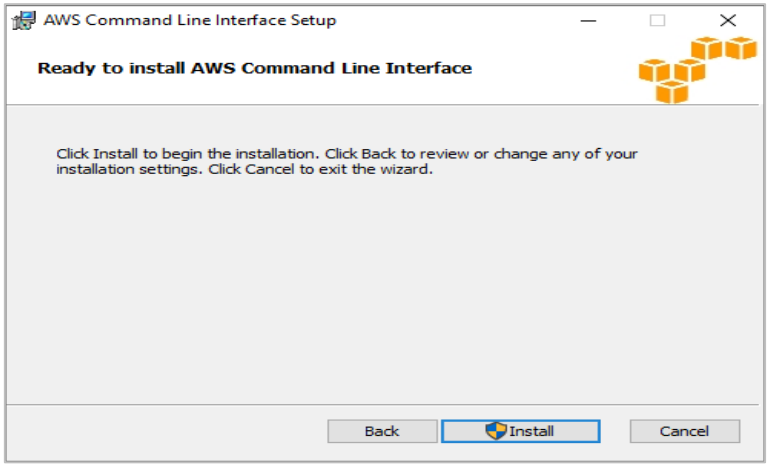
Download AWS CLI for Windows 64 Bit

You should require administrator privileges to install this package in windows machine Double click on .msi file









Click **Install**

Click **Finish**

Login back to AWS Management console and create user with programmatic access **Refer Topic 5 download** ACCESS Key and secret key

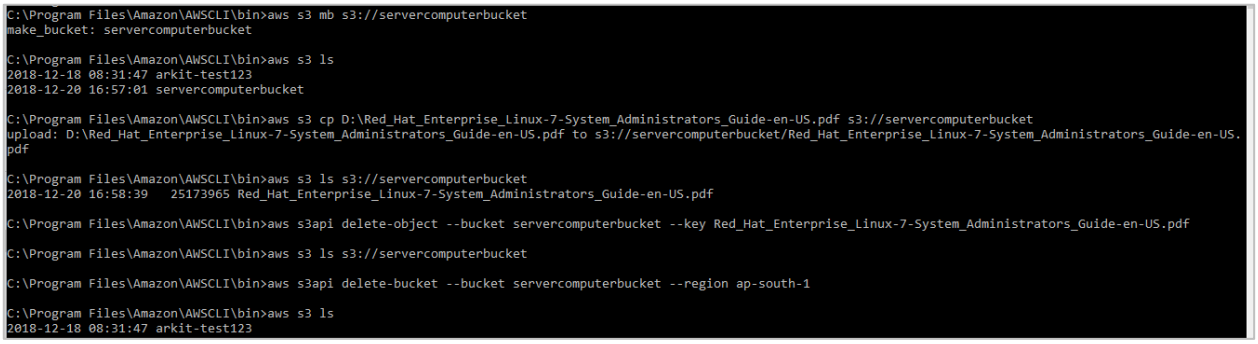
Start Menu → Run → cmd

cd C:\Program Files\Amazon\AWSCLI\bin Change your directory path to above mentioned

>aws configure

```
C:\Program Files\Amazon\AWSCLI\bin>aws configure
AWS Access Key ID [****************ULVQ]:
AWS Secret Access Key [****************XhN2]:
Default region name [ap-south-1]:
Default output format [None]:
```

Now successfully installed and configure aws cli, run few aws cli commands to manage AWS infrastructure



Create S3 Bucket

Bin> aws s3 mb s3://servercomputerbucket

make_bucket: servercomputerbucket

List S3 buckets

Bin> aws s3 ls

2018-12-18 08:31:47 arkit-test123

2018-12-20 16:57:01 servercomputerbucket

Upload Object to S3 Bucket

Bin> aws s3 cp D:\Red_Hat_Enterprise_Linux-7-System_Administrators_Guide-en-US.pdf s3://servercomputerbucket

upload: D:\Red_Hat_Enterprise_Linux-7-System_Administrators_Guide-en-US.pdf to s3://servercomputerbucket/Red_Hat_Enterprise_Linux-7-System_Administrators_Guide-en-US.pdf

List Objects in S3 Bucket

Bin> aws s3 ls s3://servercomputerbucket

2018-12-20 16:58:39 25173965 Red_Hat_Enterprise_Linux-7-System_Administrators_Guide-en-US.pdf

Delete Object from S3 bucket

Bin> aws s3api delete-object --bucket servercomputerbucket --key Red_Hat_Enterprise_Linux-7-System_Administrators_Guide-en-US.pdf

Bin> aws s3 ls s3://servercomputerbucket

Delete S3 bucket

Bin> aws s3api delete-bucket --bucket servercomputerbucket --region ap-south-1

4. Creating EC2 Instance using AWS CLI

Before creating an EC2 instance using AWS CLI collect few details

- AMI ID
- Instance Type
- Key Name (If there is no Key Pair create one)
- Security Group ID
- Subnet ID

Bin> aws ec2 run-instances --image-id ami-06bcd1131b2f55803 --count 1 --instance-type t2.micro --key-name KEYNAME --security-group-ids sg-857f92e9 --subnet-id subnet-51e5592c

Result

A custom AMI was created from the configured instance and a new instance was launched from it. A Windows instance was launched and accessed through Remote Desktop. The AWS CLI was installed and configured with access keys, S3 buckets and objects were managed from the command line, and an EC2 instance was created with the aws ec2 run-instances command.

Viva Questions

1. What does an AMI contain?

Ans. A template of the root volume, launch permissions that decide which AWS accounts can use it, and a block device mapping of the volumes attached at launch.

2. Why should an unused AMI be deregistered?

Ans. The AMI and its underlying EBS snapshot occupy storage that is billed. Deregistering the AMI and deleting the snapshot saves cost.

3. What information does the aws configure command ask for?

Ans. The access key ID, the secret access key, the default region name and the default output format.

4. What details are needed to launch an instance with the AWS CLI?

Ans. The AMI ID, the instance type, the key pair name, the security group ID and the subnet ID.

5. How is a Windows EC2 instance accessed?

Ans. Through Remote Desktop (mstsc) using the administrator password that is decrypted with the private key of the instance key pair.

Assignment Questions

1. Write AWS CLI commands to list all running instances in a region, stop an instance, start it again and terminate it. Capture the output of each command.

2. Create an AMI from a web server instance, copy the AMI to another region and launch an instance from the copied image. Explain why AMIs are region specific.

Experiment 4: Creating a Python Lambda Function with an SQS Trigger

4	a) Create a sample Python Lambda function. b) Set the Lambda Trigger as SQS and send a message to test invocations	CO2

Aim

To create a sample Python AWS Lambda function, configure an Amazon SQS queue as its trigger and send a message to the queue to test the invocation.

Description

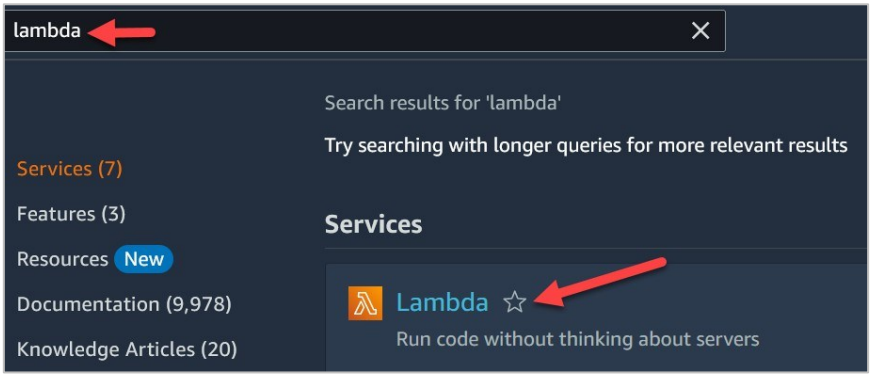
AWS Lambda runs code without provisioning or managing servers. A function is created with a runtime such as Python and is executed in response to events. Billing is based on the number of requests and the execution duration. The test feature in the Lambda console passes a sample event to the function and shows the execution result and the logs.

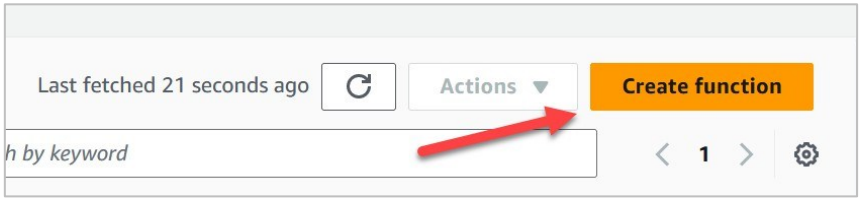
Amazon SQS is a managed message queue. When an SQS queue is configured as a Lambda trigger, the Lambda service polls the queue and invokes the function with batches of messages. After the function completes successfully the messages are deleted from the queue. This decouples the producer of the messages from the code that processes them.

Solution

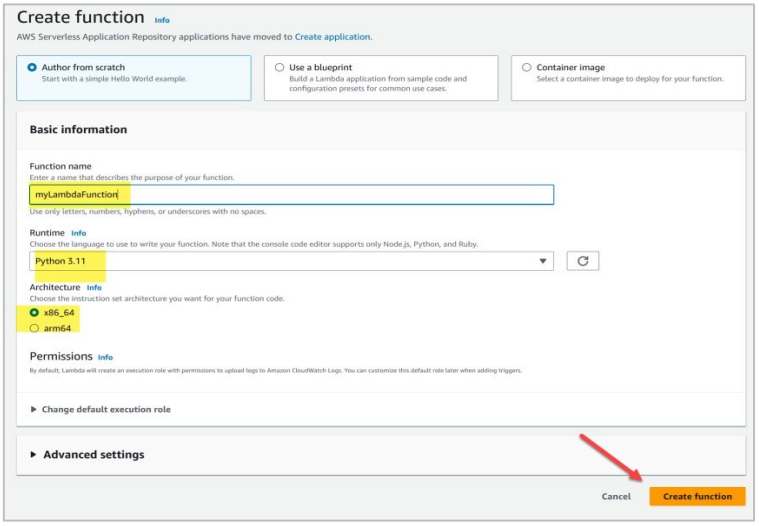
1. Create a sample Python Lambda function.
2. Set the Lambda Trigger as SQS and send a message to test invocations.

Answer

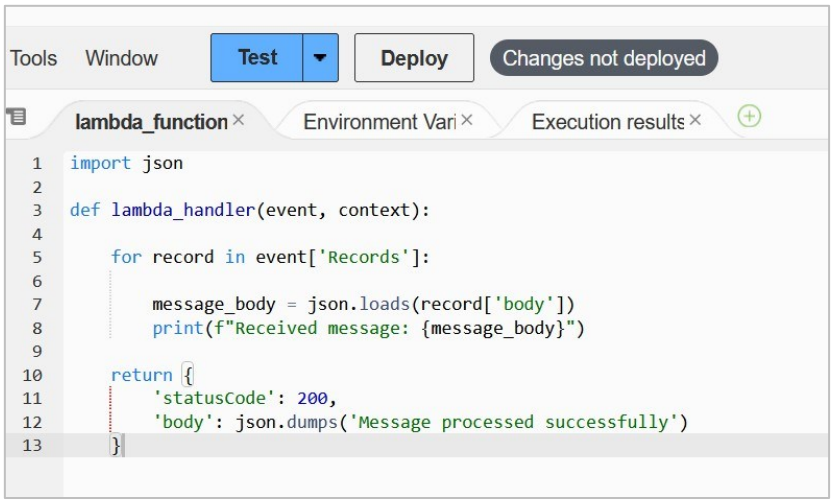




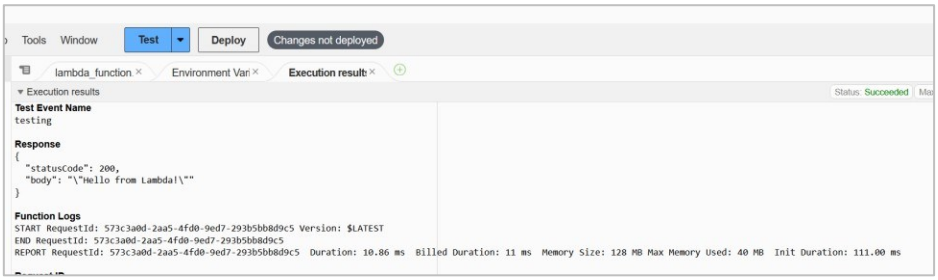
Click **Create Function**



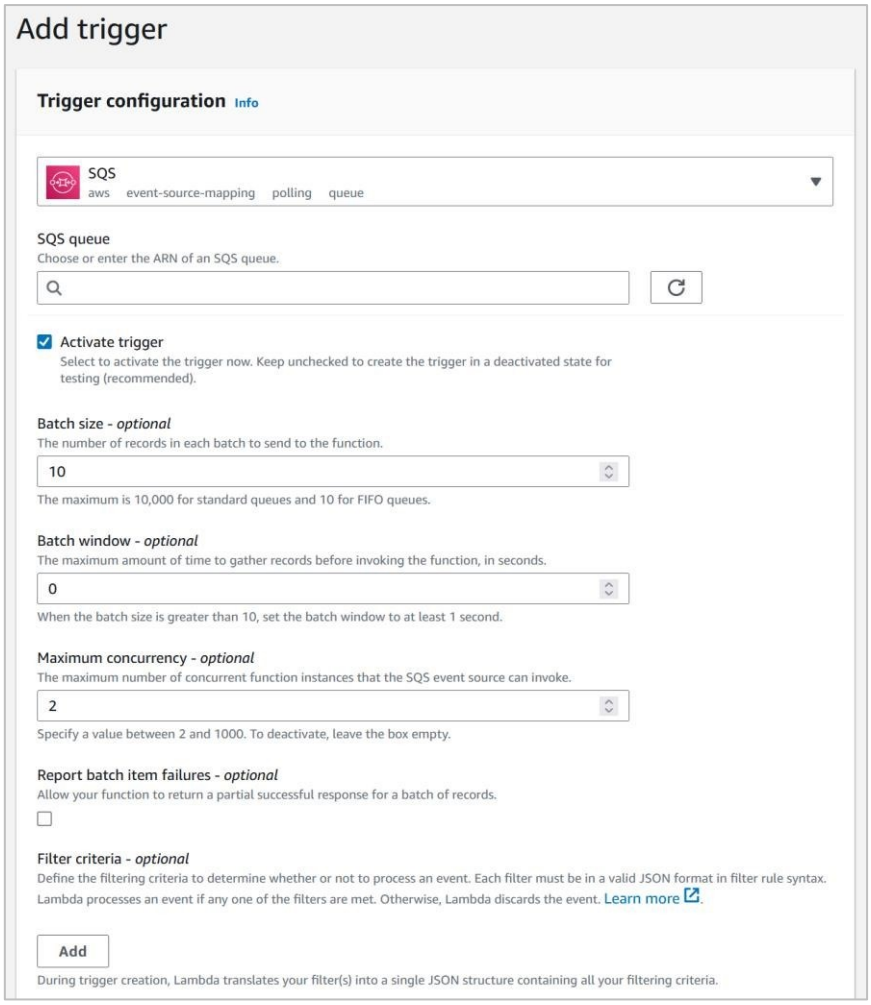
Click Create Function



Add trigger



When you click on the test above the results.



That's how you can create a Lambda function and adding the trigger.

Result

A Python Lambda function was created, an SQS queue was added as its trigger, and the function was invoked successfully when a test message was sent. The execution result and the logs were verified in the Lambda console.

Viva Questions

1. What is serverless computing?

Ans. A model in which the cloud provider runs the code and manages the servers, scaling and patching, and the user pays only for the execution time and requests.

2. Which runtimes does AWS Lambda support?

Ans. Python, Node.js, Java, Go, .NET, Ruby and custom runtimes packaged as container images.

3. What is a trigger or event source in Lambda?

Ans. A service or resource, such as SQS, S3, API Gateway or EventBridge, that invokes the function when an event occurs.

4. How does Lambda process messages from an SQS queue?

Ans. Lambda polls the queue, invokes the function with a batch of messages and deletes the messages from the queue when the function returns successfully.

5. What is the maximum execution time of a Lambda function?

Ans. 15 minutes per invocation.

Assignment Questions

1. Create a Lambda function that is triggered when an object is uploaded to an S3 bucket and writes the object name and size to CloudWatch Logs.

2. Modify the SQS-triggered function so that it fails for a specific message content and configure a dead-letter queue. Show the failed message landing in the dead-letter queue.

Experiment 5: Virtual Private Cloud (VPC) and Internet Gateway

5	a) VPC – Virtual Private Cloud (isolated Network) b) Create Internet gateway and attach to VPC	CO3

Aim

To create a Virtual Private Cloud with public and private subnets, create an internet gateway and a virtual private gateway, attach them to the VPC and configure route tables for the subnets.

Description

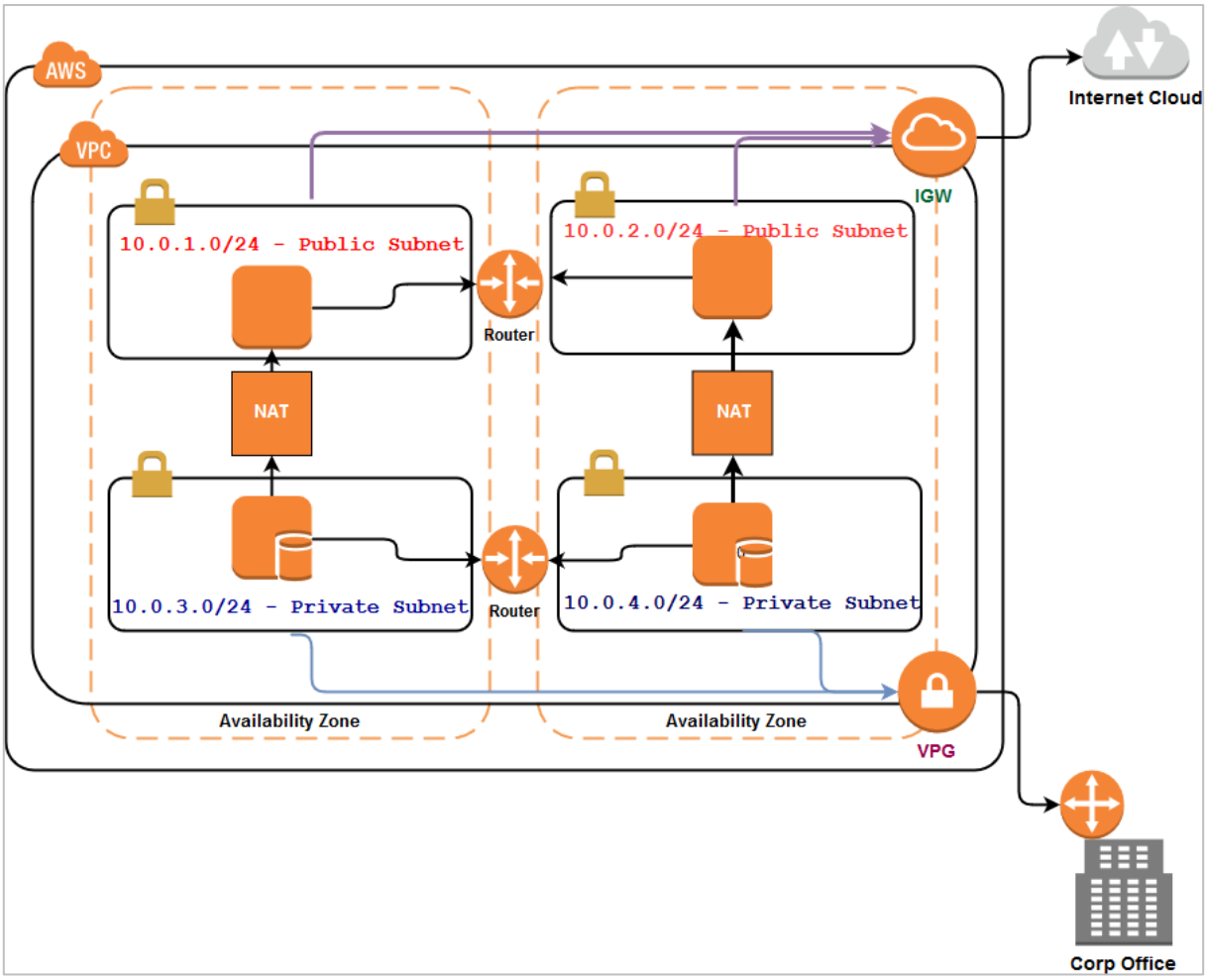
A VPC is a logically isolated virtual network inside an AWS region. It is defined with an IPv4 CIDR block, for example 10.0.0.0/16, and divided into subnets, each of which lies in one availability zone. An internet gateway is a horizontally scaled, highly available VPC component that lets instances in the VPC communicate with the internet. A virtual private gateway is the AWS side of a VPN connection to a corporate network.

Route tables contain routes that decide where the traffic from a subnet is sent. A subnet whose route table has a route for 0.0.0.0/0 pointing to the internet gateway is a public subnet. A subnet without that route is a private subnet. Each subnet must be associated with exactly one route table.

Solution

5.a VPC – Virtual Private Cloud (isolated Network)

A **virtual private cloud** (VPC) is a virtual network dedicated to your AWS account. It is logically isolated from other virtual networks in the AWS Cloud. You can launch your AWS resources, such as Amazon EC2 instances, into your VPC.



Picture: 1.1 Typical VPC Example

- EC2 Instance
- Virtual Private Gateway
- Router
- Customer Gateway
- Internet Gateway
- Availability Zone
- VPC subnet

Architecture Explanation:

- AWS in single region
- Two Availability zones
- One Virtual Private Cloud

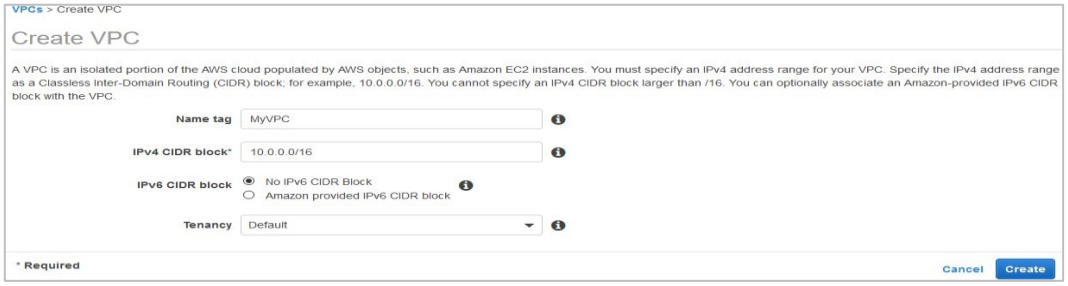
- Four Subnets Two Are Public and Two Are Private subnets
- Four instances Two App Servers, Two Database Servers
- One Internet Gateway to access internet
- One Virtual Private Gateway to Connect Corporate Office
- Two routers one is connected to private subnets, another is connected to public subnets

We would like to host web application with two web app servers and two Database servers. Two Tier architecture. Web app servers will serve to public, from public facing subnets. Database servers are in private network and only have access to app servers and corporate network (VPG).

When Database servers want to download any kind of files/patches from internet it routes through NAT Gateway and get the internet data from web app servers.



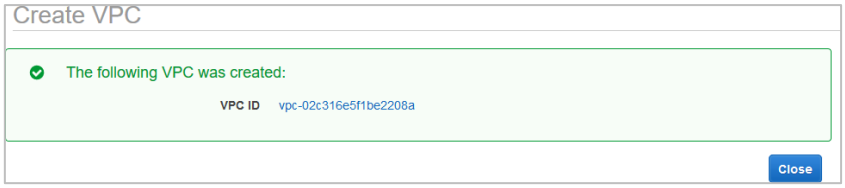
AWS Console → Services → Networking & Content Delivery → VPC → Your VPCs



- **VPC Name:** MyVPC
- **IPv4 CIDR Block:** 10.0.0.0/16 (Use this CIDR Calculator)

Click **Create**

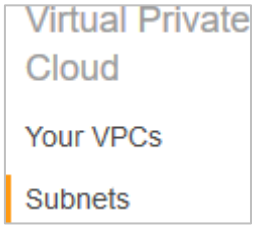
Result	
CIDR Range	10.0.0.0/16
Netmask	255.255.0.0
Widlcard Bits	0.0.255.255
First IP	10.0.0.0
Last IP	10.0.255.255
Total Host	65536
CIDR	
10.0.0.0/16	



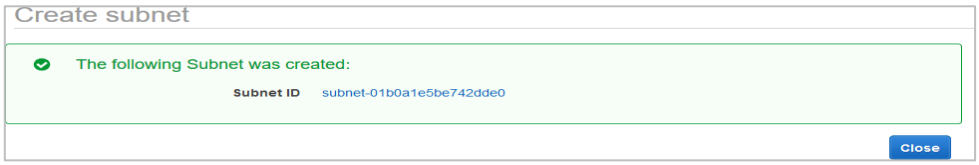
Your VPC created successfully.

Create subnets

Inside VPC to divide smaller blocks and separation







In Similar way, create all four subnets

Subnet Name	Availability Zone	CIDR Block	Private/Public
S1-Private	Us-east-2a	10.0.1.0/24	Private
S2-Private	Us-east-2b	10.0.2.0/24	Private
S3-Public	Us-east-2a	10.0.3.0/24	Public
S4-Public	Us-east-2b	10.0.4.0/24	Public

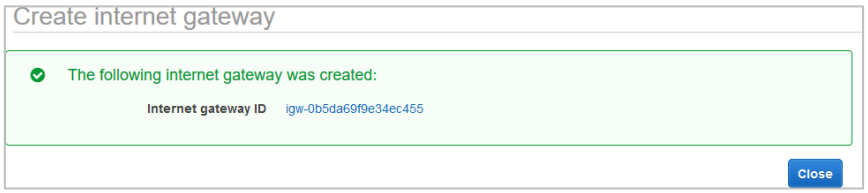
Name	Subnet ID	State	VPC server-computer.com	IPv4 CIDR	Available IPv4	IPv6 CIDR	Availability Zone	Availability Zone ID
S1-Private	subnet-01b0a1e5be742dde0	available	vpc-02c316e5f1be2208a | MyVPC	10.0.1.0/24	251	-	us-east-2a	use2-az1
S2-Private	subnet-0415e767640ae4ef9	available	vpc-02c316e5f1be2208a | MyVPC	10.0.2.0/24	251	-	us-east-2b	use2-az2
S3-Public	subnet-01f8724bb68578a99	available	vpc-02c316e5f1be2208a | MyVPC	10.0.3.0/24	251	-	us-east-2a	use2-az1
S4-Public	subnet-09d6c82e020a61325	available	vpc-02c316e5f1be2208a | MyVPC	10.0.4.0/24	251	-	us-east-2b	use2-az2

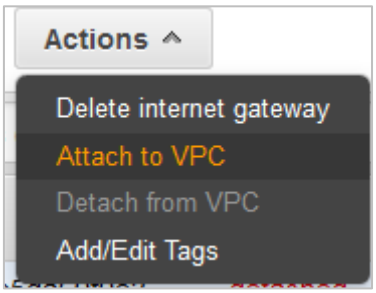
8.b Create Internet gateway and attach to VPC

Internet Gateways. An internet gateway is a horizontally scaled, redundant, and highly available VPC component that allows communication between instances in your VPC and the internet. It therefore imposes no availability risks or bandwidth constraints on your network traffic.

Attach to S3 and S4, after attach S3 and S4 become public subnets.







Now attach Internet Gateway to VPC



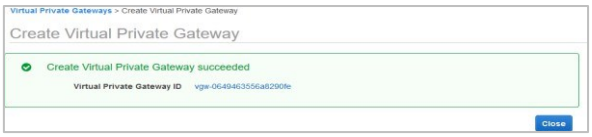
Select MyVPC in drop down menu Click **Attach**

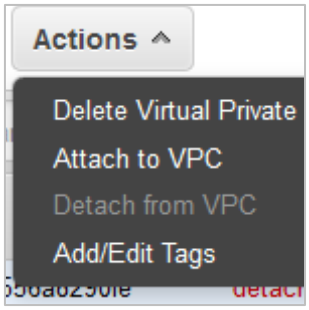
C Create Virtual Private Gateway and Attach to VPC

It can be a physical or software appliance. The anchor on the AWS side of the VPN connection is called a virtual private gateway. The following diagram shows your network, the customer gateway, the VPN connection that goes to the virtual private gateway, and the VPC.

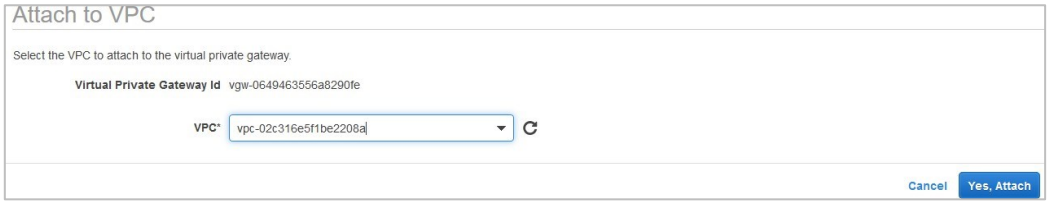








Attach VGW to MyVPC



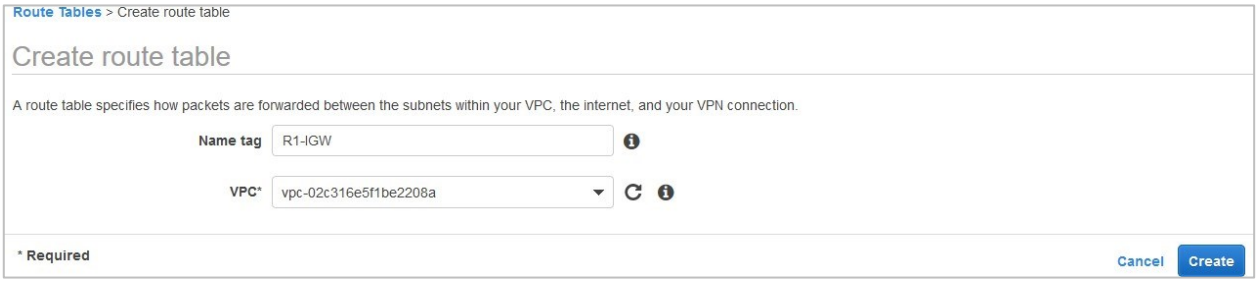
D Create route tables and attach to subnets

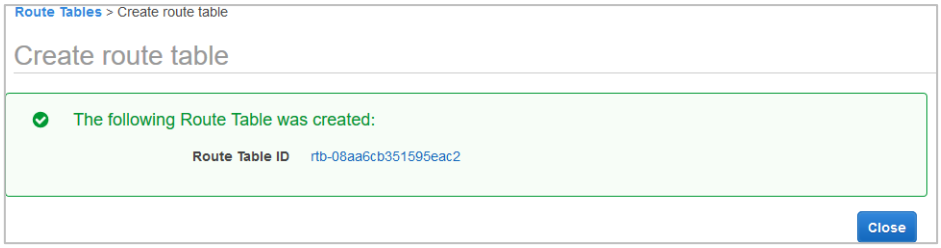
Route Tables. A route table contains a set of rules, called routes that are used to determine where network traffic is directed. Each subnet in your VPC must be associated with a route table; the table controls the routing for the subnet.

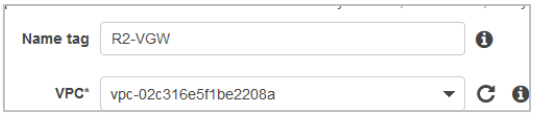
One route for Internet gateway, another for Virtual private gateway (R1-IGW and R2-VGW)

- Route - 0.0.0.0/0 to IGW
- Route - 192.168.0.0/16 to VGW

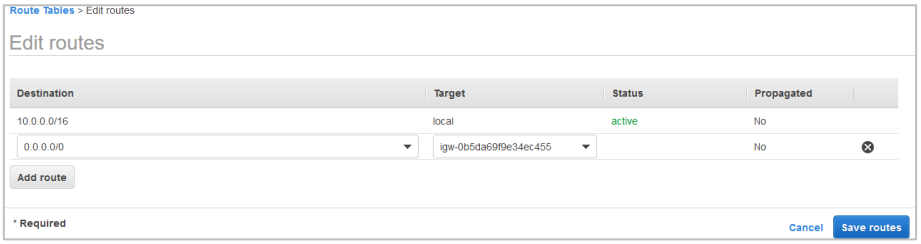


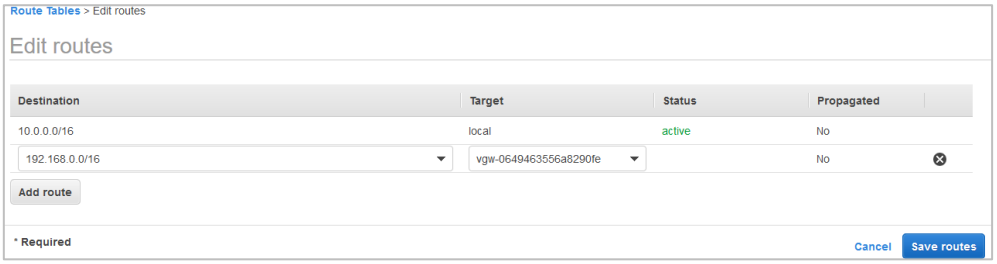




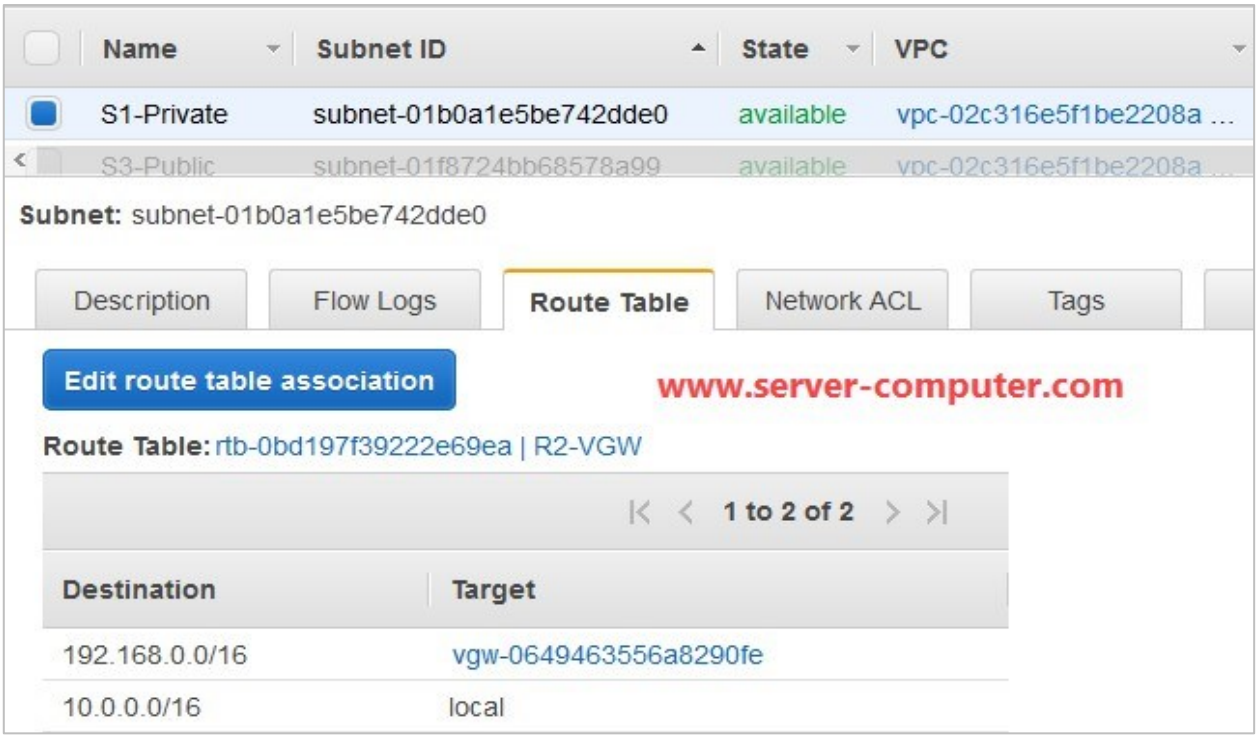


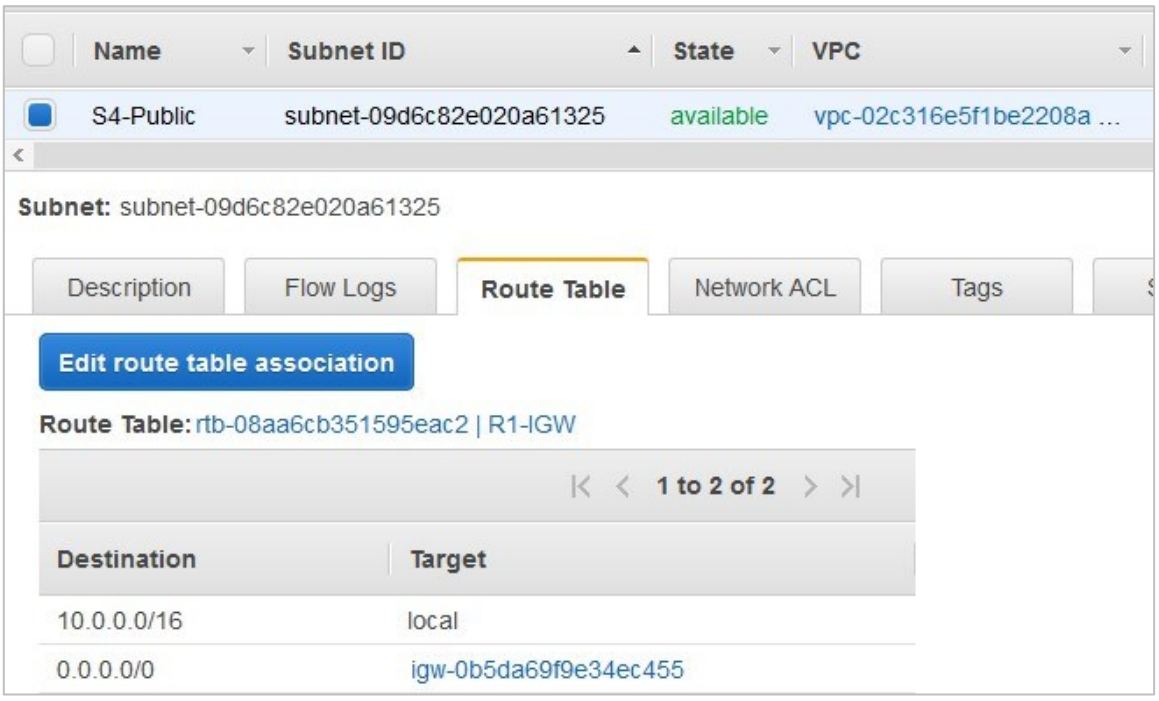
Now edit R1-IGW and add routing rule as mentioned below





Attach routing tables to subnets. R1-IGW to S3-Public and S4-Public, public network required to have internet access. Attach R2-VGW to S1-Private and S2-Private (No internet become a private subnets)





Result

The VPC MyVPC was created with four subnets, two public and two private. An internet gateway and a virtual private gateway were created and attached to the VPC, and two route tables were configured and associated with the public and private subnets respectively.

Viva Questions

1. What is a VPC?

Ans. A logically isolated virtual network dedicated to one AWS account in which AWS resources such as EC2 instances are launched.

2. What is a CIDR block?

Ans. A notation such as 10.0.0.0/16 that defines a range of IP addresses. The number after the slash is the number of fixed network prefix bits.

3. What makes a subnet public?

Ans. Its route table contains a route to an internet gateway, and the instances in it have public IP addresses.

4. What is the difference between an internet gateway and a virtual private gateway?

Ans. The internet gateway provides connectivity to the internet. The virtual private gateway terminates a VPN connection to an on-premises network.

5. How many availability zones can a subnet span?

Ans. One. Each subnet resides entirely within a single availability zone.

Assignment Questions

1. Launch one EC2 instance in a public subnet and one in a private subnet of MyVPC. Show that the public instance is reachable from the internet and that the private instance can be reached only from the public instance.

2. Create a network ACL that denies inbound SSH from the internet to the public subnet while the security group still allows it, and explain the result.

Experiment 6: VPC with Public and Private Subnets and NAT Gateway

week 6	a) Create a VPC with 120.0.0.0/16 CIDR block. b) Create 1 public subnet and 2 private subnets and make sure you connect a NAT gateway for internet connectivity to a private subnet

Aim

To create a VPC with the 120.0.0.0/16 CIDR block, create one public subnet and two private subnets, attach an internet gateway, and connect a NAT gateway so that the private subnets get outbound internet connectivity.

Description

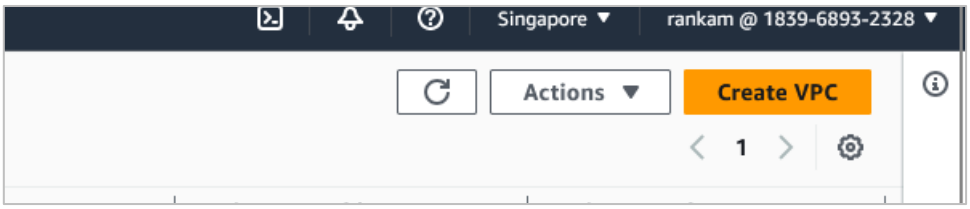
Instances in a private subnet do not have public IP addresses and cannot be reached from the internet. They still need outbound access for software updates and patches. A NAT gateway is a managed network address translation service placed in a public subnet with an Elastic IP address. The private subnet route table sends 0.0.0.0/0 traffic to the NAT gateway, which forwards it to the internet through the internet gateway and returns the responses.

Separate route tables are used for the public subnet, which routes to the internet gateway, and for the private subnets, which route to the NAT gateway. A NAT gateway is charged per hour and per GB of data processed, so it should be deleted when the lab is over.

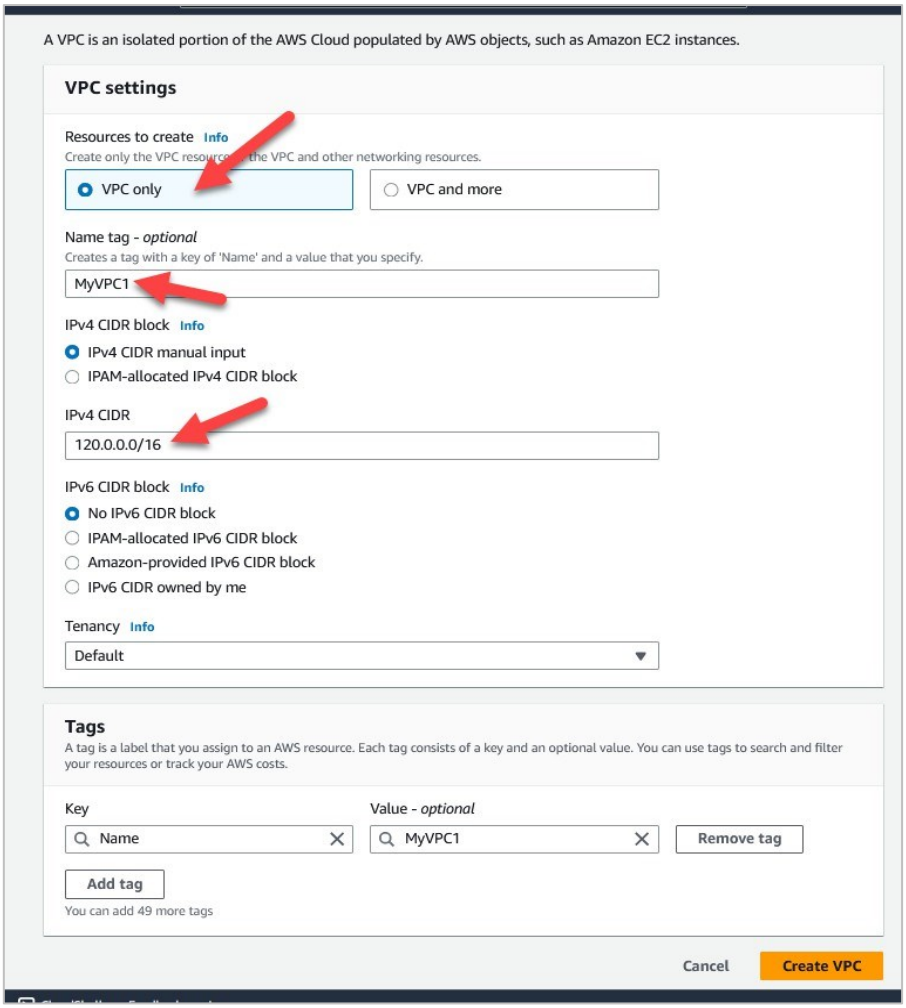
Solution

1. Create a VPC with 120.0.0.0/16 CIDR block.
2. Create 1 public subnet and 2 private subnets and make sure you connect a NAT gateway for internet connectivity to a private subnet

Answer:



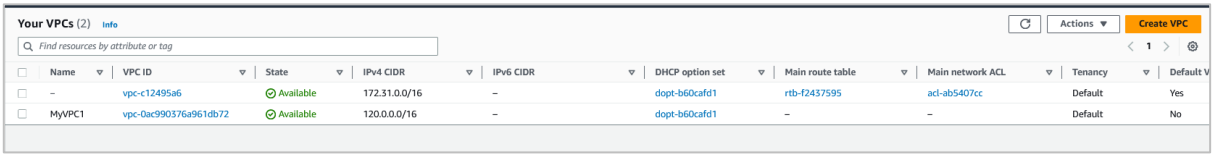
Login to the AWS Management console Services select **VPC → Create VPC**



Select VPC Only option to create VPC with customized options

Provide a VPC name

Select IPv4 CIDR manual input (Currently we are targeting for IPv4 only)

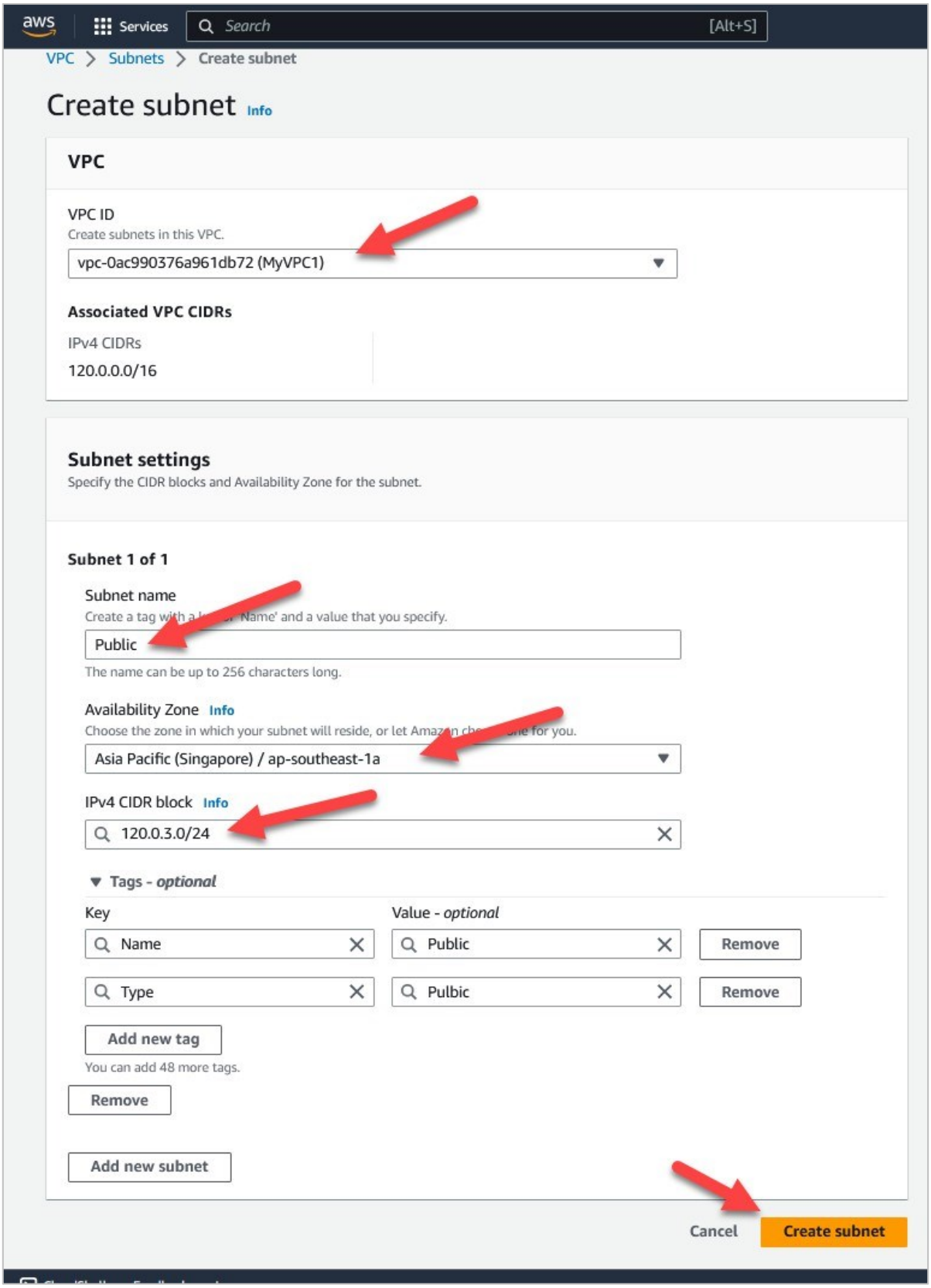


1. Select Default tenancy (Shared resources) Click "**Create VPC**"

MyVPC1 is created successfully. Now create the subnets as per the requirement.

Creating Subnets

In **VPC service → Click on subnets → Create subnet**



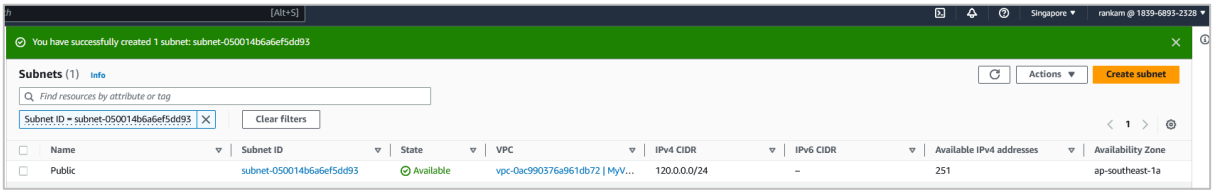
Select the correct VPC.

Provide a subnet Name i.e., Public

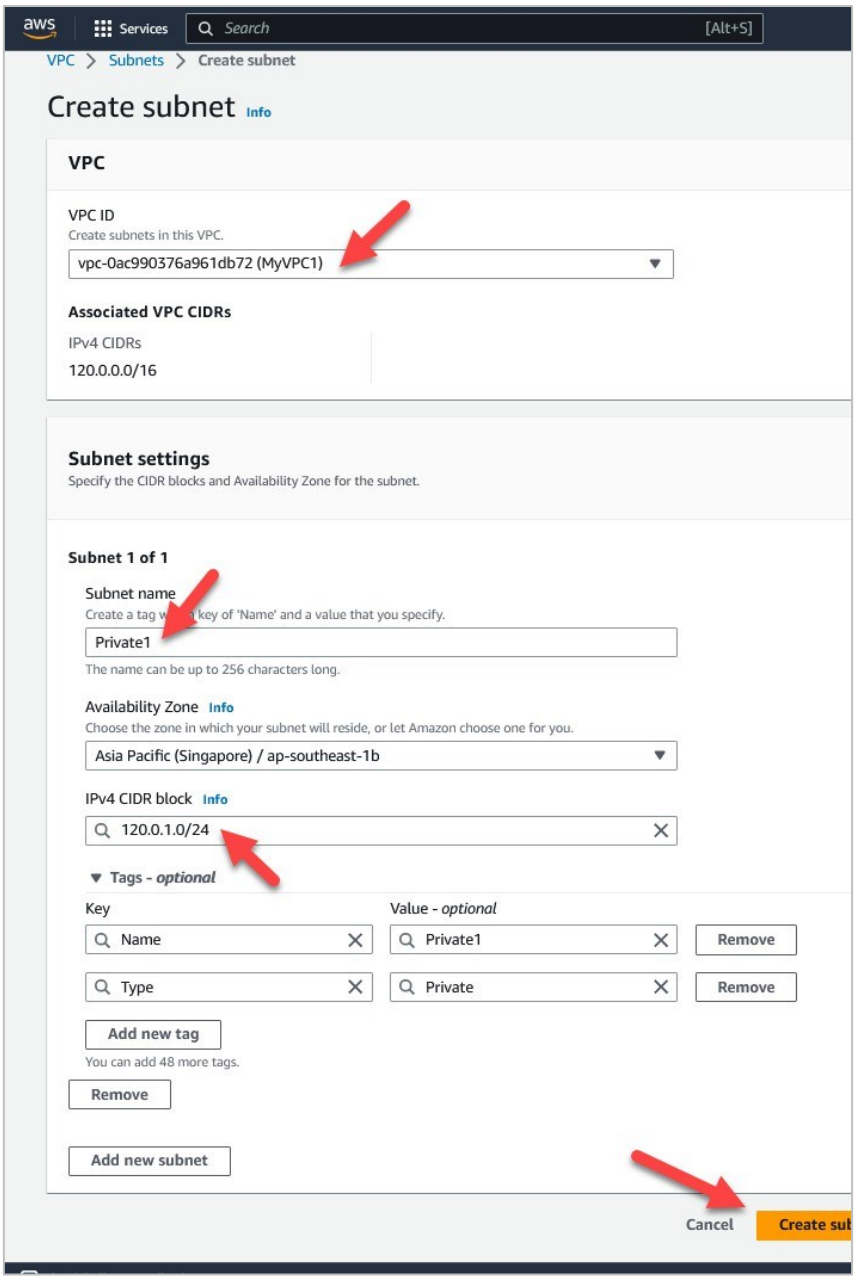
Assign the IPv4 CIDR block for this subnet 120.0.3.0/24.

Provide Tags for easy tracking and identification.

Click "**Create subnet**"



The public subnet has been created successfully.

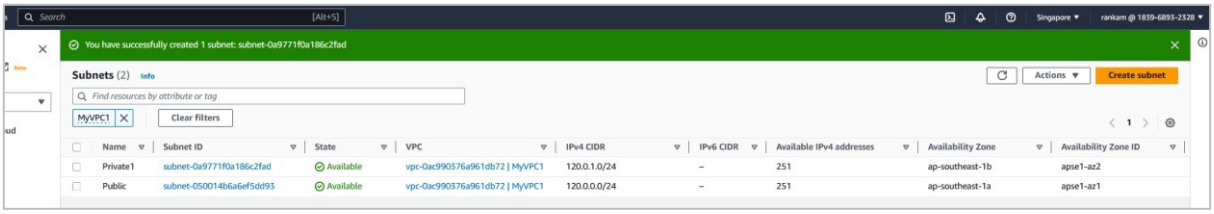


Click "**Create subnet**"

Select the appropriate VPC.

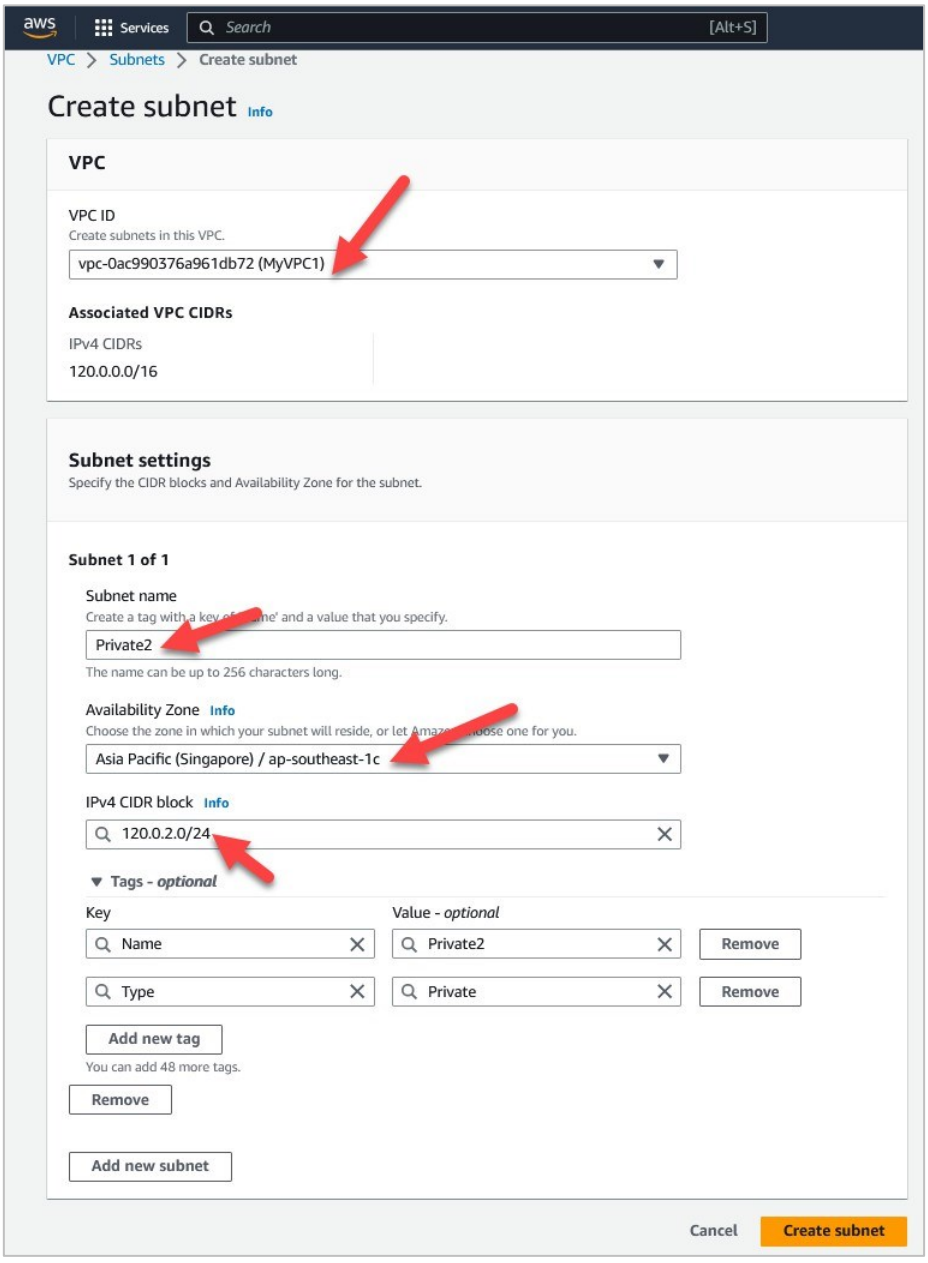
Provide a subnet name i.e., Private1.

1. Select the AZ (Availability Zone) and select a different AZ than another subnet for redundancy.



2. Provide IPv4 CIDR block i.e., 120.0.1.0/24. Click "**Create subnet**"

Private1 subnet created successfully.

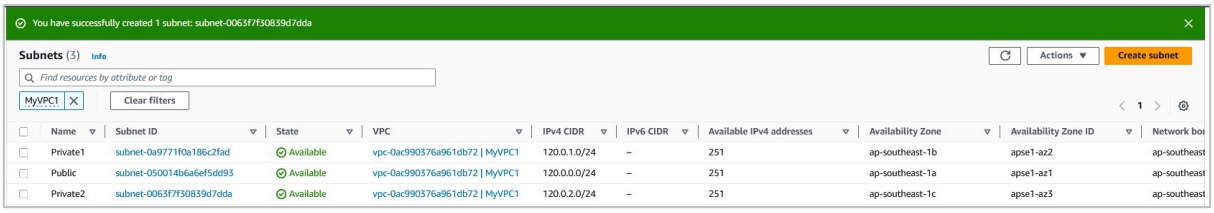


Click Create subnet

Select appropriate VPC,

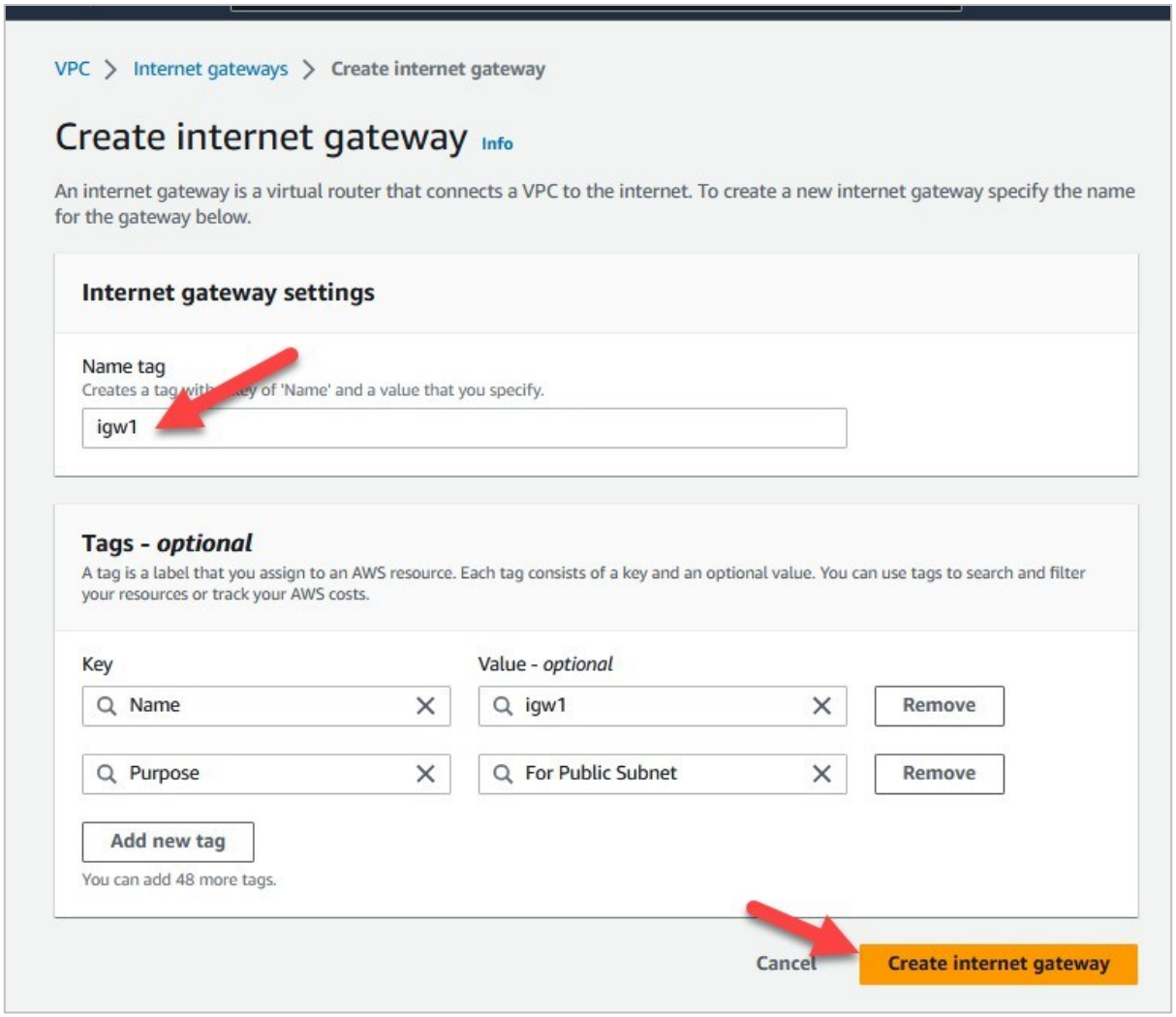
Provide a subnet name i.e., Private2.

1. Select the AZ (Availability Zone) and select a different AZ than another subnet for redundancy.



2. Provide IPv4 CIDR block i.e., 120.0.2.0/24. Click "**Create subnet**"

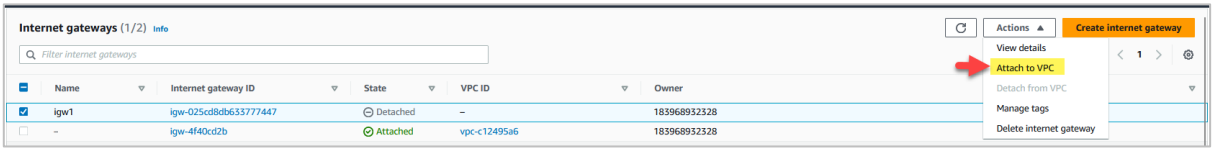
Create Internet Gateway



Click on "Create Internet Gateway"

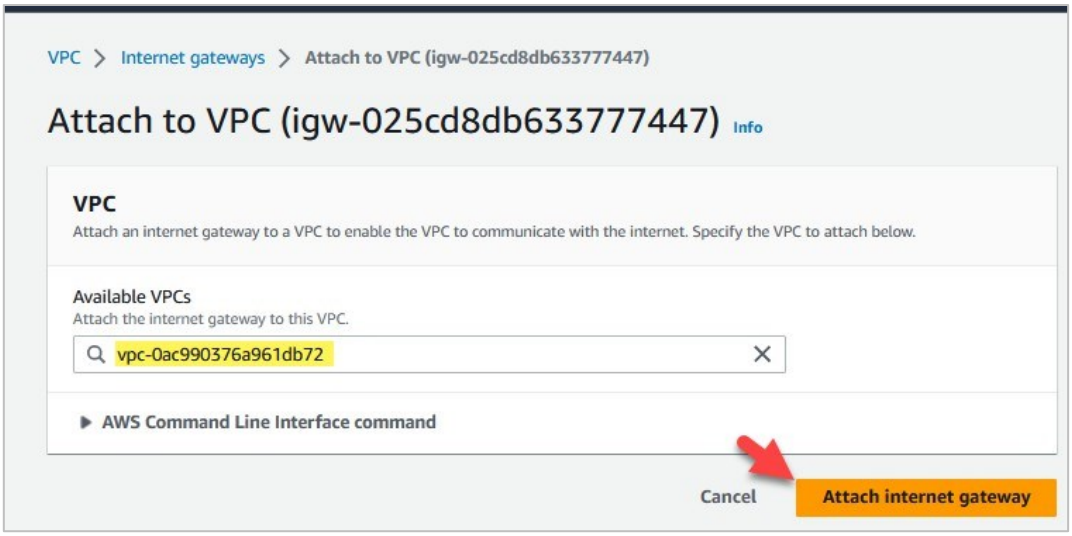
Provide a Internet Gateway a Name "**igw1**"

Provide Tags for later identification



Click "**Create internet Gateway**" IGW is created successfully.

Select the "igw1" which is newly created, **Actions → Attach to VPC**



Select "MyVPC1" which is a newly created then click on "**Attach internet gateway**"

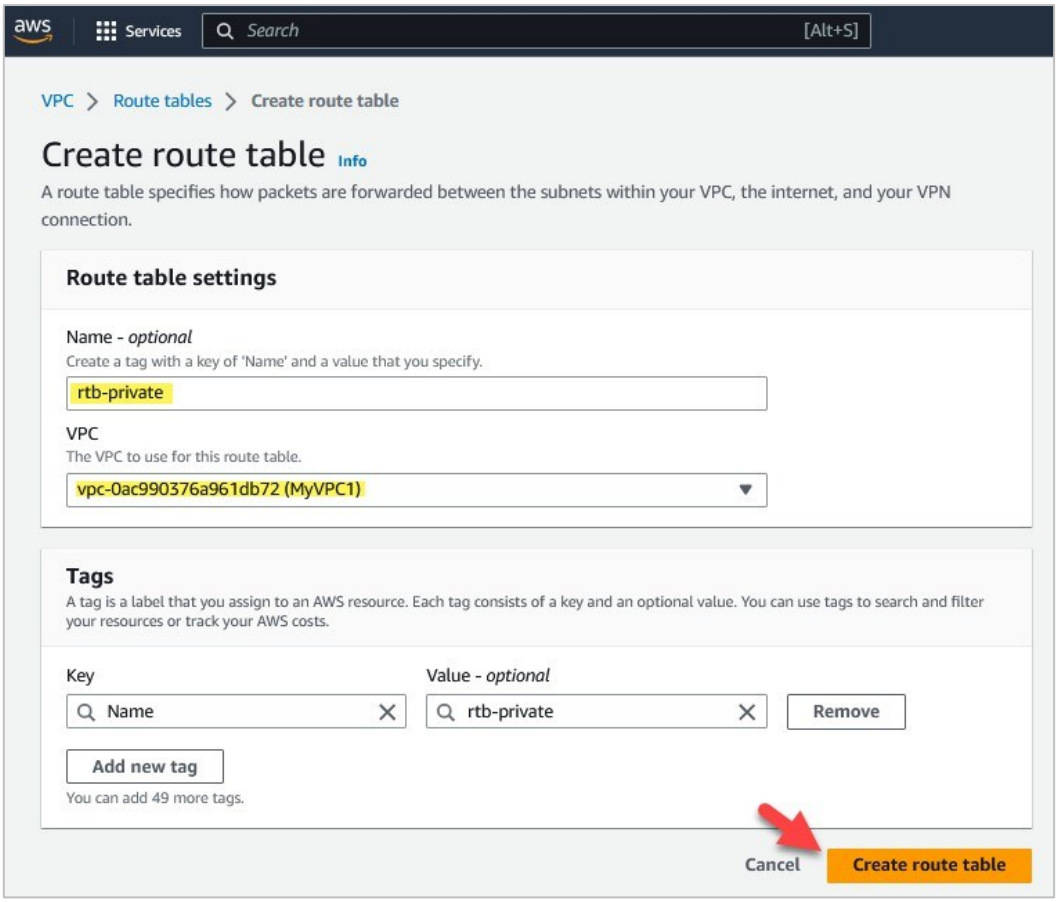
Enable Internet Route to Public Subnet

We require multiple route tables to add routes to them. Since we have a single route table I am going to create another route table for the private subnet.

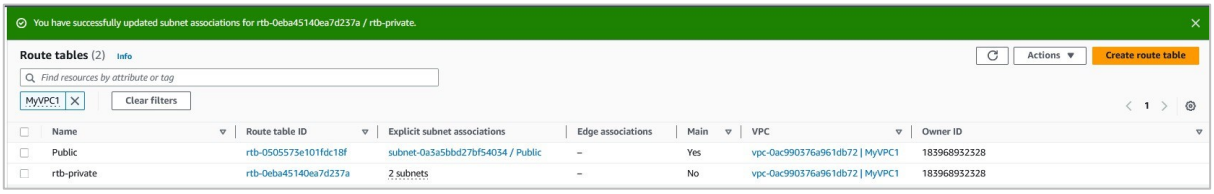
1. De-associate private subnets from existing subnets (to avoid having IGW and route table)

Associate private subnets to the "rtb-private" subnet to have different routes.

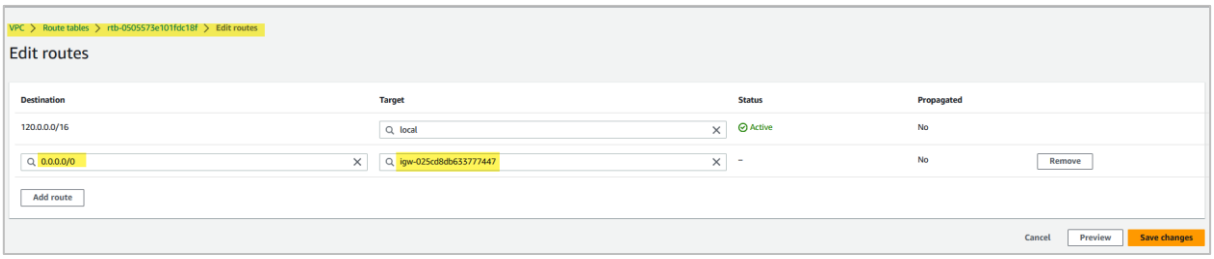
In **VPC service → route tables → Create route table**



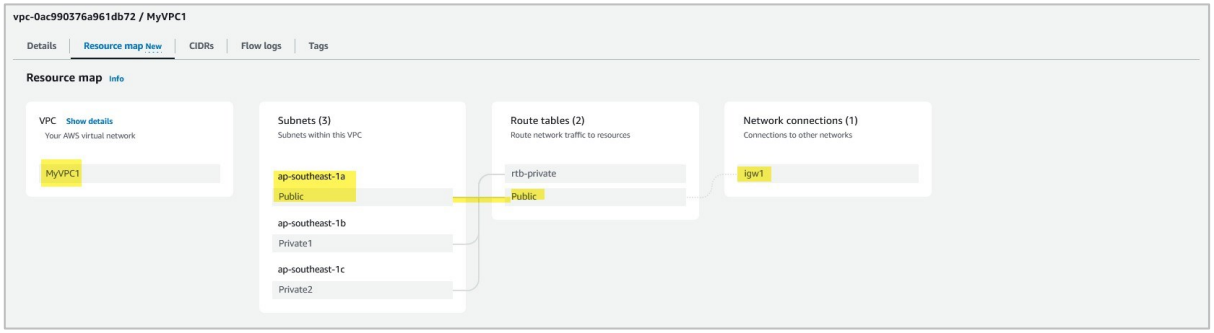
Click "**Create route table**"



Edit Public route table and add internet route

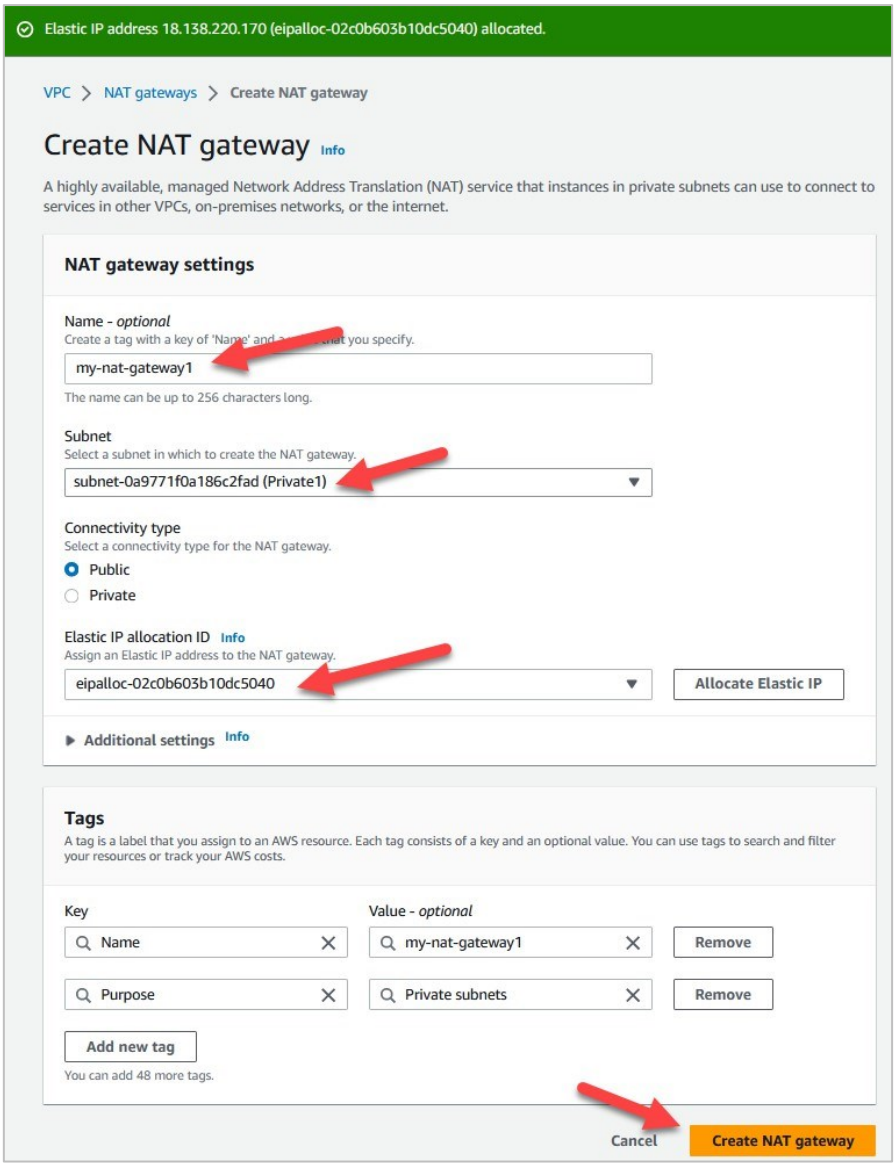


Now Public subnet have internet access.



Create NAT Gateway

VPC Service → NAT gateways → Create NAT gateways →



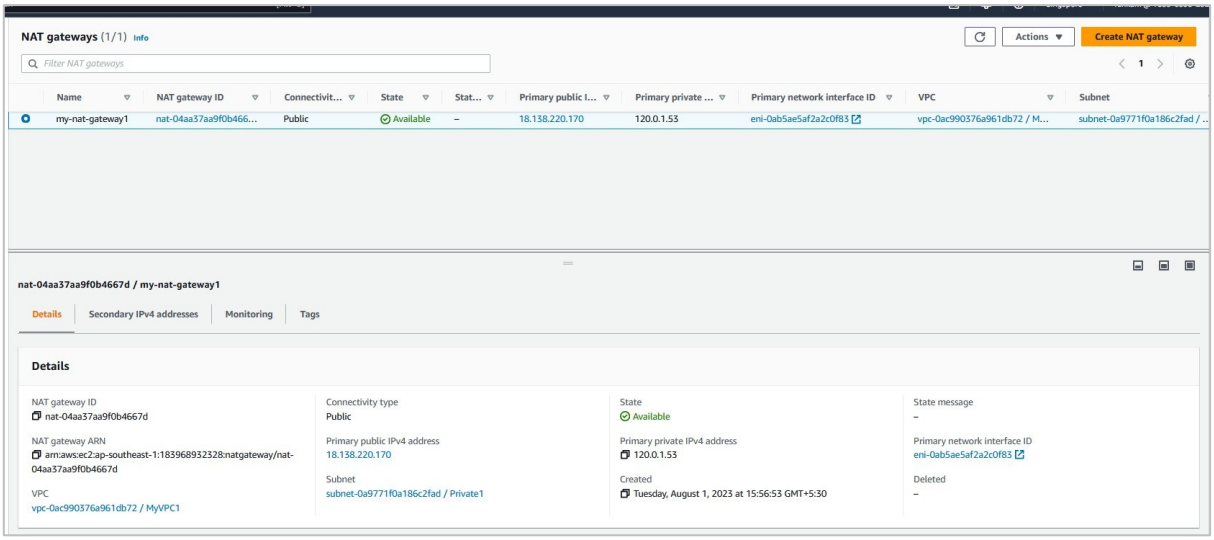
Provide a NAT gateway name i.e., my-nat-gateway1.

Select the subnets.

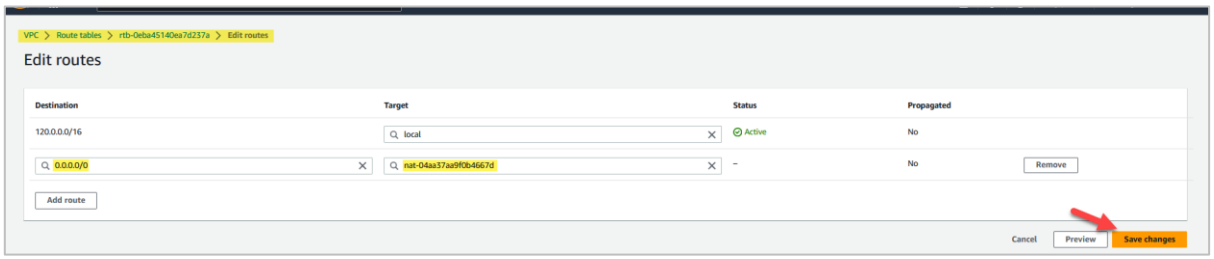
Connectivity type Public

Assign Elastic IP

Click "**Create NAT Gateway**"



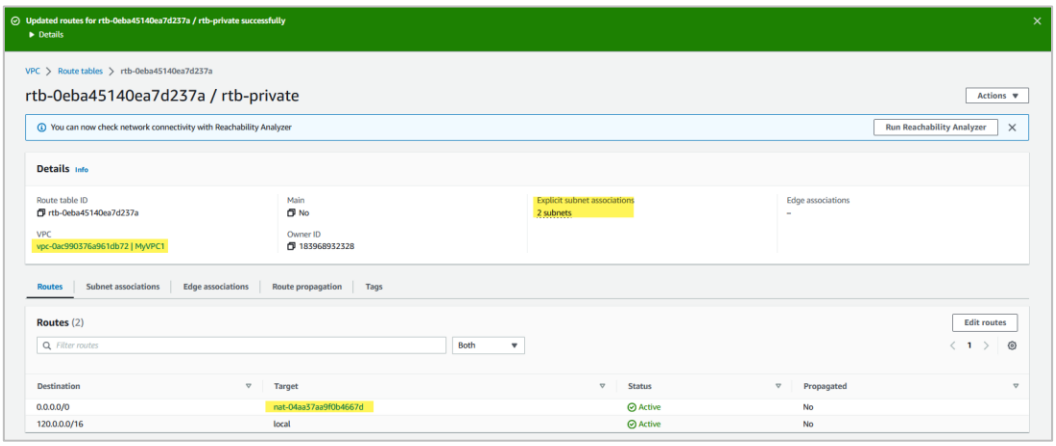
NAT Gateway is created successfully.



VPC Service → Route tables → Select Private route table → Edit route table

Add another route Destination: 0.0.0.0/0 Target: NAT-GATEWAY

Click "**Save Changes**"



Activity Completed.

Result

The VPC MyVPC1 (120.0.0.0/16) was created with the subnets Public (120.0.3.0/24), Private1 (120.0.1.0/24) and Private2 (120.0.2.0/24). The internet gateway igw1 was attached, the NAT gateway was created in the public subnet with an Elastic IP, and the private route table was updated with a 0.0.0.0/0 route to the NAT gateway.

Viva Questions

1. What is a NAT gateway?

Ans. A managed service that lets instances in a private subnet start outbound connections to the internet while blocking inbound connections from the internet.

2. In which subnet is the NAT gateway created and why?

Ans. In a public subnet, because it needs a route to the internet gateway and an Elastic IP address.

3. What is an Elastic IP address?

Ans. A static public IPv4 address allocated to the account that can be associated with an instance or a NAT gateway.

4. Why is a separate route table needed for the private subnets?

Ans. The private subnets need a default route to the NAT gateway while the public subnet routes to the internet gateway. One route table cannot hold two different default routes.

5. How many usable IP addresses are there in a /24 subnet in AWS?

Ans. 256 addresses minus the 5 reserved by AWS, which leaves 251 usable addresses.

Assignment Questions

1. Launch an instance in Private1, connect to it through a bastion host in the public subnet and run sudo yum update to prove that the NAT gateway provides outbound internet access.

2. Replace the NAT gateway with a NAT instance (an EC2 instance with the source/destination check disabled) and compare the two in terms of cost, availability and bandwidth.

Experiment 7: EC2 Instance with EBS Volume Attachment and Resize

7	a) Launch a Linux EC2 instance. b) Create an EBS volume with 20 GB of storage and attach it to the created EC2 instance. c) Resize the attached volume and make sure it reflects in the connected instance

Aim

To launch a Linux EC2 instance, create a 20 GB EBS volume, attach it to the instance, create a file system on it, and resize the volume so that the new size is reflected in the instance.

Description

Amazon Elastic Block Store provides persistent block storage volumes for EC2 instances. A volume lives in one availability zone and can be attached to an instance in the same zone. After attachment the volume appears in the operating system as a block device such as /dev/xvdb. A file system is created on it with mkfs and the device is mounted on a directory.

EBS volumes can be modified while they are in use. The size, the volume type and the IOPS can be changed from the console. After the volume is modified, the file system has to be grown inside the operating system, using growpart, xfs_growfs or resize2fs depending on the partition layout and the file system type, so that the added space becomes usable.

Solution

Module-1: EBS Assignment

Problem Statement:

You work for XYZ Corporation. Your corporation wants to launch a new web-based application using AWS Virtual Machines, and configure the resources accordingly with appropriate storage for the tasks.

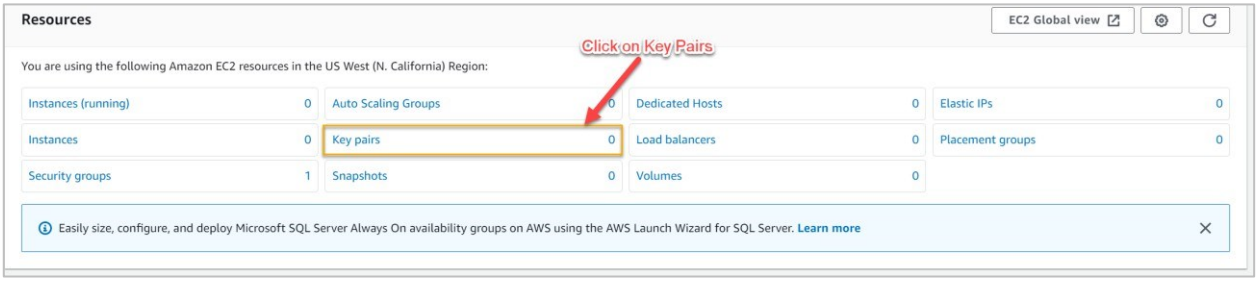
You are asked to perform the following tasks:

1. Launch a Linux EC2 instance.
2. Create an EBS volume with 20 GB of storage and attach it to the created EC2 instance.
3. Resize the attached volume and make sure it reflects in the connected instance.

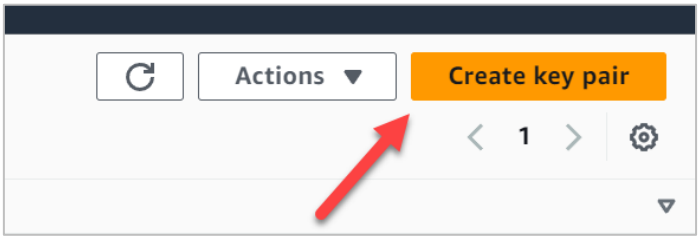
Answer:

Login to AWS Console using your credentials https://aws.amazon.com/console/

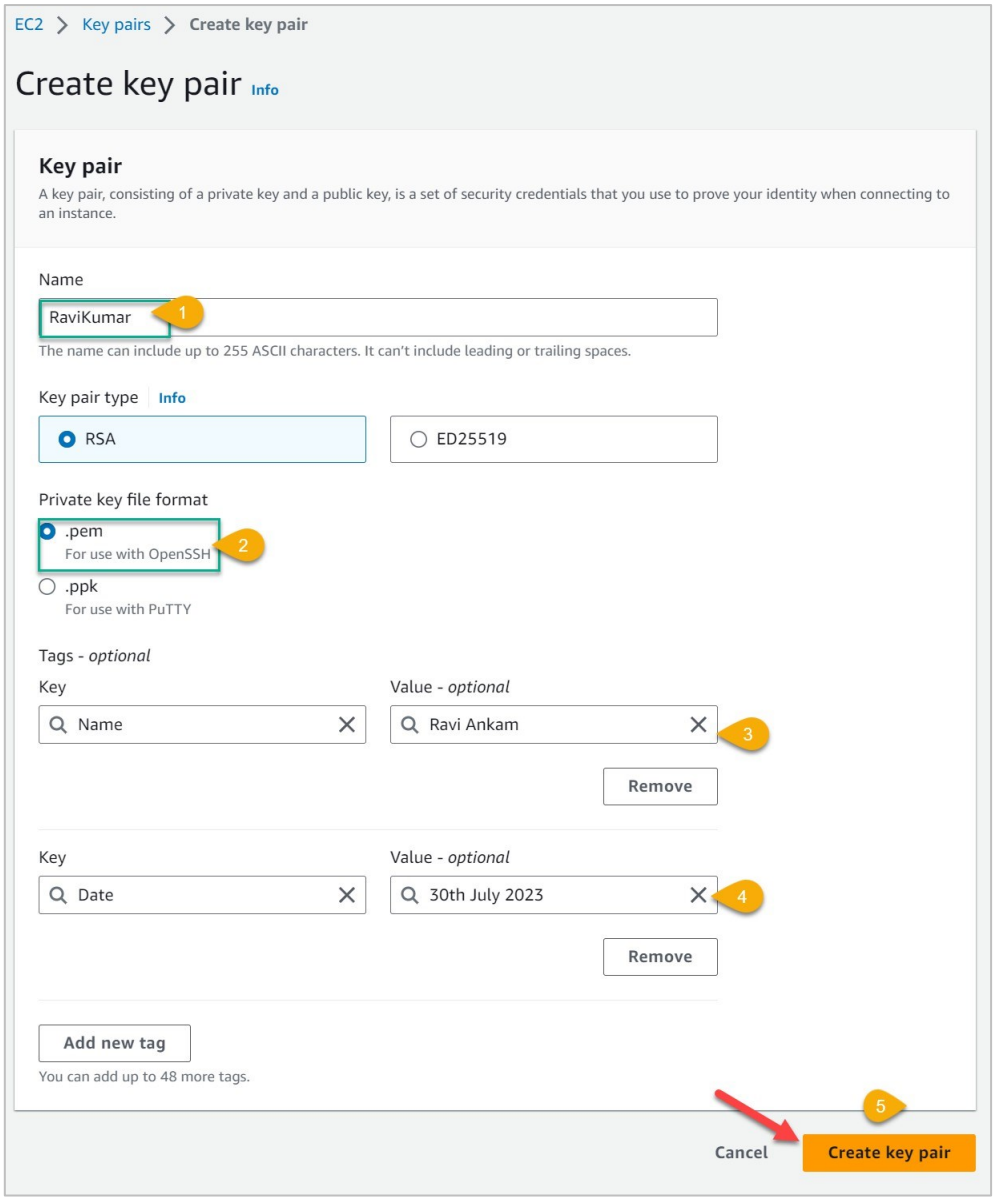
Navigate to EC2 service or search for EC2 in search bar click on it.



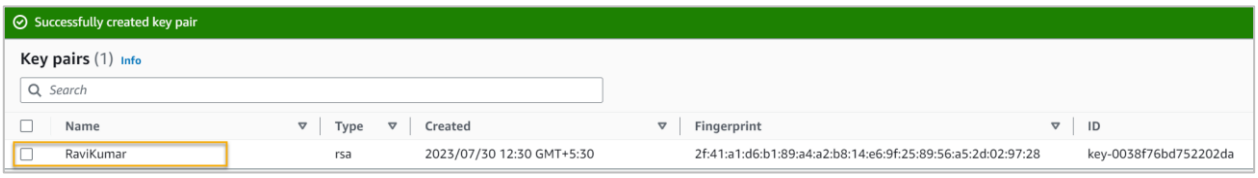
Step1: Create Key Pair



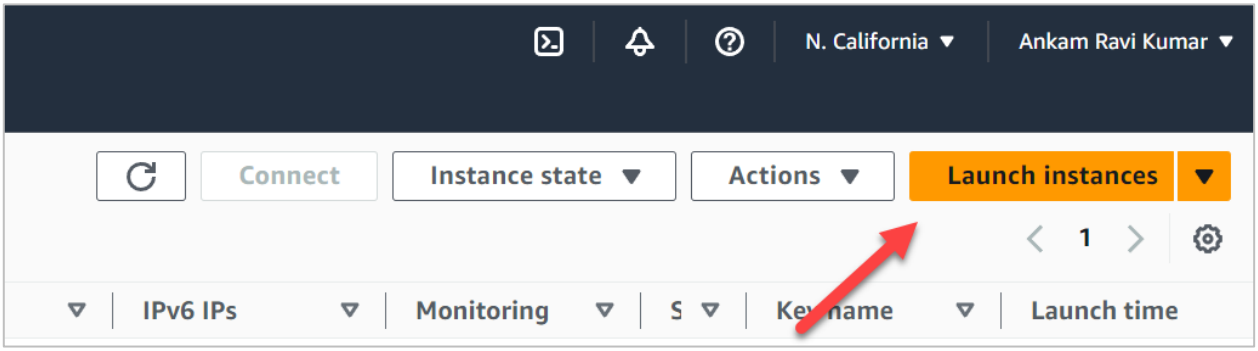
Click on “Create Key Pair”



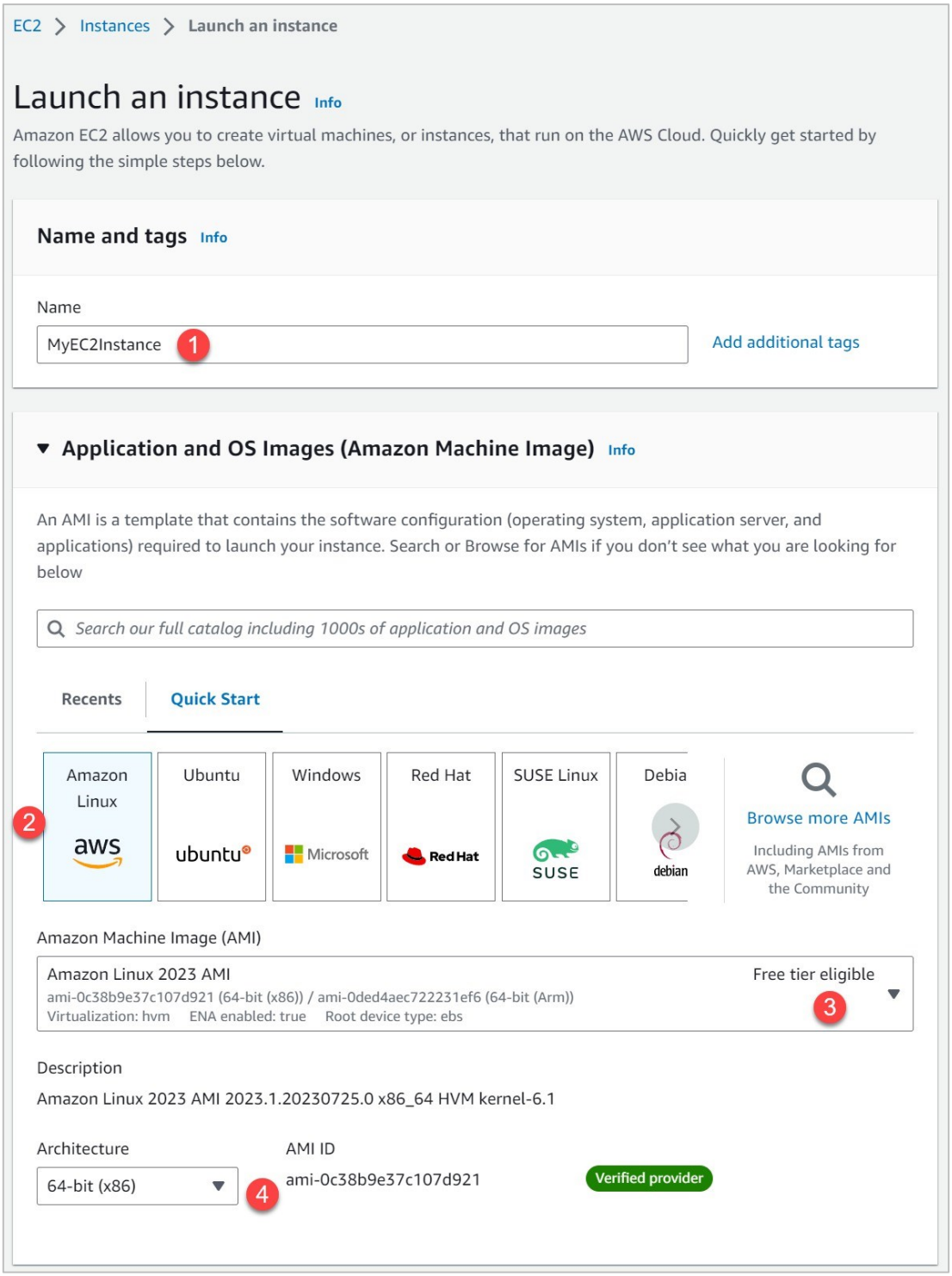
- Provide Name
- Select type
- Select the key format



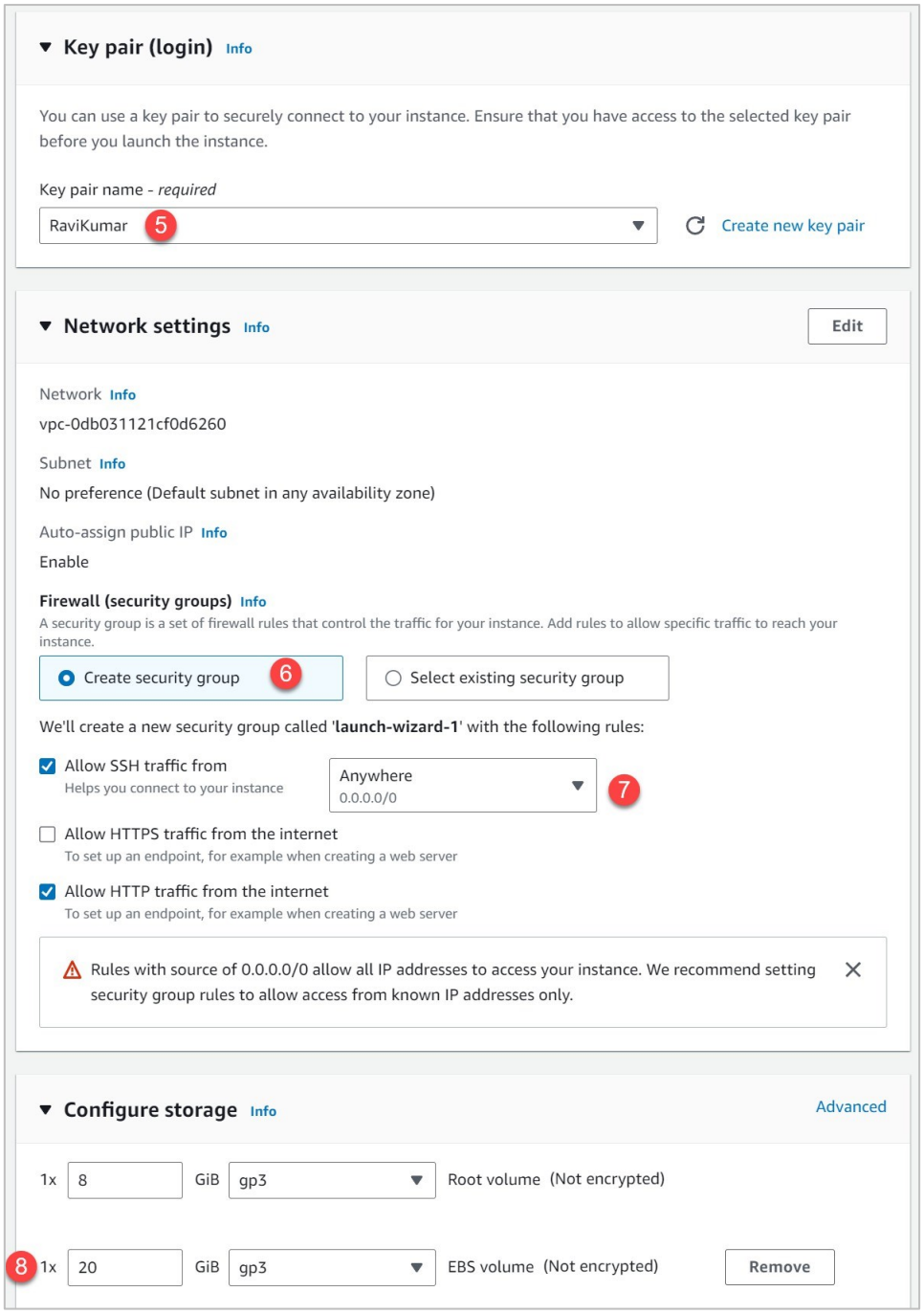
- Click on “Create key pair”



Click on instances and “Launch instances”

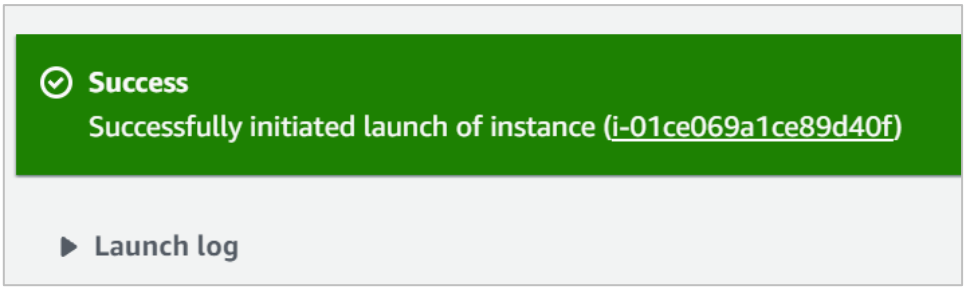


1. Provide EC2 instance with a name
2. Select the Operating system
3. Select instance type
4. Select the Architecture of the instance



5. Select the Key pair that we have created in the first step
6. Create a Security Group
7. Allow SSH (port 22) from anywhere for now
8. Create a new 20GB EBS Volume

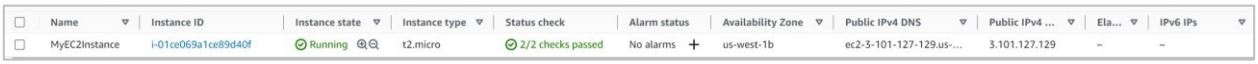




The instance is created successfully.

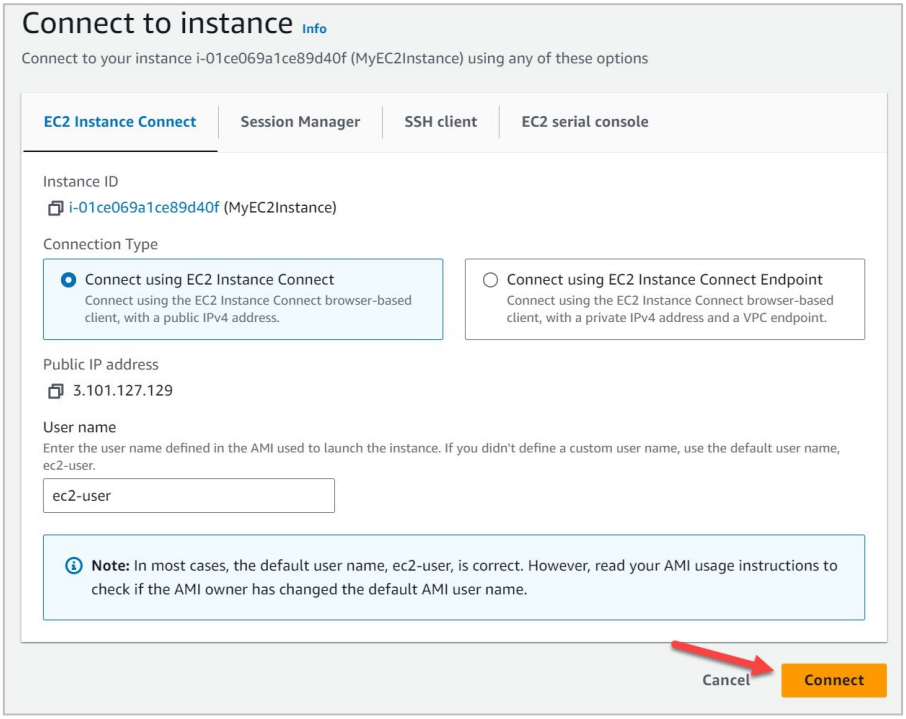


Note: Wait for few minutes to complete the initialization



Checks completed

Connect to the EC2 instance now



Click “Connect”

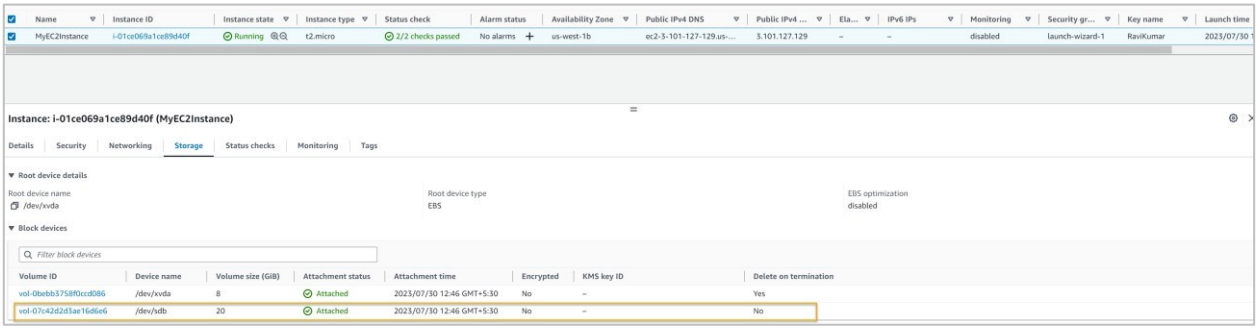
```
[ec2-user@ip-172-31-27-80 ~]$ sudo -s
[root@ip-172-31-27-80 ec2-user]# lsblk
NAME      MAJ:MIN RM SIZE RO TYPE MOUNTPOINTS
xvda      202:0    0   8G  0 disk
├─xvda1   202:1    0   8G  0 part /
├─xvda127 259:0    0   1M  0 part
└─xvda128 259:1    0  10M  0 part
xvdb      202:16   0  20G  0 disk
[root@ip-172-31-27-80 ec2-user]# mkfs.xfs /dev/xvdb
meta-data=/dev/xvdb              isize=512    agcount=4, agsize=1310720 blks
         =                       sectsz=512   attr=2, projid32bit=1
         =                       crc=1        finobt=1, sparse=1, rmapbt=0
         =                       reflink=1    bigtime=1 inobtcount=1
data     =                       bsize=4096   blocks=5242880, imaxpct=25
         =                       sunit=0      swidth=0 blks
naming   =version 2              bsize=4096   ascii-ci=0, ftype=1
log      =internal log           bsize=4096   blocks=16384, version=2
         =                       sectsz=512   sunit=0 blks, lazy-count=1
realtime =none                   extsz=4096   blocks=0, rtextents=0
[root@ip-172-31-27-80 ec2-user]# mkdir /data
[root@ip-172-31-27-80 ec2-user]# mount /dev/xvdb /data
[root@ip-172-31-27-80 ec2-user]# df -h /data
Filesystem      Size  Used Avail Use% Mounted on
/dev/xvdb        20G  175M   20G   1% /data
[root@ip-172-31-27-80 ec2-user]#
```

Check the disk is present

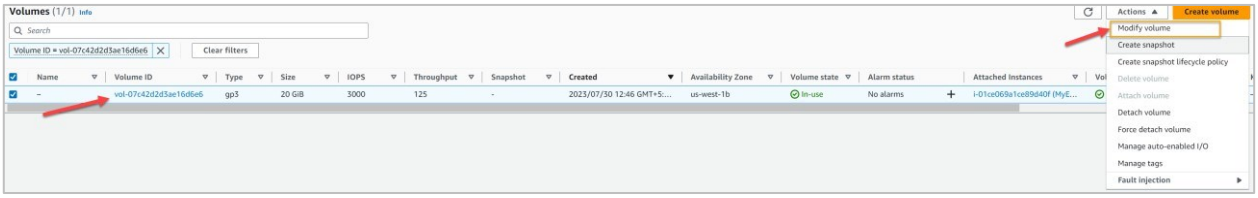
Create the file system in the disk using the below command

mkfs.xfs /dev/xvdb

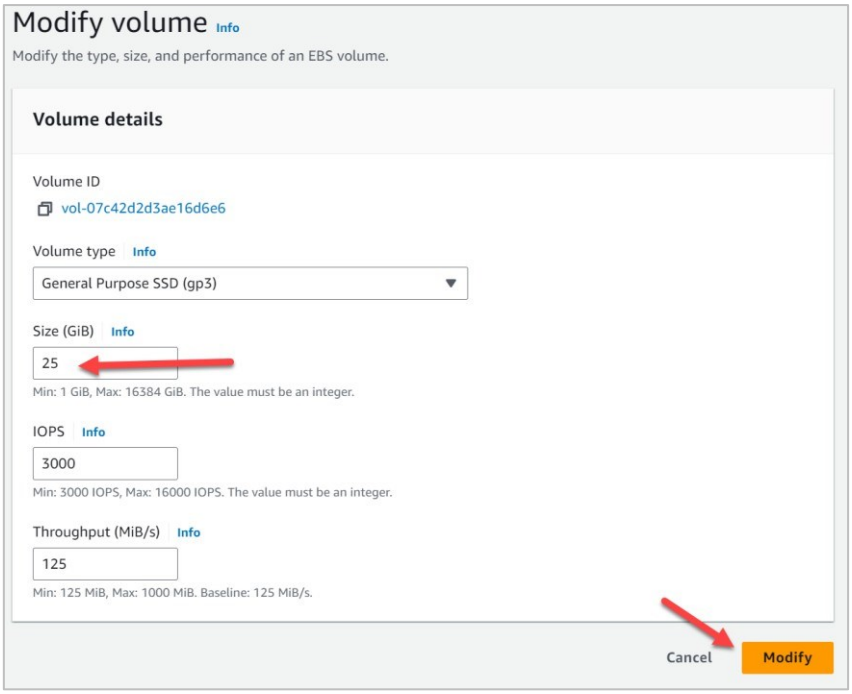
mount the partition



mount /dev/xvdb /data



Note down the volume ID then go to EC2 → volumes



Now update the size value from 20GB to 25GB click on Modify

```
[root@ip-172-31-27-80 ec2-user]# xfs_growfs /data/
meta-data=/dev/xvdb              isize=512    agcount=4, agsize=1310720 blks
         =                       sectsz=512   attr=2, projid32bit=1
         =                       crc=1        finobt=1, sparse=1, rmapbt=0
         =                       reflink=1    bigtime=1 inobtcount=1
data     =                       bsize=4096   blocks=5242880, imaxpct=25
         =                       sunit=0      swidth=0 blks
naming   =version 2              bsize=4096   ascii-ci=0, ftype=1
log      =internal log           bsize=4096   blocks=16384, version=2
         =                       sectsz=512   sunit=0 blks, lazy-count=1
realtime =none                   extsz=4096   blocks=0, rtextents=0
data blocks changed from 5242880 to 6553600
[root@ip-172-31-27-80 ec2-user]# df -h /data/
Filesystem      Size  Used Avail Use% Mounted on
/dev/xvdb        25G  211M   25G   1% /data
[root@ip-172-31-27-80 ec2-user]#
```

EBS volume is expanded successfully.

Result

A Linux instance was launched with a 20 GB EBS volume attached. An XFS file system was created on /dev/xvdb and mounted on /data. The volume was then modified from 20 GB to 25 GB and the new size was reflected in the instance.

Viva Questions

1. What is Amazon EBS?

Ans. A block level storage service that provides persistent volumes which can be attached to EC2 instances.

2. Can an EBS volume be attached to an instance in a different availability zone?

Ans. No. The volume and the instance must be in the same availability zone.

3. What does the command mkfs.xfs /dev/xvdb do?

Ans. It creates an XFS file system on the attached block device so that it can be mounted and used to store files.

4. After increasing the EBS volume size, why does df not show the new size immediately?

Ans. The file system still has the old size. It has to be extended with xfs_growfs or resize2fs to use the added space.

5. Name the EBS volume types.

Ans. General Purpose SSD (gp2, gp3), Provisioned IOPS SSD (io1, io2), Throughput Optimized HDD (st1), Cold HDD (sc1) and Magnetic (standard).

Assignment Questions

1. Create a snapshot of the 25 GB volume, create a new volume from the snapshot in another availability zone and attach it to a new instance. Verify that the data is present.

2. Add an entry in /etc/fstab so that the EBS volume is mounted automatically on reboot, reboot the instance and confirm that /data is available.

Experiment 8: Launching RDS Instance and Accessing MySQL Using Workbench

8	a) Launching RDS Instance b) Accessing MySQL Instance Using Workbench	CO4

Aim

To launch an Amazon RDS MySQL database instance, connect to it remotely using MySQL Workbench, and create a database and a table on it.

Description

Amazon RDS is a managed relational database service that takes care of provisioning, patching, backups and replication. It supports the MySQL, PostgreSQL, MariaDB, Oracle, SQL Server and Amazon Aurora engines. While creating an instance the DB instance class, the storage type and size, Multi-AZ deployment, the master user name and password, the VPC, public accessibility, the security group, the database name, the port and the backup retention period are chosen.

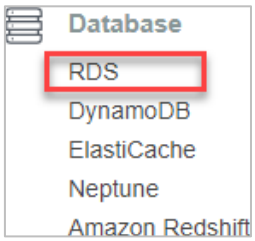
The database is reached through its endpoint host name on port 3306. MySQL Workbench is a desktop client that connects to the endpoint after the security group allows inbound traffic on port 3306 from the client machine. SQL statements are then run against the database from the Workbench editor.

Solution

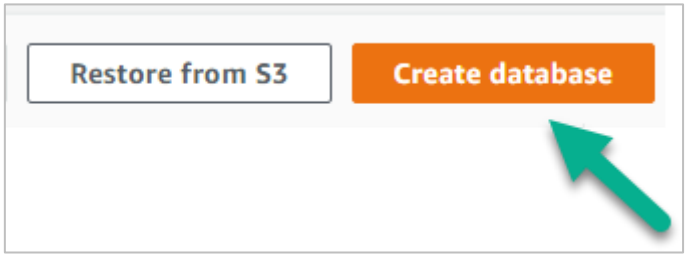
8.a Launching RDS Instance

Amazon Relational Database Service (Amazon RDS) is a web service that makes it easier to set up, operate, and scale a relational database in the cloud. It provides cost-efficient, resizable capacity for an industry-standard relational database and manages common database administration tasks.

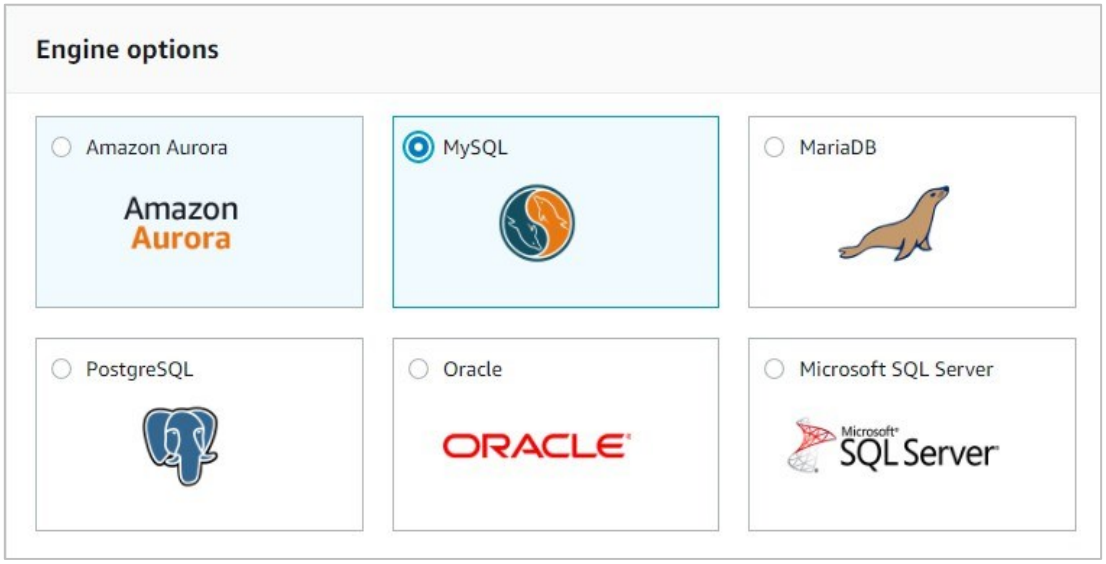
Login to **AWS Console** and Click on **services** to list all services. Navigate to **Database → RDS**



Now we are going to create a new Database instance with empty database

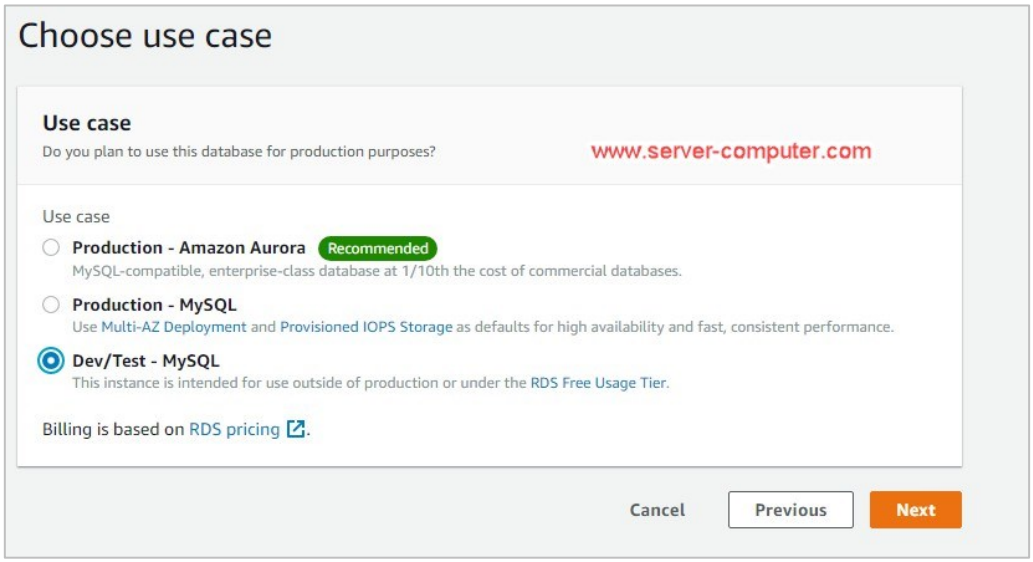


Amazon will support below 5 types of Relational database engines as managed services

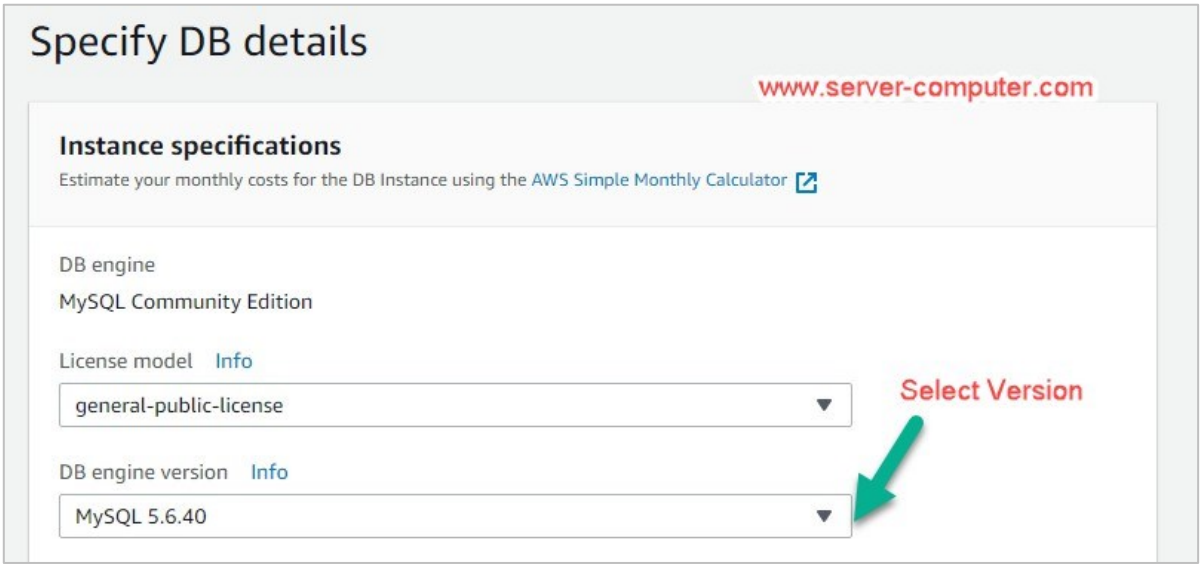


Select any one of the database engine, which you want to launch and Click **Next**

Note: Careful if you are using free tier account. MSSQL and Oracle are charged.

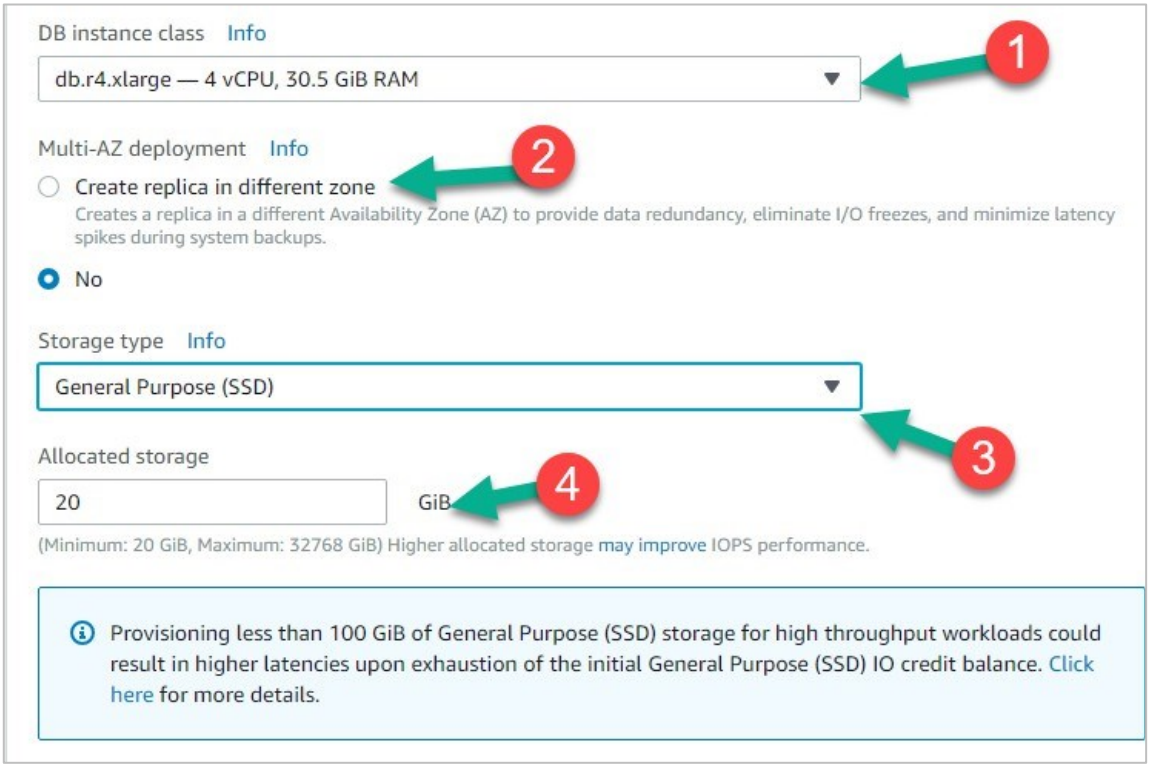


Choose appropriate usage of your instance. In this scenario, I am using Dev/Test instance Click **Next**



In drop down, select appropriate and required MySQL Version.

Note: If you select Free Tier. Selected version and options will overwritten free options.

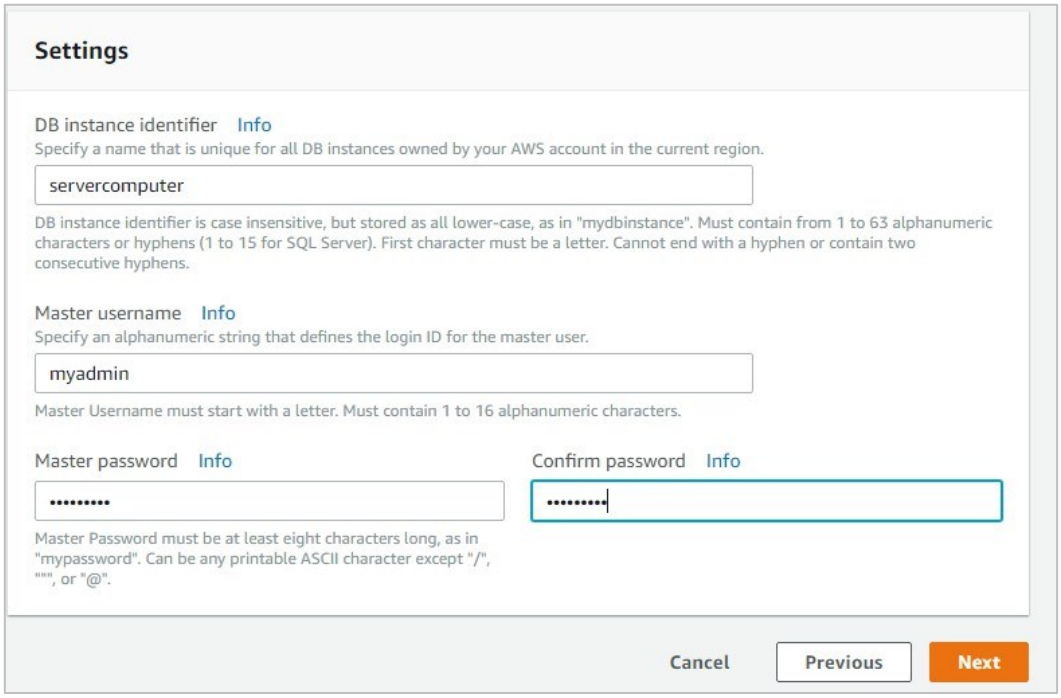


1. Select DB Instance class like required CPU Cores and RAM.
2. Create Replica in Different Zone. (Which means database will be replicated to another available zone for redundant(data protection))
3. General purpose (SSD) or provisioned IOPS (SSD)

General purpose is for low through put applications

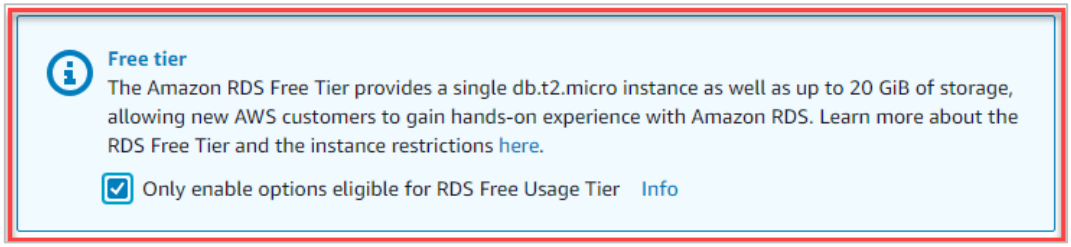
Provisioned IOPS is for most read/write operations

4. Size of the storage

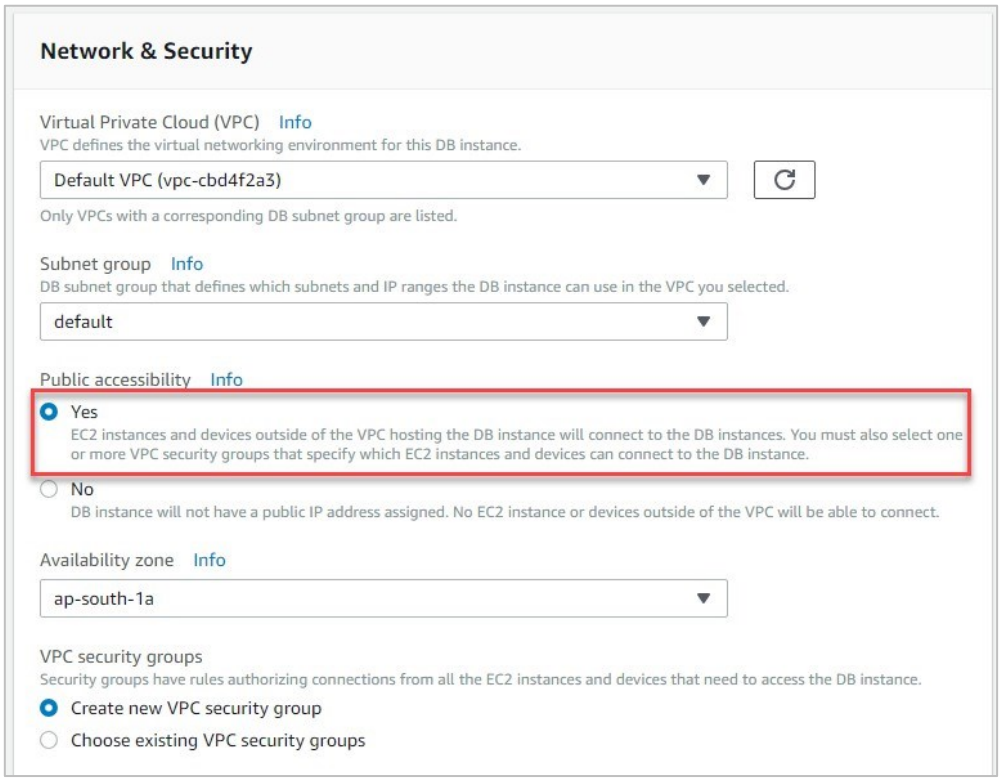


Provide

- Instance name should be unique
- Master username anything you can give without special characters
- Provide master password and remember



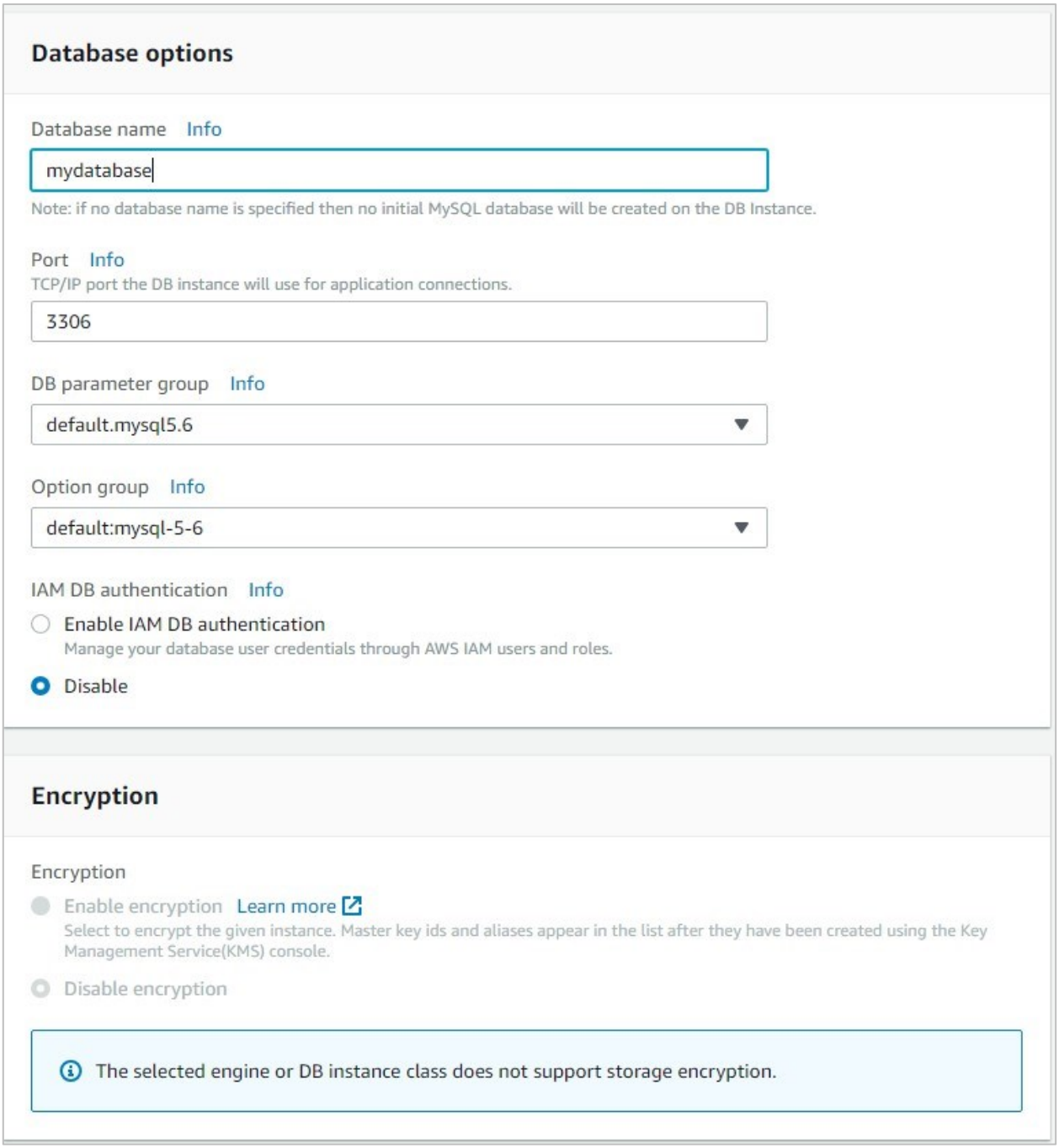
DO NOT FORGOT TO SELECT IF YOU'RE USING FREE TIER OTHERWISE YOU WILL BE CHARGED



Select appropriate VPC and Subnet group (If any)

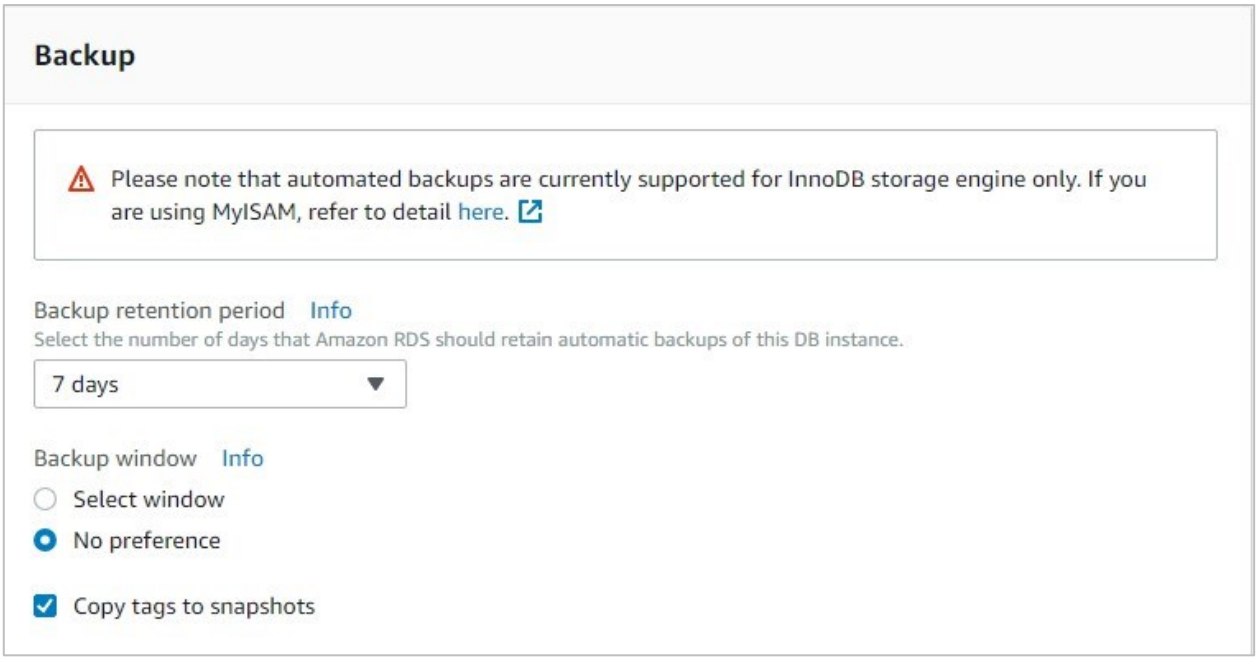
If you want access database from remote machine put "Public Accessibility" **Yes**

Choose existing VPC security groups if you have already or it will create new security group for this instance access.



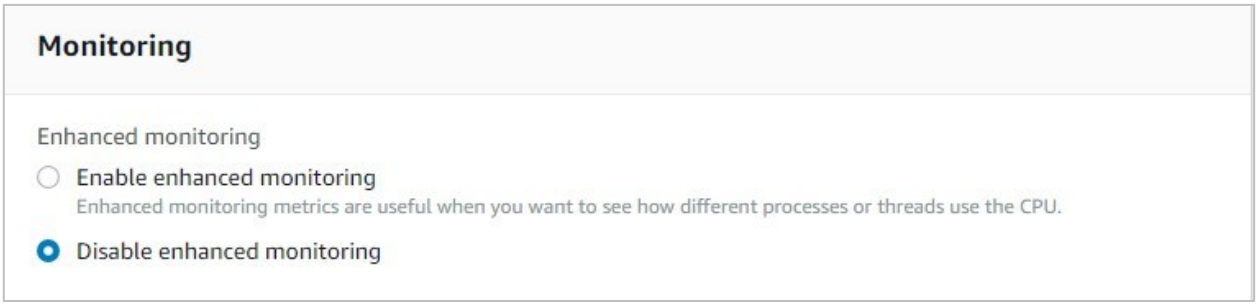
Provide database name, default port number is 3306 you can even customize the port number if you want. Enabling IAM DB Authentication. IAM Users also can access your instance based on IAM policies.

For free tier encryption option is disabled

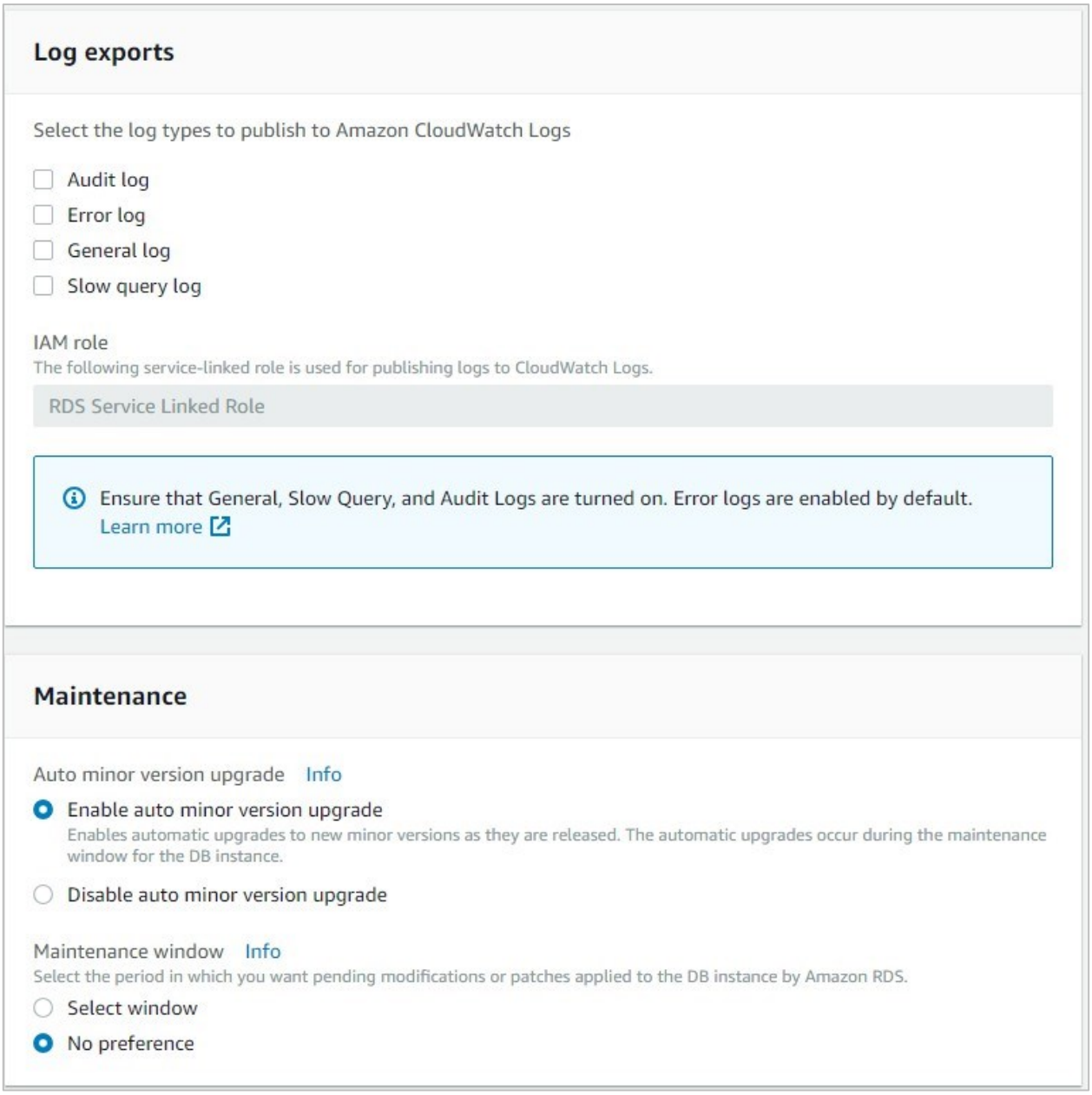


If you want database backups select, the retention max is **35 Days**

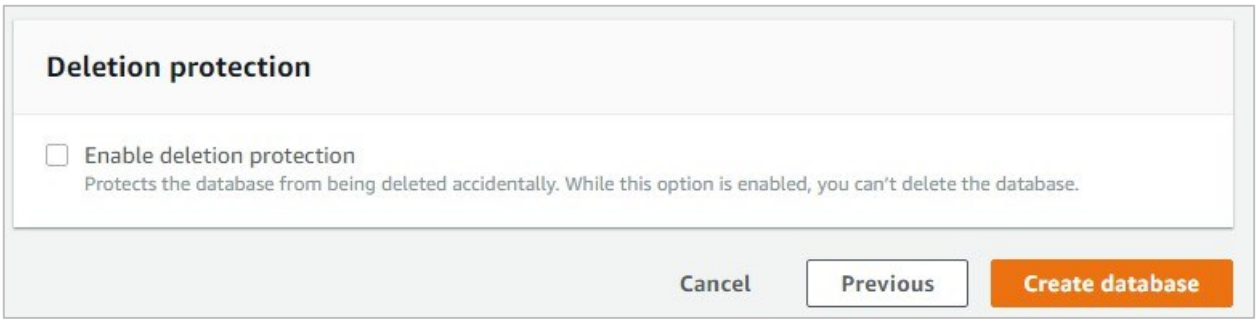
If you have particular backup window for database select it otherwise leave it default.



Enhanced monitoring will charged



Select the options you required

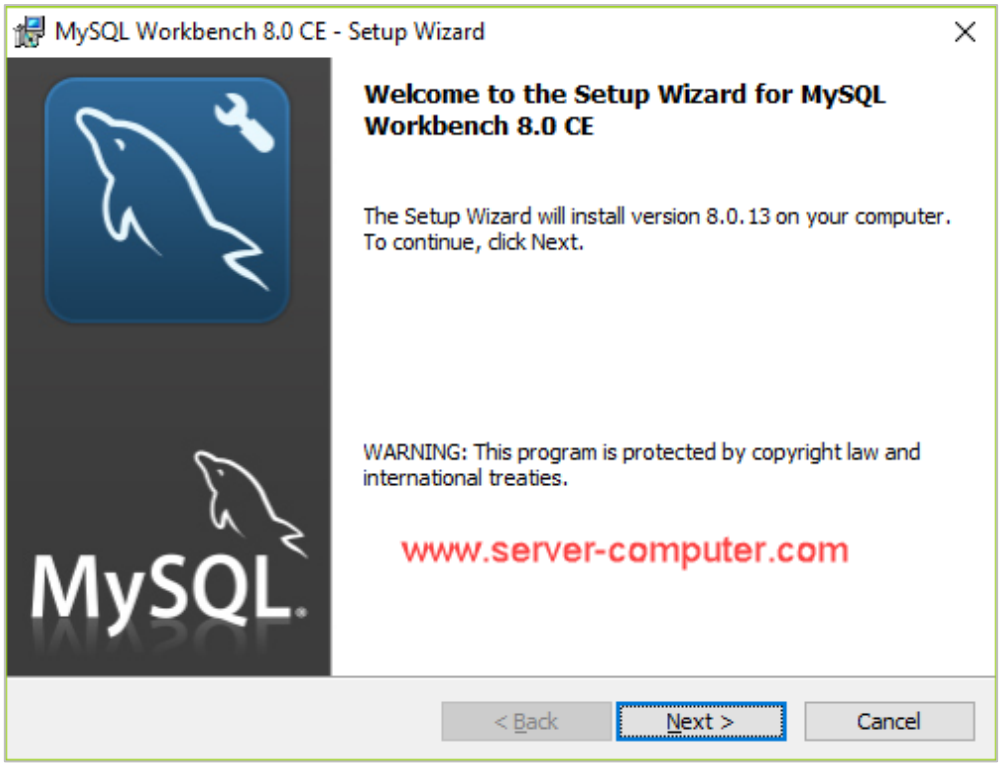


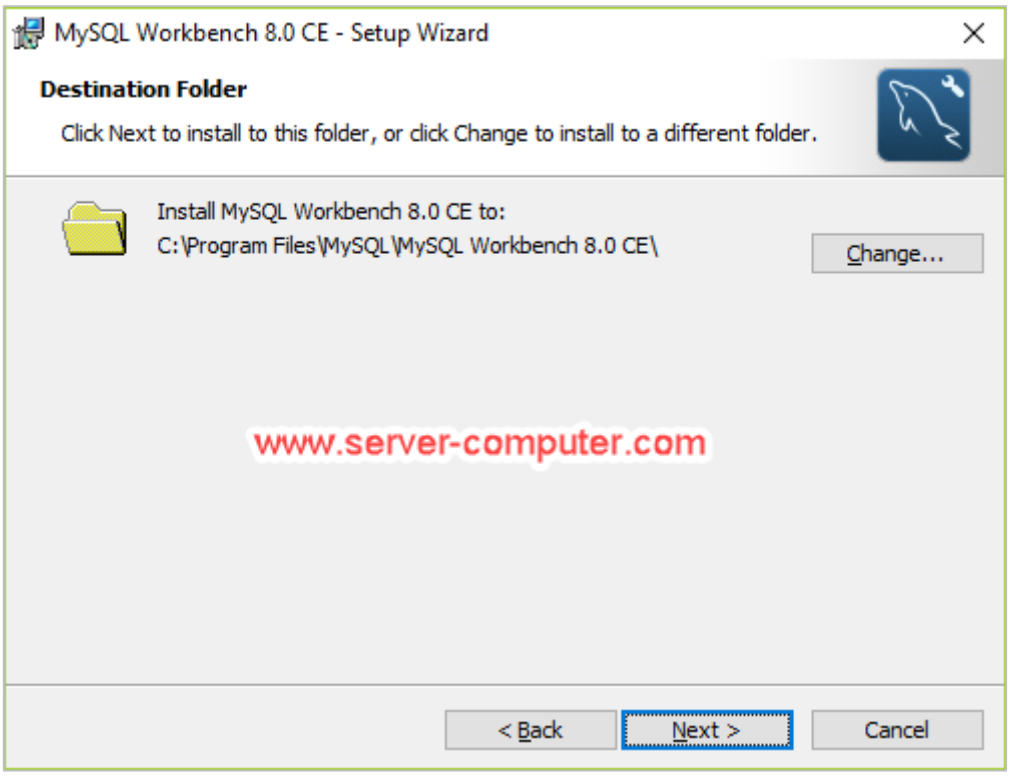
Enabling database protection, you cannot delete database Click **Create Database**

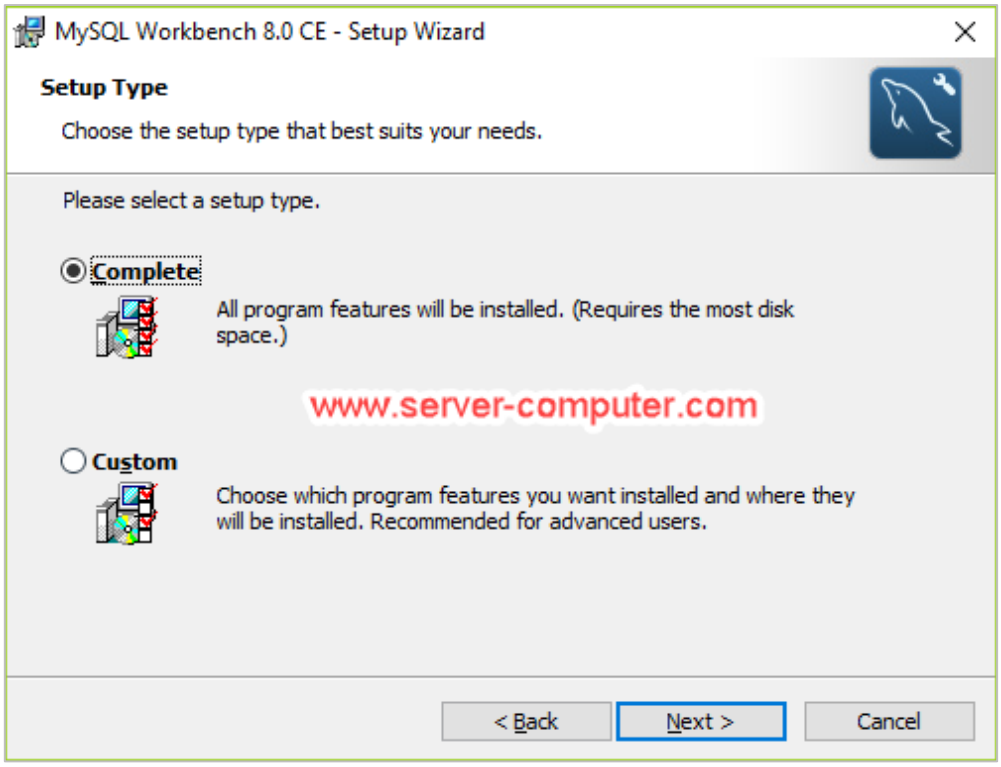
Note: Database instance creation will take at least 10minutes.

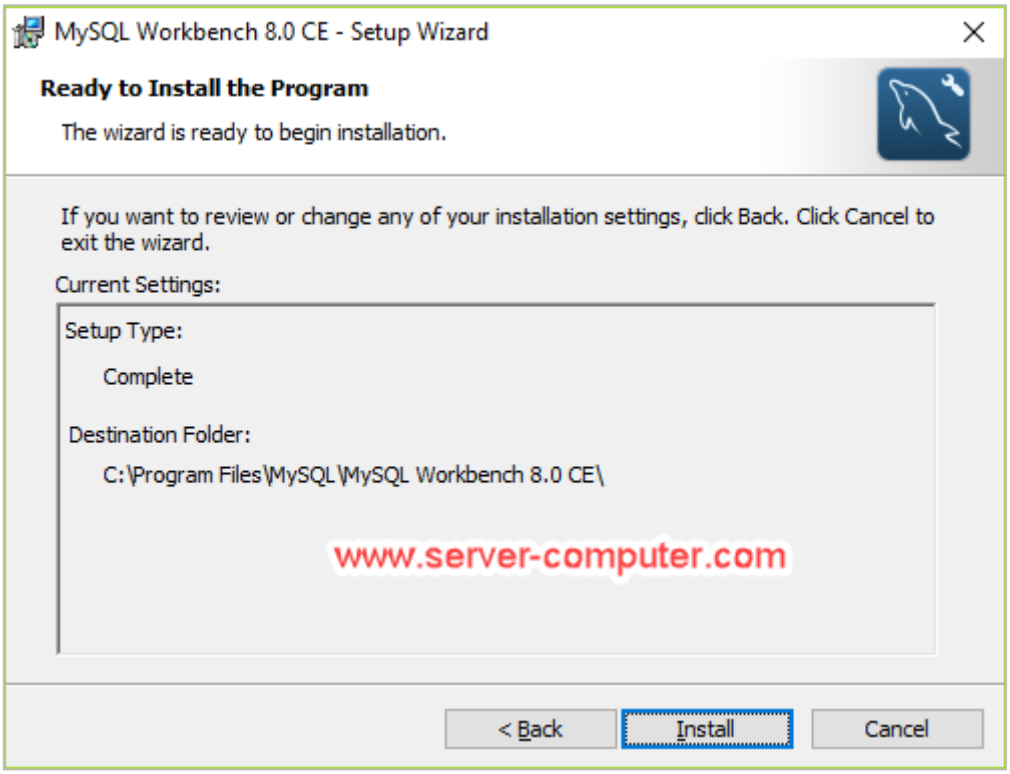
8.b Accessing MySQL Instance Using Workbench

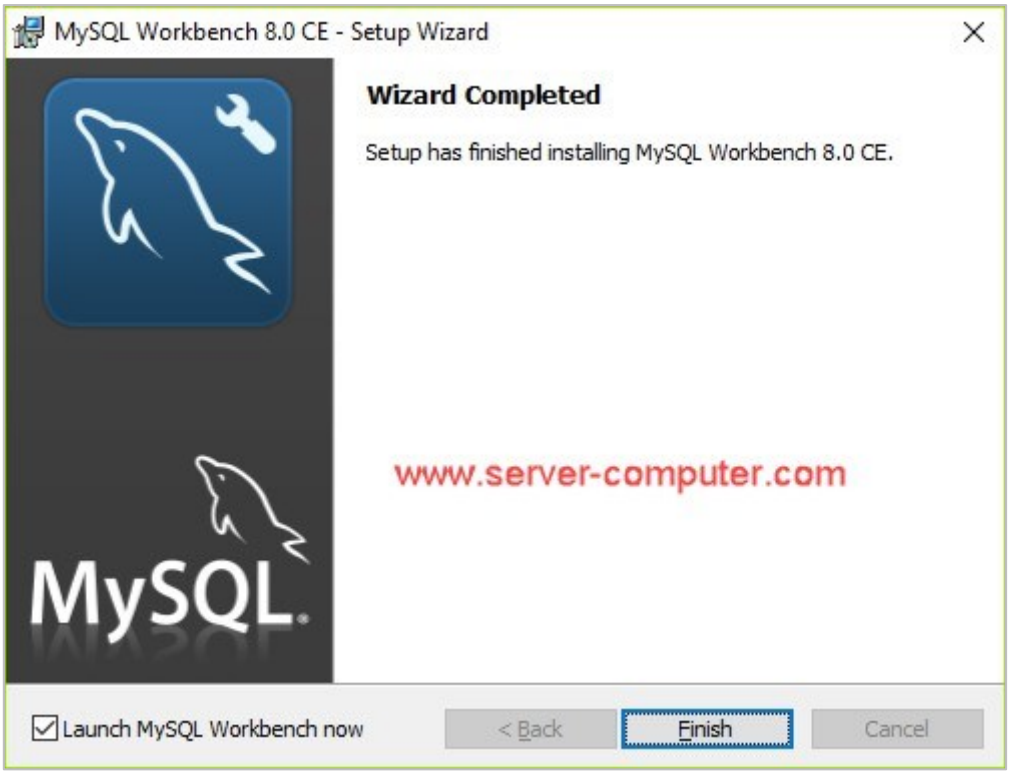
Download MySQL Workbench to access MySQL instance remotely
https://dev.mysql.com/downloads/workbench/



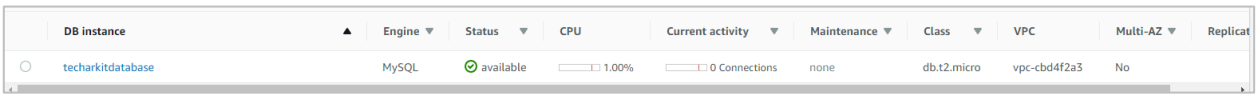



MySQL Workbench 8.0 CE - Setup Wizard
Setup Type
Choose the setup type that best suits your needs.
Please select a setup type.
Complete
All program features will be installed. (Requires the most disk space.)
www.server-computer.com
Custom
Choose which program features you want installed and where they will be installed. Recommended for advanced users.
< Back
Next >
Cancel


MySQL Workbench 8.0 CE - Setup Wizard
Ready to Install the Program
The wizard is ready to begin installation.
If you want to review or change any of your installation settings, click Back. Click Cancel to exit the wizard.
Current Settings:
Setup Type:
Complete
Destination Folder:
C:\Program Files\MySQL\MySQL Workbench 8.0 CE\
www.server-computer.com
< Back
Install
Cancel

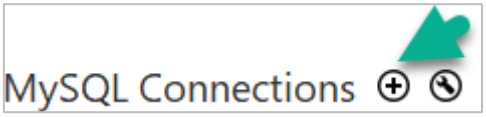


After successful creation you see like below

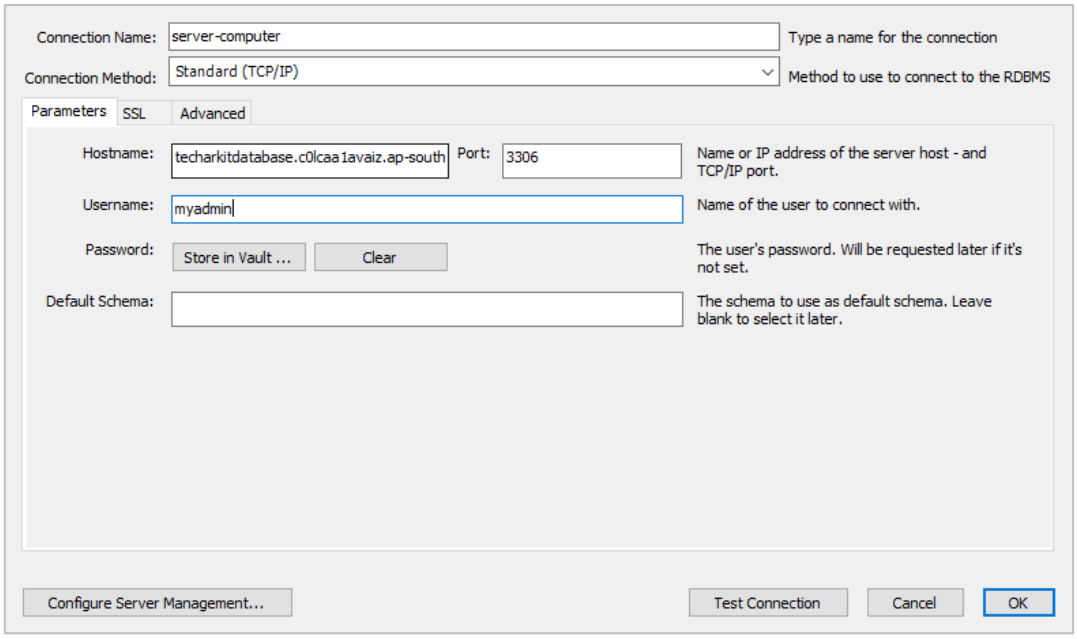


Click on Database name and come down copy the **Endpoint URL**

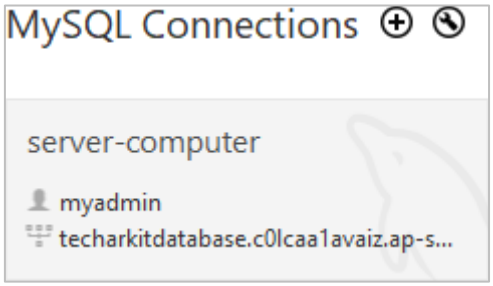
Open your MySQL workbench and create connection



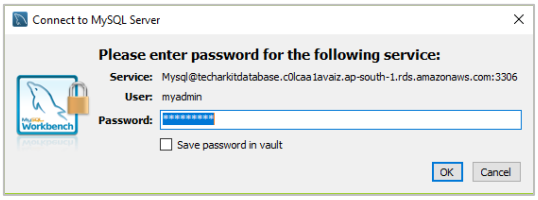
Click on Plus (+) sign to create a New MySQL Connection

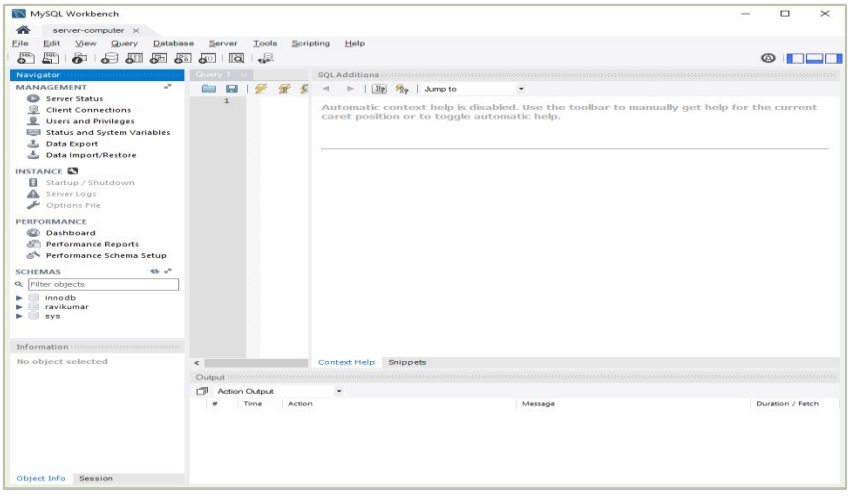


Click **OK**



After successful creation, Click on Connection it will ask you for the password





Successfully launched MySQL RDS Instance and accessed via MySQL Work bench. Run below queries to create database and some tables on it.

create database 'DBNAME'; use DBNAME;

Create Table using below query

create table students(

student_id INT NOT NULL AUTO_INCREMENT, student_title VARCHAR(100) NOT NULL, student_author VARCHAR(40) NOT NULL, submission_date DATE,

PRIMARY KEY (student_id)

);

show databases; use DBNAME; show tables;

If you know much more database queries like select, insert and delete statement try doing more. Good Luck.

Result

A MySQL RDS instance was launched with the Free Tier options, its endpoint was copied and a connection was created in MySQL Workbench. The database and the students table were created with SQL statements and verified with the show databases and show tables commands.

Viva Questions

1. What is Amazon RDS?

Ans. A managed web service for setting up, operating and scaling relational databases in the cloud.

2. Which database engines are supported by RDS?

Ans. MySQL, PostgreSQL, MariaDB, Oracle, Microsoft SQL Server and Amazon Aurora.

3. What is a Multi-AZ deployment?

Ans. A synchronous standby replica of the database is maintained in another availability zone and RDS fails over to it automatically when the primary instance fails.

4. What is the default port of MySQL and what must be allowed for Workbench to connect?

Ans. Port 3306. The security group of the RDS instance must allow inbound TCP 3306 from the client IP address and the instance must be publicly accessible.

5. What is the maximum backup retention period in RDS?

Ans. 35 days.

Assignment Questions

1. Create a read replica of the RDS instance, insert rows through the primary endpoint and read them from the replica endpoint. Note the replication lag you observe.

2. Take a manual snapshot of the RDS instance, delete the instance and restore a new instance from the snapshot. Verify that the students table is present in the restored instance.

Experiment 9: CloudWatch Dashboard

week 9	CloudWatch Dashboard : Create a dashboard that lets you check the CPU utilization and networking for a particular EC2 instance.

Aim

To create an Amazon CloudWatch dashboard that shows the CPU utilization and the networking metrics of a particular EC2 instance.

Description

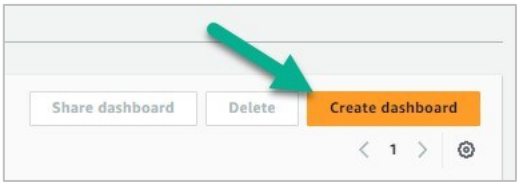
Amazon CloudWatch collects metrics from AWS services. EC2 publishes CPUUtilization, NetworkIn, NetworkOut, disk operation and status check metrics for every instance at five minute intervals by default, or at one minute intervals when detailed monitoring is enabled. Memory and disk space usage are not published unless the CloudWatch agent is installed on the instance.

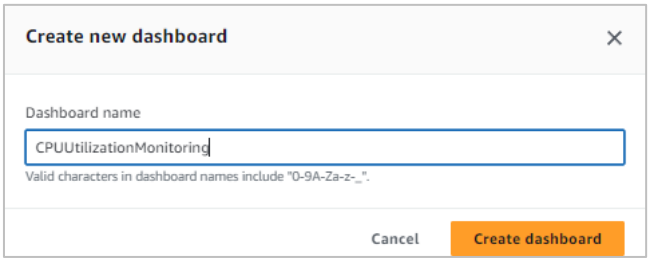
A dashboard is a customizable page in the CloudWatch console that holds widgets such as line graphs, stacked areas, numbers and text. Each widget is built from one or more metrics selected from a namespace, for example EC2 Per-Instance Metrics filtered by the instance ID. A dashboard can show metrics from different regions on one page and can be shared with other users.

Solution

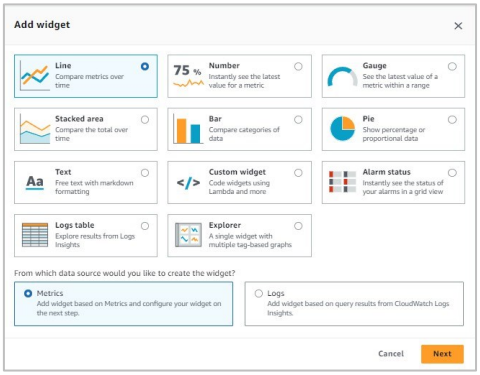
Create a dashboard that lets you check the CPU utilization and networking for a particular EC2 instance.

Login to AWS Management console → **CloudWatch → Click Dashboards → Create dashboard**



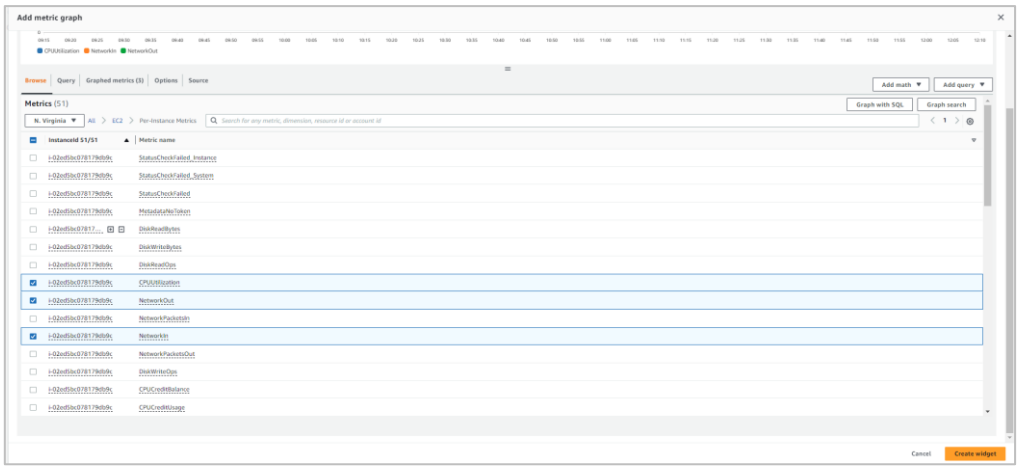


"Create dashboard"

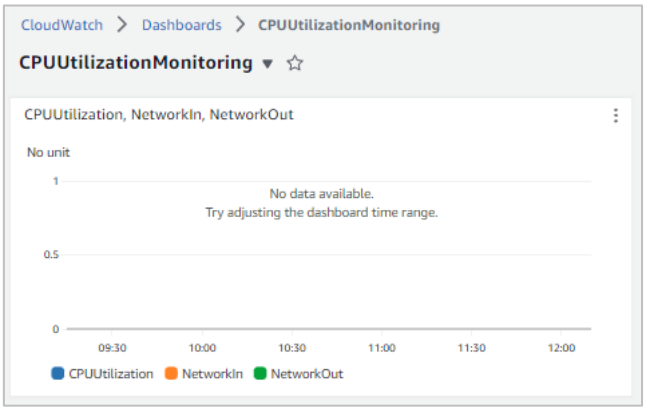


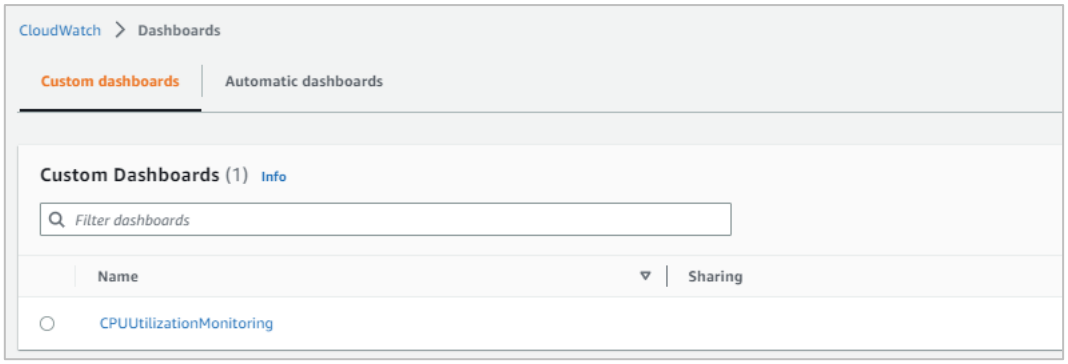
Select the widget you like Click **Next**

EC2 → Per-Instance Metrics



Create Widget





The dashboard has been created successfully

Result

A CloudWatch dashboard was created with widgets for the CPU utilization and the network in and out metrics of the selected EC2 instance, and the graphs were verified to display the data.

Viva Questions

1. What is Amazon CloudWatch?

Ans. A monitoring service that collects metrics, logs and events from AWS resources and applications.

2. What is a metric and what is a namespace?

Ans. A metric is a time ordered set of data points such as CPUUtilization. A namespace is a container for metrics, for example AWS/EC2.

3. What is the default monitoring interval for EC2 metrics?

Ans. Five minutes with basic monitoring, and one minute when detailed monitoring is enabled.

4. Which EC2 metrics are available without installing an agent?

Ans. CPU utilization, network in and out, disk read and write operations and status check metrics. Memory and disk space usage need the CloudWatch agent.

5. What types of widgets can be added to a dashboard?

Ans. Line, stacked area, number, gauge, bar, pie, text, logs and alarm status widgets.

Assignment Questions

1. Install the CloudWatch agent on the EC2 instance and add memory utilization and disk usage widgets to the dashboard.

2. Create a dashboard that shows metrics from two instances in two different regions on the same page and export the dashboard definition as JSON.

Experiment 10: CloudWatch Alarm Creation

week 10	Cloud Watch Alarm Creation: a) Create a CloudWatch billing alarm that goes off when the estimated charges go above $500. b) Create a CloudWatch alarm which goes off to an Alarm state when the CPU utilization of an EC2 instance goes above 65%. Also, add an SNS topic so that it notifies the person when the threshold is crossed.

Aim

To create a CloudWatch billing alarm that triggers when the estimated charges exceed $500, and a CloudWatch alarm that enters the ALARM state when the CPU utilization of an EC2 instance exceeds 65%, with an SNS topic that notifies the user by email.

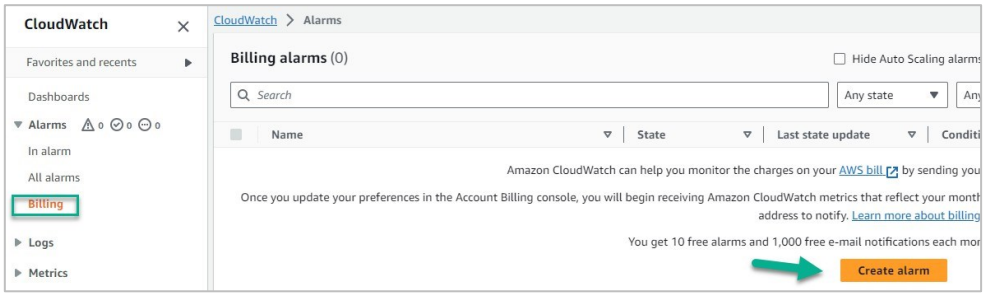
Description

A CloudWatch alarm watches a single metric over a period and performs actions when the metric crosses a threshold for a number of evaluation periods. An alarm has three states: OK, ALARM and INSUFFICIENT_DATA. Billing alarms use the EstimatedCharges metric, which is published in the us-east-1 region after billing alerts are enabled in the account billing preferences.

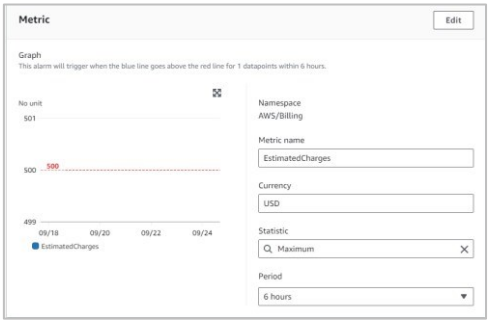
The CPU alarm uses the CPUUtilization metric of the selected instance with a static threshold. Amazon SNS is used for notifications. The alarm action publishes a message to an SNS topic and the email subscriber receives it after confirming the subscription.

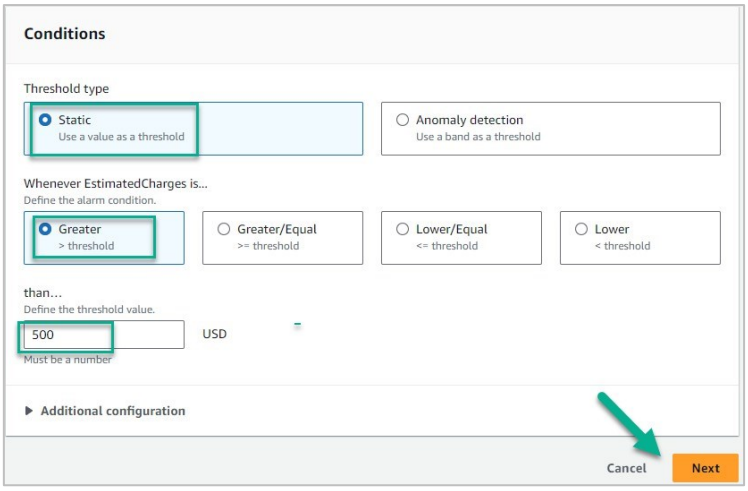
Solution

Login to the AWS Management console then search for Cloudwatch



CloudWatch → Alarms → Billing → Create Alarm





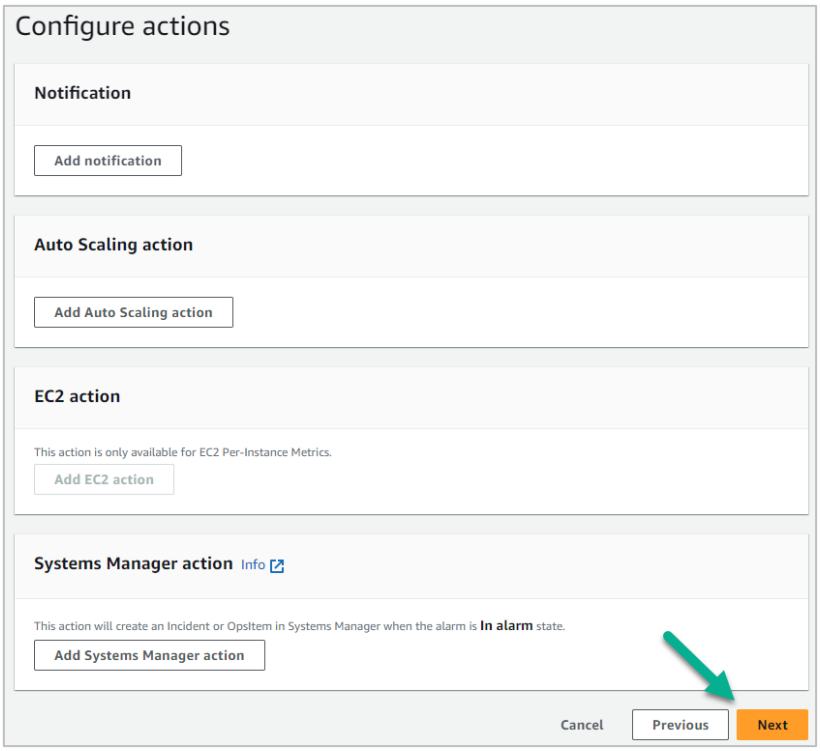
Select the conditions based on your requirement

Type: Static

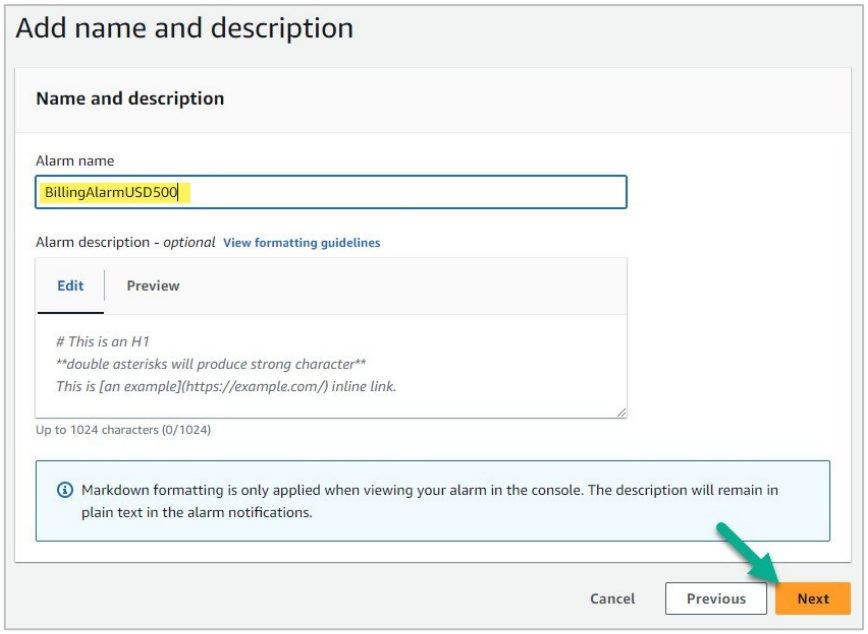
Estimated Charges: Greater than

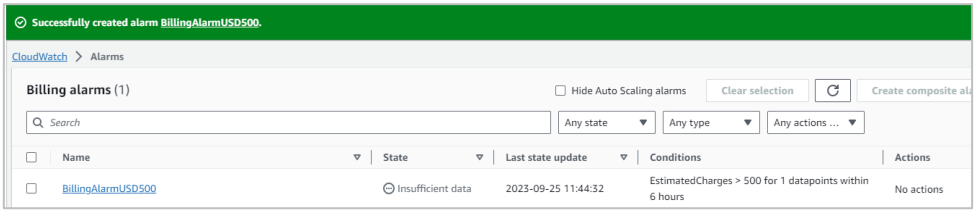
$500

Click **Next**



Click **Next**



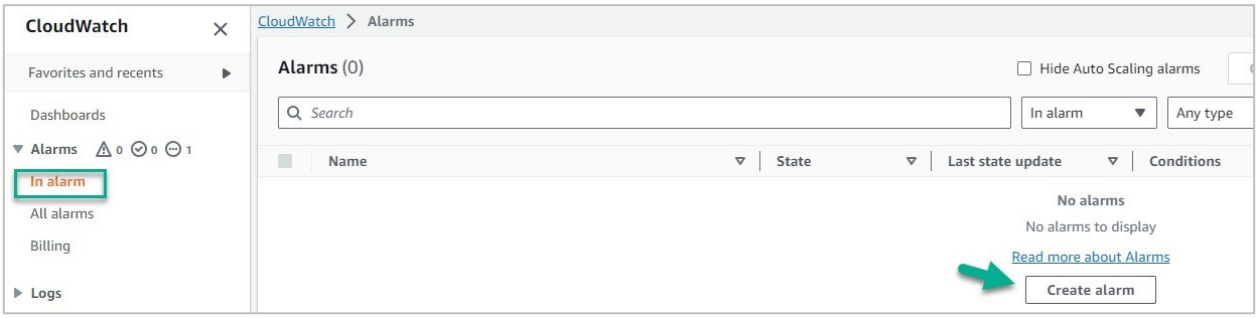


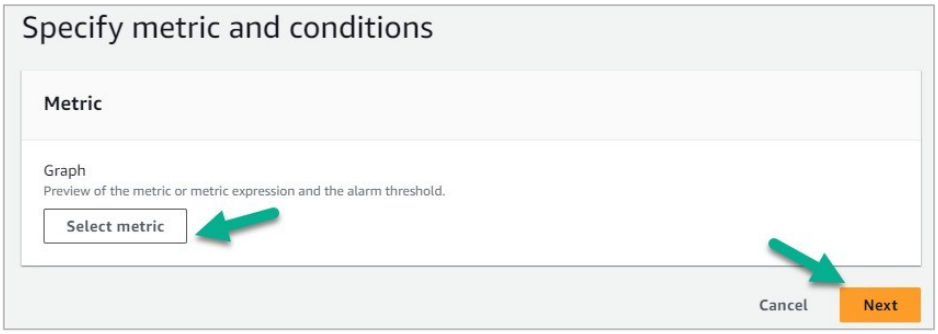
Preview and confirm the alarm "**Create Alarm**"

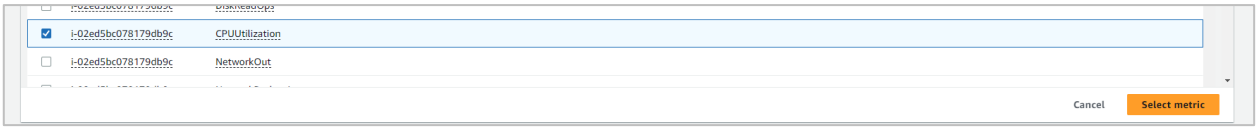
The alarm was Created successfully as per the requirement. Which will trigger the alarm when the usage of the AWS account reaches $500.

Creating an Alarm for CPU Utilization notification

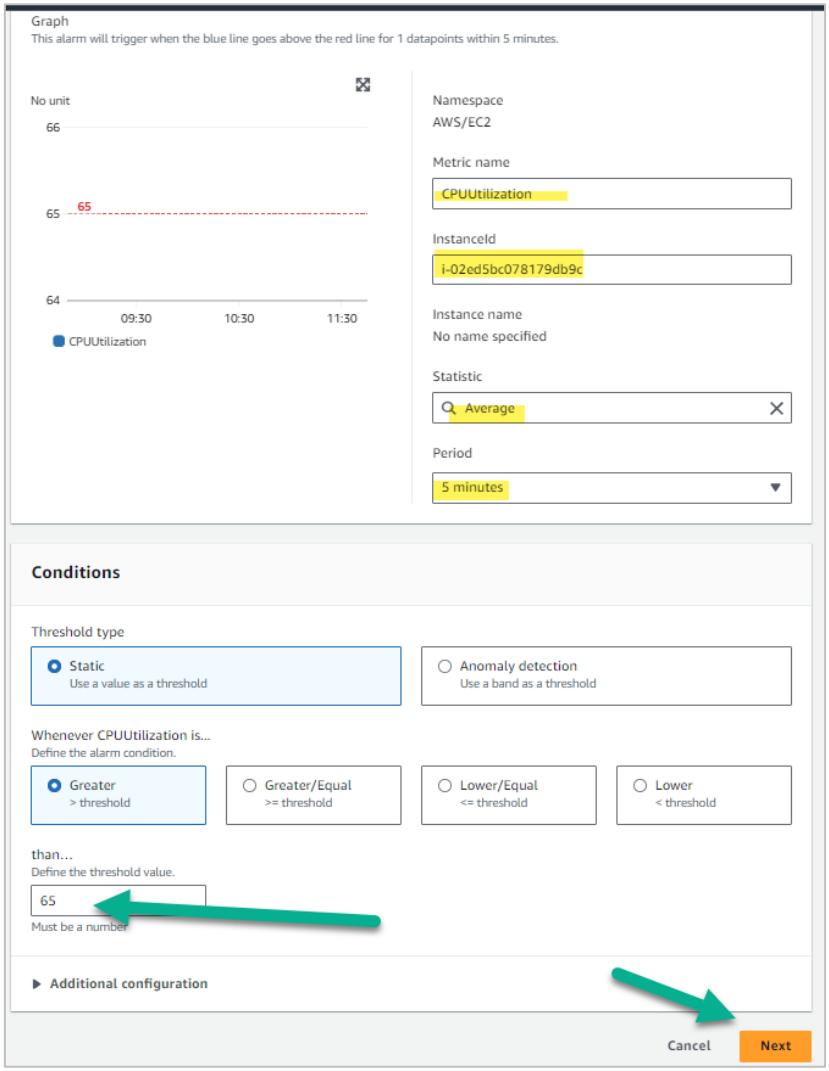
CloudWatch → In Alarm → Create Alarm



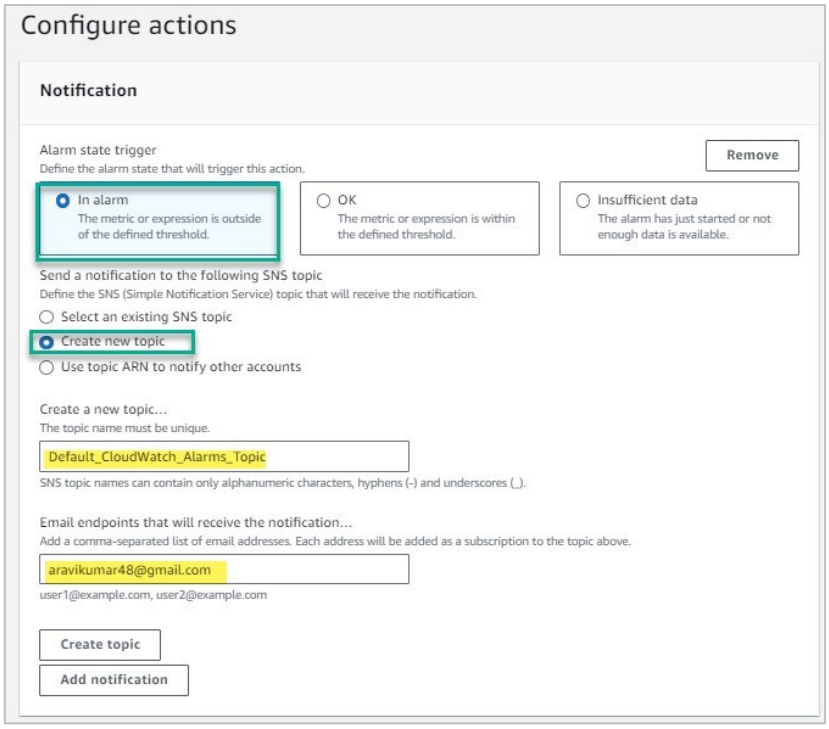




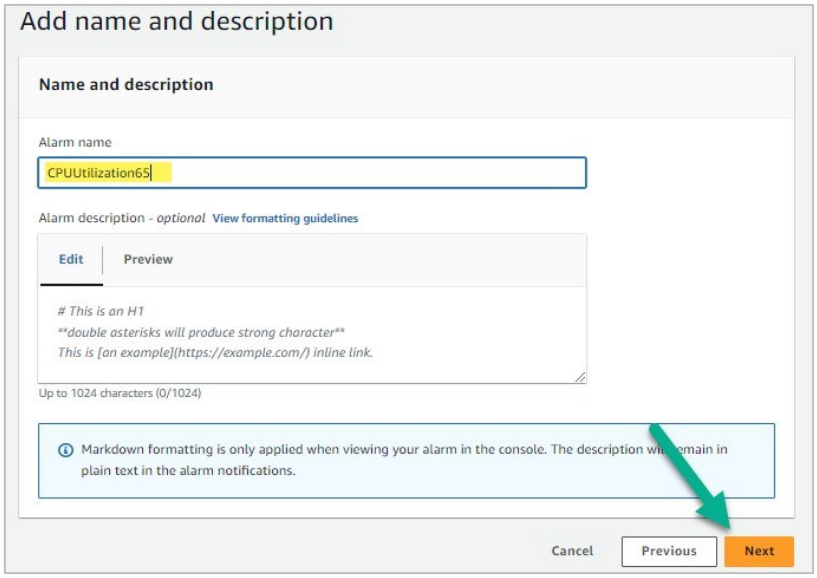
Select metric → EC2 → Per-Instance Metrics → CPU Utilization



Select the threshold Greater/Equal => 65% Click **Next**



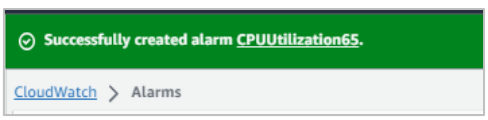
I do not have an SNS topic already created so I am creating a new SNS topic, if you have already a topic just select the existing SNS topic



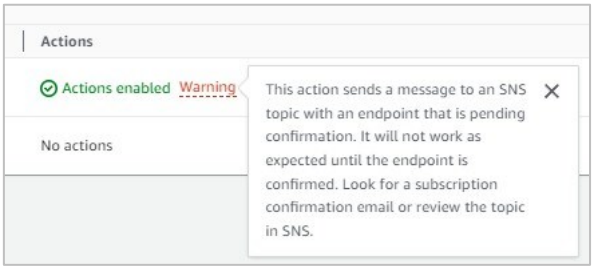
Click "**Create Topic**" Click **Next**

Provide the Alarm Name, Click **Next**

Preview and Create the Alarm



In order to get the email notifications you have to confirm the subscription from your email address



Result

A billing alarm with a $500 threshold and a CPU utilization alarm with a 65% threshold were created. An SNS topic with an email subscription was attached to the CPU alarm and the subscription was confirmed, so an email is received when the alarm state is reached.

Viva Questions

1. What are the three states of a CloudWatch alarm?

Ans. OK, ALARM and INSUFFICIENT_DATA.

2. In which region are billing metrics available?

Ans. US East (N. Virginia), us-east-1, after billing alerts are enabled in the billing preferences.

3. What is an evaluation period?

Ans. The number of most recent periods over which the metric is compared with the threshold before the alarm changes its state.

4. What is Amazon SNS?

Ans. A managed publish and subscribe messaging service that delivers notifications to subscribers such as email, SMS, SQS, Lambda and HTTP endpoints.

5. Why must the email subscription be confirmed?

Ans. SNS sends a confirmation link to the address and delivers notifications only after the owner confirms it. This prevents unwanted messages.

Assignment Questions

1. Create a composite alarm that goes to the ALARM state only when both the CPU utilization alarm and a NetworkIn alarm are in the ALARM state.

2. Configure an alarm action that stops the EC2 instance automatically when the CPU utilization stays below 5% for 30 minutes, and test it.

Experiment 11: IAM Users and Groups

11	a) Create 4 IAM users named "Dev1", "Dev2", "Test1", and "Test2". b) Create 2 groups named "Dev Team" and "Ops Team". c) Add Dev1 and Dev2 to the Dev Team. d) Add Dev1, Test1 and Test2 to the Ops Team

Aim

To create four IAM users named Dev1, Dev2, Test1 and Test2, create two groups named Dev Team and Ops Team, and add the users to the groups as required.

Description

AWS Identity and Access Management (IAM) controls who can sign in to the account and what they can do. An IAM user is an identity with long term credentials: a console password and, optionally, access keys for programmatic access. An IAM group is a collection of users. Permissions are attached to users or groups as policies written in JSON, and a user receives the combined permissions of all the groups the user belongs to.

Managing permissions through groups is the recommended practice because adding or removing a user from a group changes the permissions in one step. A user can be a member of several groups. In this experiment Dev1 is a member of both the Dev Team and the Ops Team.

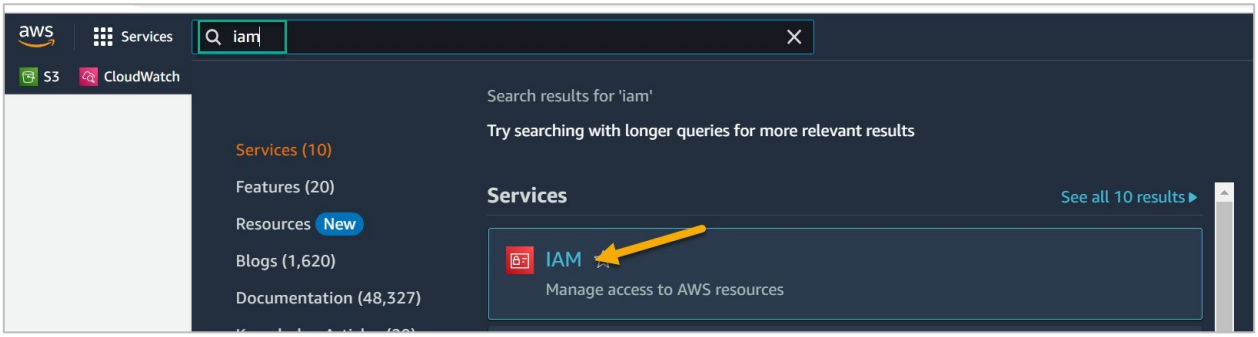
Solution

IAM Assignment

Tasks to Be Performed:

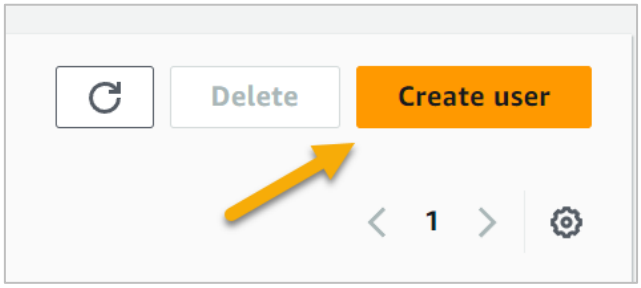
1. Create 4 IAM users named "Dev1", "Dev2", "Test1", and "Test2".
2. Create 2 groups named "Dev Team" and "Ops Team".
3. Add Dev1 and Dev2 to the Dev Team.
4. Add Dev1, Test1 and Test2 to the Ops Team

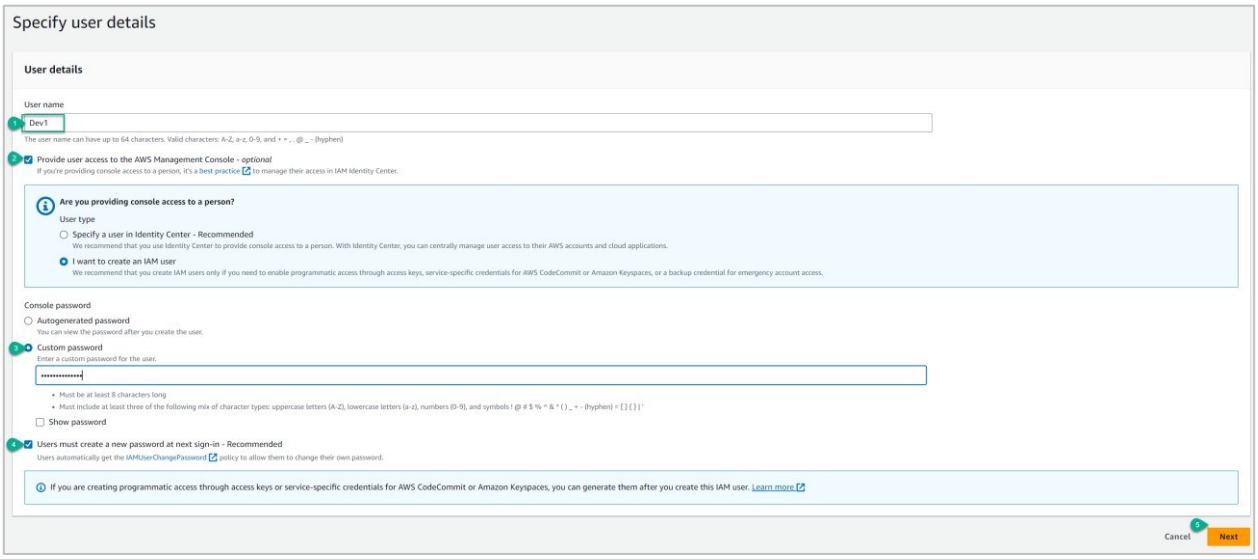
Answer:



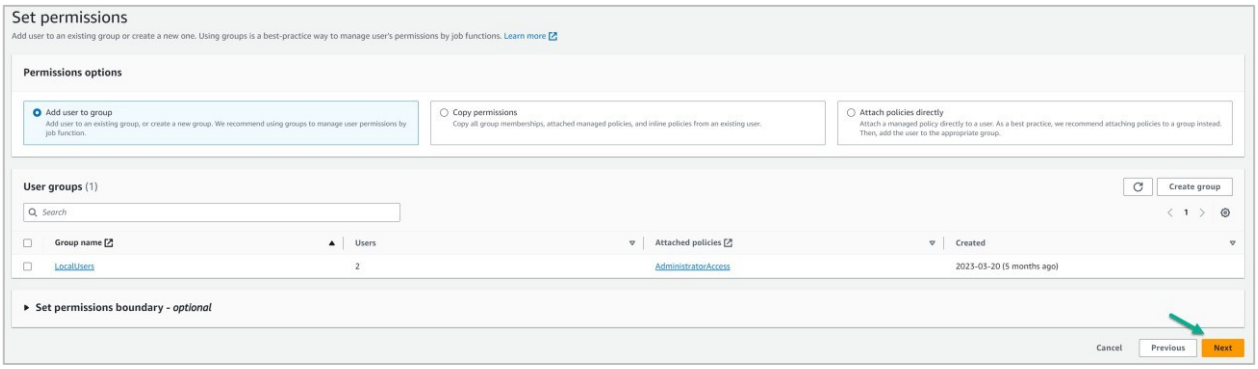
Login to the AWS Console providing your credentials

Search for IAM in search box and Select “IAM Service” Click on IAM Console → Users → Create User



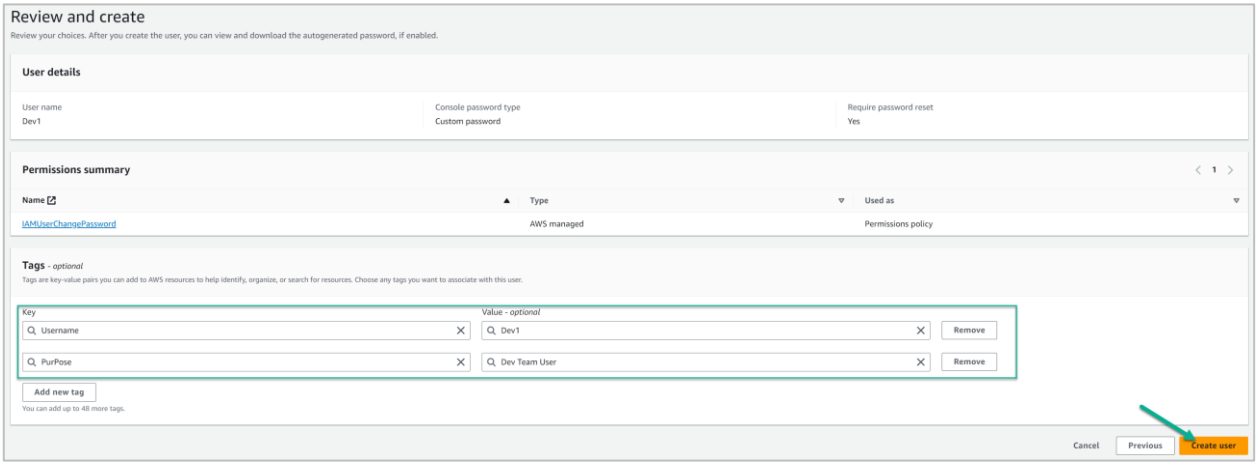


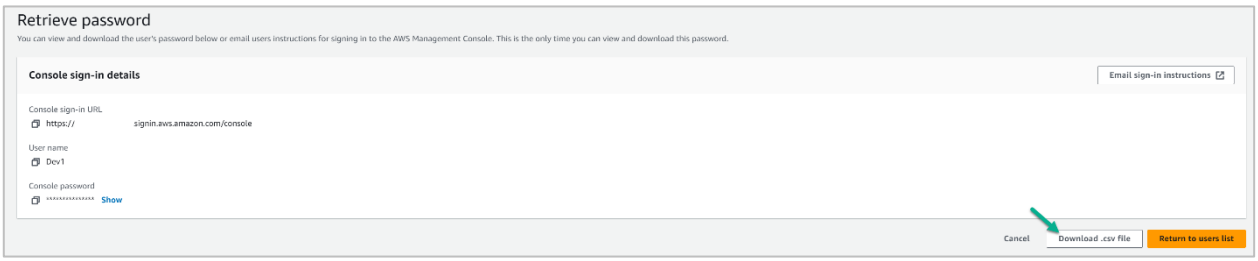
1. Provide Username
2. Tick mark “Provide User Access to the AWS Management Console” since the user is going to log in to the AWS Console.
3. Password (Either Automatic or Custom)
4. Users Must Create a new password at the next sign-in – to change users password at the first login



5. Click Next to Continue

Click Next Do not add to any Group





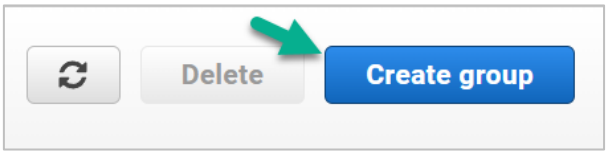
Click on Create User

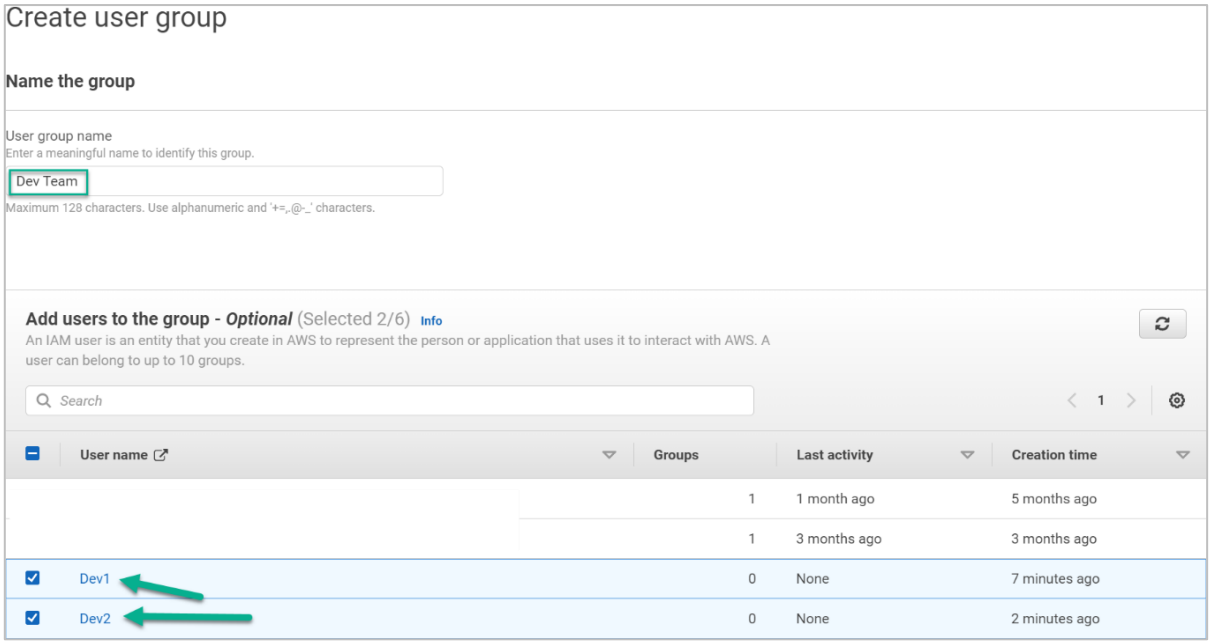
Download the .csv File and keep it safe.

Now repeat the same process for creating the other three users.

Create Groups

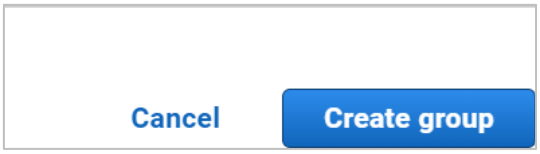
IAM Console → Create Group



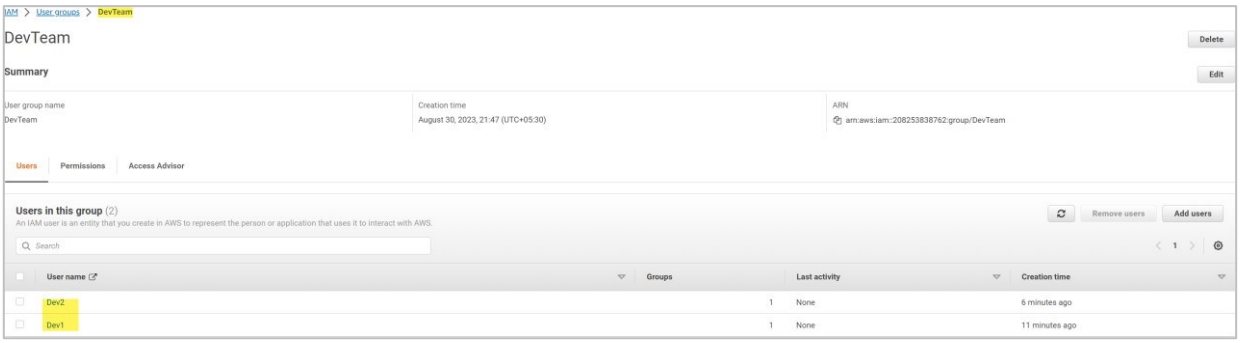


Provide Group Name as “DevTeam” (No Space)

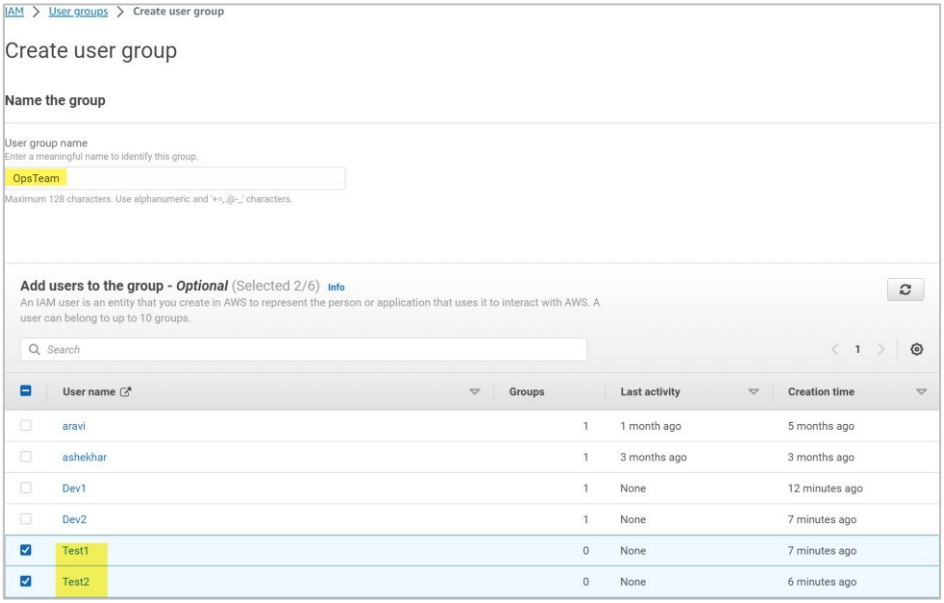
Select Dev1 and Dev2 users to include them in the “Dev Team” group



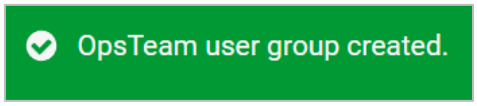
Click on Create Group



That's it DevTeam group is completed, in similar way create OpsTeam



Click on "Create Group"





Done.

Result

The IAM users Dev1, Dev2, Test1 and Test2 were created with console access. The groups DevTeam and OpsTeam were created, Dev1 and Dev2 were added to DevTeam, and Dev1, Test1 and Test2 were added to OpsTeam.

Viva Questions

1. What is IAM?

Ans. AWS Identity and Access Management, the service that manages users, groups, roles and their permissions for an AWS account.

2. What is the difference between an IAM user and an IAM role?

Ans. A user has permanent credentials and represents a person or an application. A role is assumed temporarily by a user, a service or an application and has no long term credentials.

3. Can a user belong to more than one group?

Ans. Yes. A user can be a member of up to ten groups.

4. What is an IAM policy?

Ans. A JSON document that lists the allowed or denied actions on resources. It is attached to users, groups or roles.

5. What is the principle of least privilege?

Ans. Granting only the permissions that are needed to perform a task and nothing more.

Assignment Questions

1. Attach the AmazonS3ReadOnlyAccess managed policy to the Dev Team and the AmazonEC2FullAccess policy to the Ops Team. Sign in as Dev1 and as Test1 and record which actions each user can and cannot perform.

2. Write a custom IAM policy that allows a user to start and stop only the instances tagged Environment=Dev, attach it to a new group and test it.

Experiment 12: DynamoDB Table Creation, Backup and Deletion

12	a) Create a DynamoDB table with a partition key as ID. b) Add 5 items to the DynamoDB table. c) Take a backup and delete the table.	CO6

Aim

To create a DynamoDB table with a partition key named ID, add five items to the table, take a backup of the table and delete the table.

Description

Amazon DynamoDB is a fully managed NoSQL key-value and document database. A table requires a primary key, which is either a partition key alone or a partition key together with a sort key. Items are added through the console, the CLI, the SDK or by importing data from S3. Apart from the key attributes, each item can have its own set of attributes.

DynamoDB offers on-demand backups that capture the full table at a point in time without affecting its performance, and point-in-time recovery for continuous backups. A deleted table can be restored from an on-demand backup under a new table name.

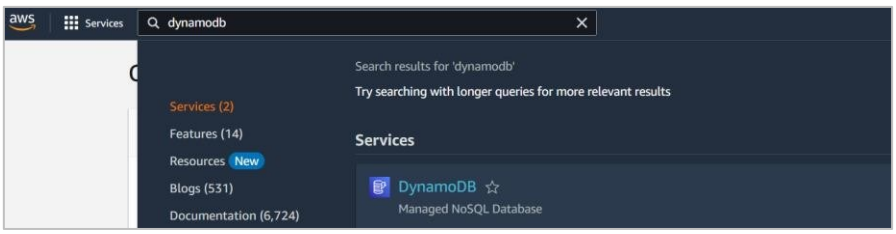
Solution

While migrating, you are asked to perform the following tasks:

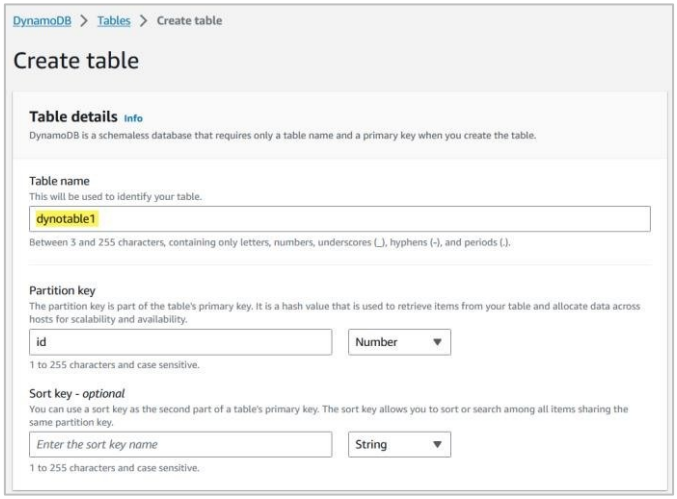
1. Create a DynamoDB table with a partition key as ID.
2. Add 5 items to the DynamoDB table.
3. Take a backup and delete the table.

Answer

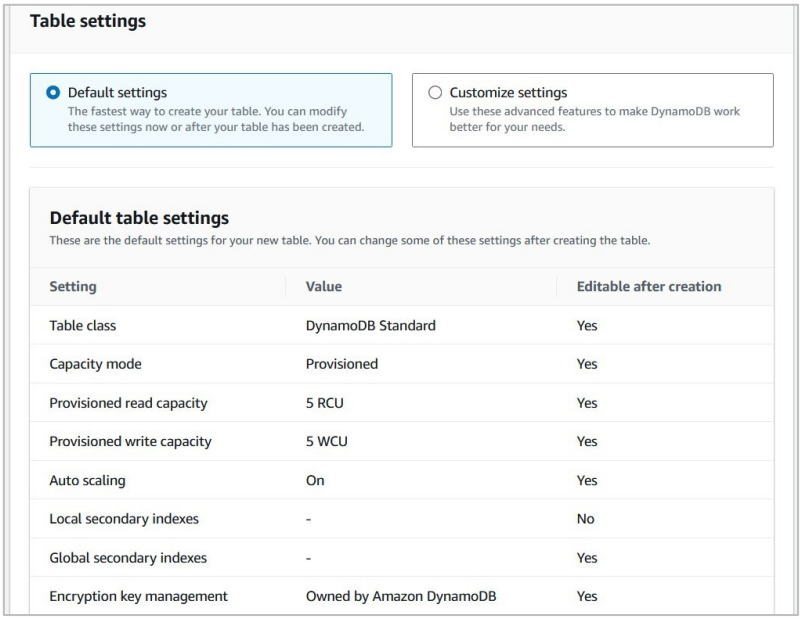
Login to AWS Management Console to to DynamoDB service



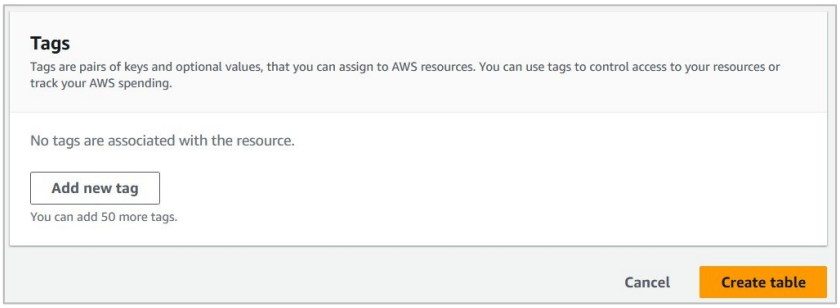
Click DynamoDB



Provide the DynamoDB table name and partition key details



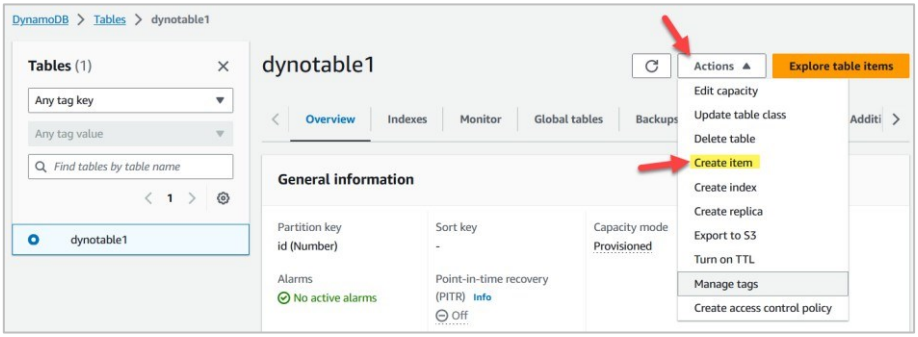
Default settings



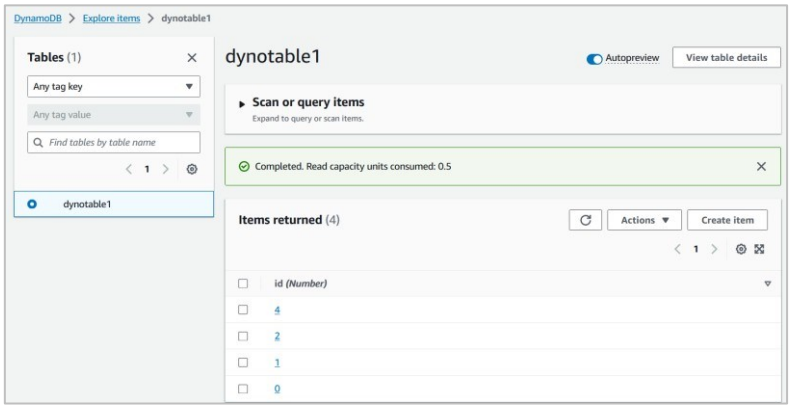
Click “Create table”

Once the Table is created successfully.

Select the table → Actions → create an item

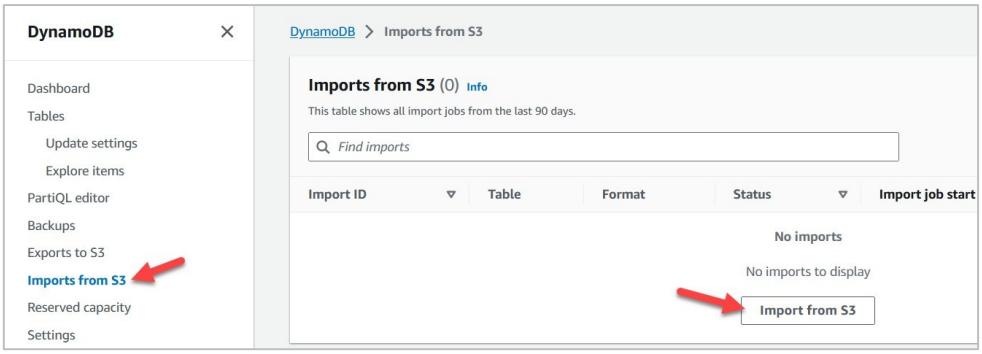




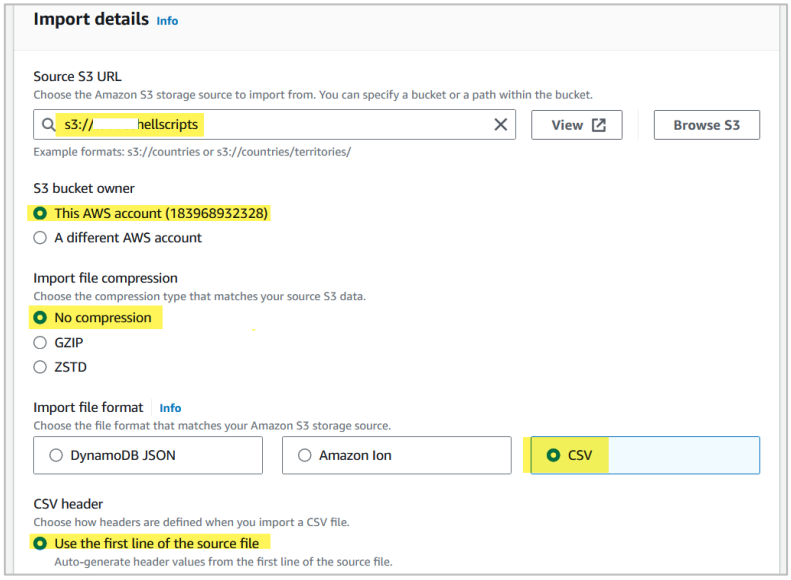


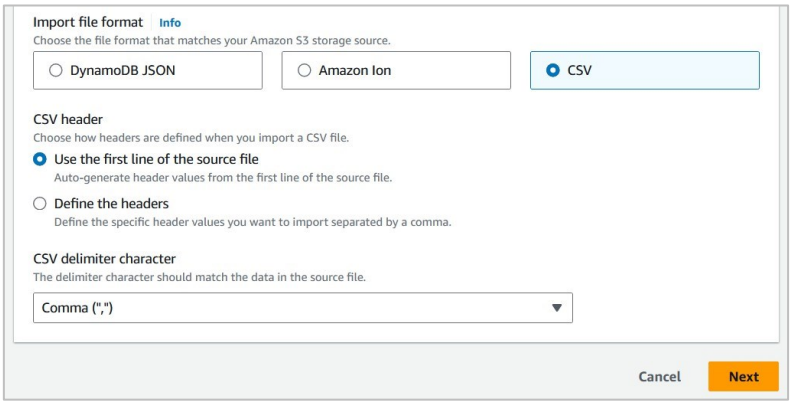
You can see 4 Objects are Created successfully.

Import from S3 is also possible using the below method.

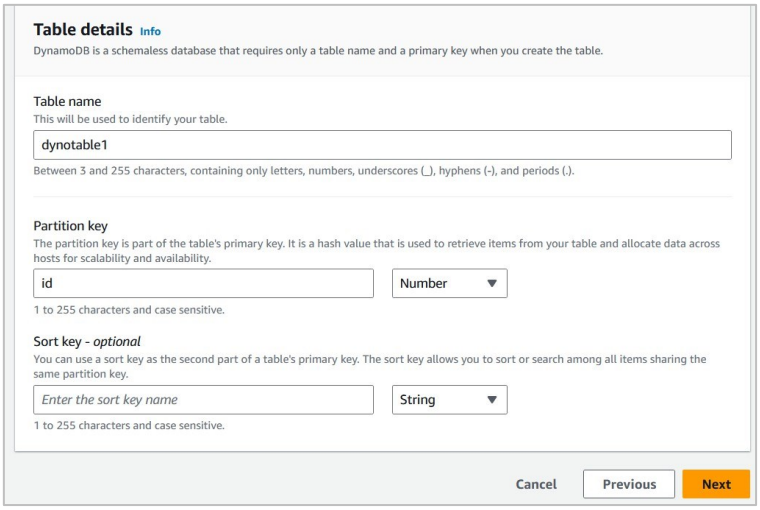


Import from S3





Click Next

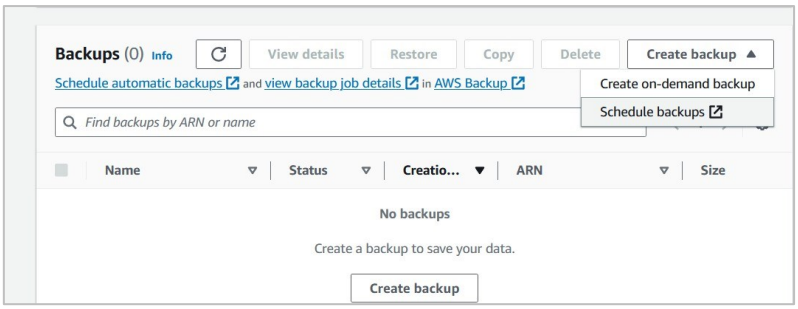


Provide the table details, Click **Next**

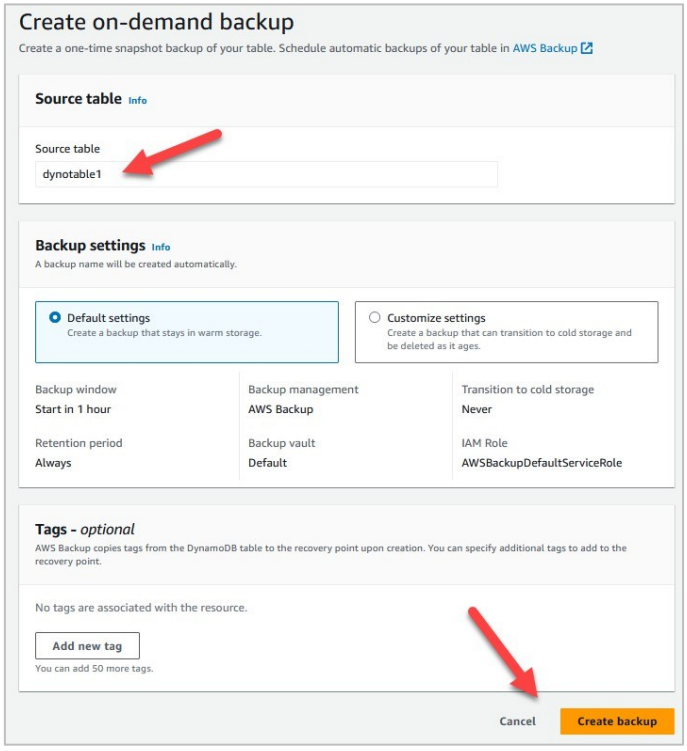
Default Settings Click **Next**

Now Click **Import**

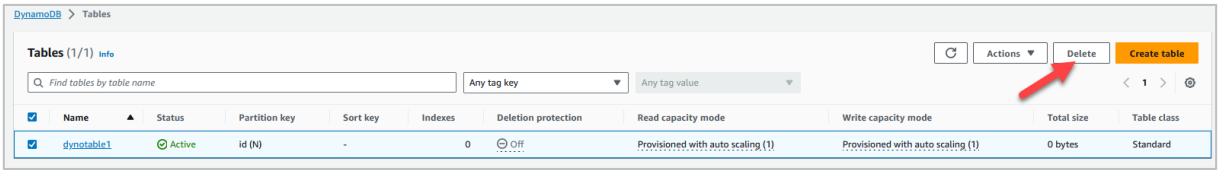
Creating DynamoDB Table Backup



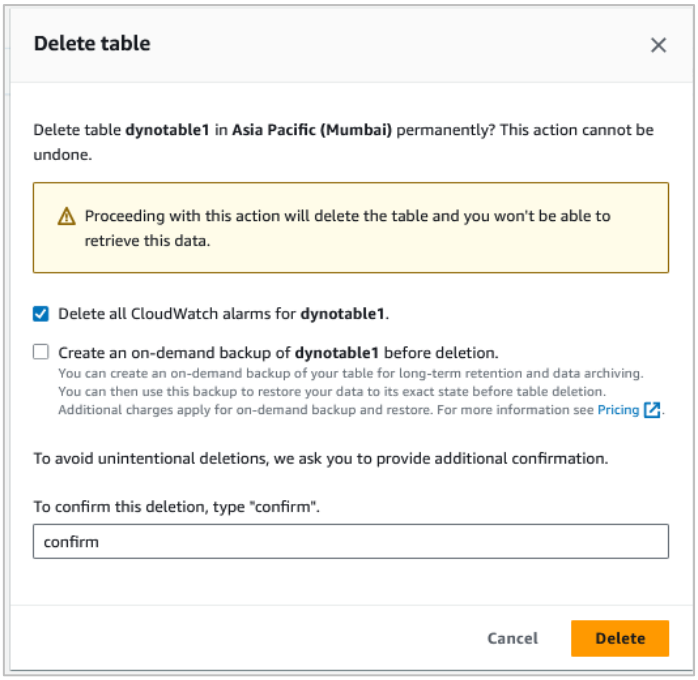
Go to **DynamoDB Table → Backups → Create Backup**



That will create the table backup.



Select the table click Delete





Result

The DynamoDB table was created with ID as the partition key, items were added to it, an on-demand backup of the table was created and the table was deleted. The backup remained available for restoring the table.

Viva Questions

1. What is DynamoDB?

Ans. A fully managed NoSQL database service that provides single digit millisecond performance at any scale.

2. What is a partition key?

Ans. The attribute whose hashed value decides the partition in which an item is stored. It must be unique for each item when there is no sort key.

3. What is the difference between a partition key and a sort key?

Ans. The partition key distributes items across partitions. The sort key orders the items that share the same partition key, and together they form a composite primary key.

4. What are the two capacity modes in DynamoDB?

Ans. Provisioned capacity with read and write capacity units, and on-demand capacity that charges per request.

5. What is an on-demand backup?

Ans. A full backup of a table taken at a point in time that can be used later to restore the table.

Assignment Questions

1. Restore the deleted table from the backup under a new name and verify that all items are present. Then enable point-in-time recovery on the restored table.

2. Create a table with a composite primary key (StudentID as the partition key and Subject as the sort key), add marks for three students and query the marks of one student using the console query feature.

Extra lab Experiment

The following experiments extend the course beyond the twelve sessions listed above. They follow the same structure and can be used for practice or for the capstone project.

Extra Experiment 1: AWS Elastic Load Balancer (ELB)

Aim

To launch two web server instances in different availability zones, create a Classic Load Balancer, register both instances and verify that the traffic is distributed between them.

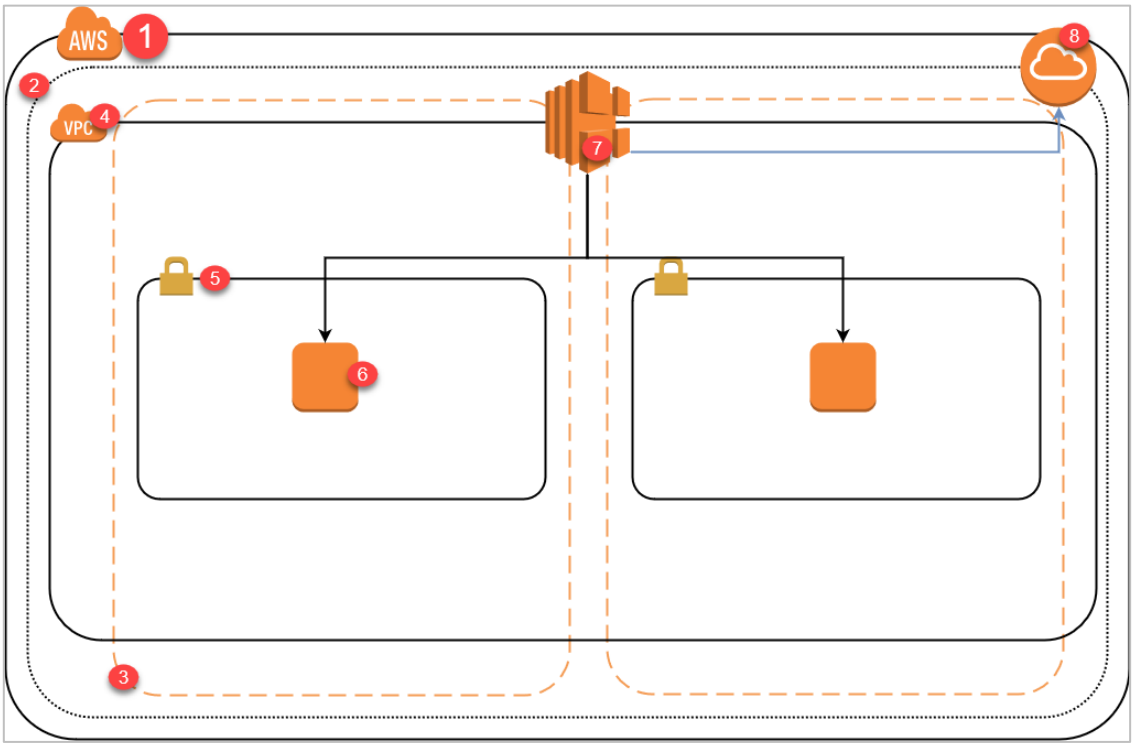
Description

Elastic Load Balancing distributes incoming traffic across targets such as EC2 instances in multiple availability zones. It performs health checks on the targets and sends traffic only to healthy instances. Three types are used in practice: the Application Load Balancer for HTTP and HTTPS at layer 7, the Network Load Balancer for TCP, UDP and TLS at layer 4, and the Classic Load Balancer for the older generation of applications.

A user data script runs when an instance starts for the first time. In this experiment it installs the web server and writes an index page, so that each server shows a different message. The load balancer gets a public DNS name, and refreshing the page in the browser alternates the response between the two servers.

Solution

AWS Elastic Load Balancer (ELB)



1. Elastic Load Balancer Typical Architecture
    1. AWS Cloud
    2. Region
    3. Availability Zone
    4. VPC – Virtual Private Cloud
    5. VPC Subnet
    6. EC2 Instance Running Webserver
    7. Elastic Load Balancer
    8. Internet Gateway

Elastic Load Balancing (ELB) is a load-balancing service for Amazon Web Services (AWS) deployments. ELB automatically distributes incoming application traffic and scales resources to meet traffic demands.

A Managed Load Balancing service

- Distributes load incoming application traffic across multiple targets, such as amazon EC2 instances, containers, and IP Addresses
- Recognizes and responds to unhealthy instances
- Can be public or internal-facing
- Uses HTTP, HTTPS, TCP, and SSL Protocols
- Each Load Balancer is given a public DNS name
    - Internet-facing load balancers have DNS names which publicly resolve to the public IP Addresses of the load balancer of the load balancers nodes
    - Internal load balancers have DNS names, which publicly resolve to the private IP Addresses of the load balancers nodes.

Types of ELB

1. Application Load Balancer
2. Network Load Balancer
3. Classic Load Balancer

ELB Practical

- Launch two EC2 instances in different AZs
- Enable Web services
- Launch Load Balancer
- Add both instances under load balancer now check traffic

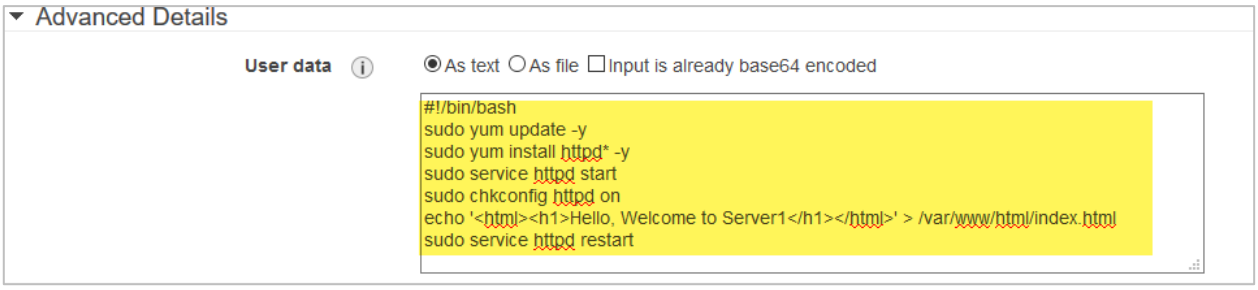
Follow **EC2 Linux instance launch steps** however in step two (configure Instance) go to down to the bottom in advanced section add below script will create auto webserver

https://github.com/techtutorials/aws-lab-guide/blob/aws/webserver.sh

#!/bin/bash

sudo yum update -y

sudo yum install httpd* -y sudo service httpd start sudo chkconfig httpd on

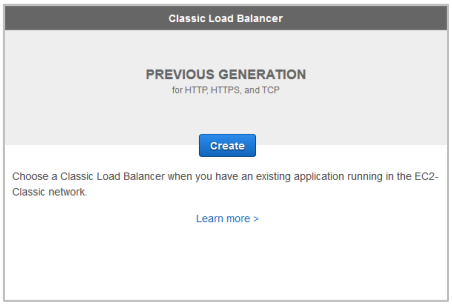


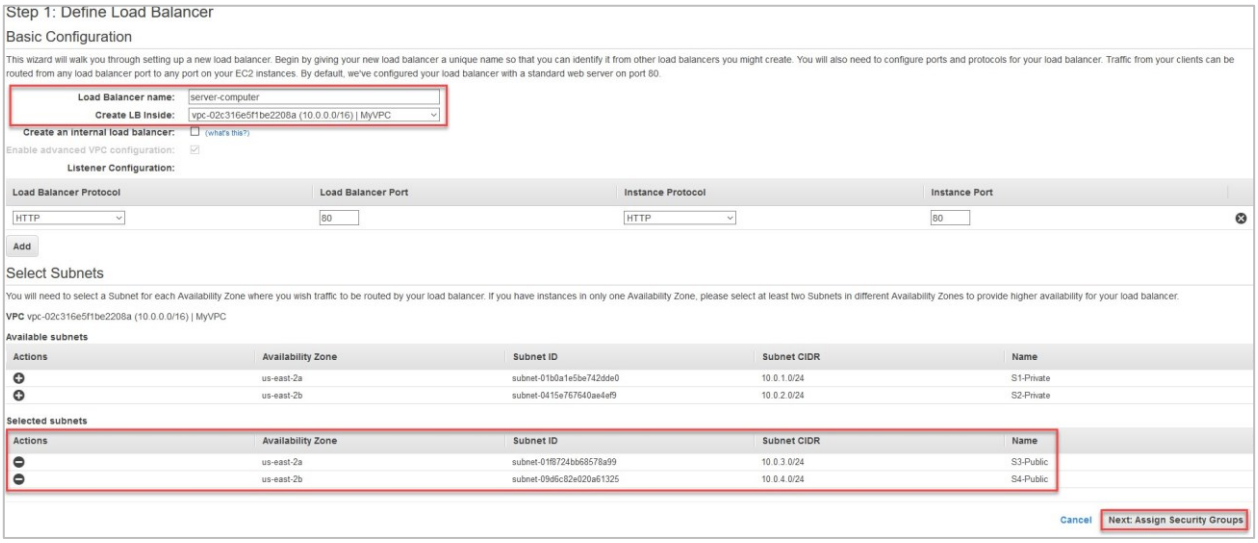
echo '<html><h1>Hello, Welcome to Server1</h1></html>' > /var/www/html/index.html sudo service httpd restart

Note: while launching second instance change echo statement to server2

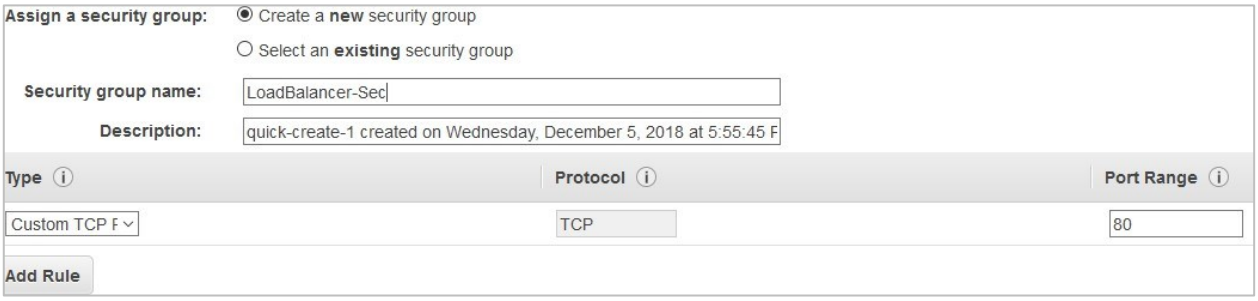
echo '<html><h1>Hello, Welcome to Server2</h1></html>' > /var/www/html/index.html

Creating Classic Elastic Load Balancer



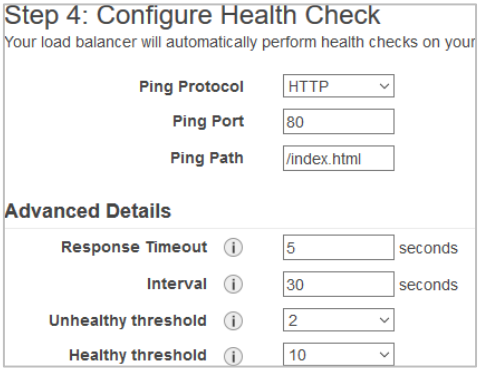


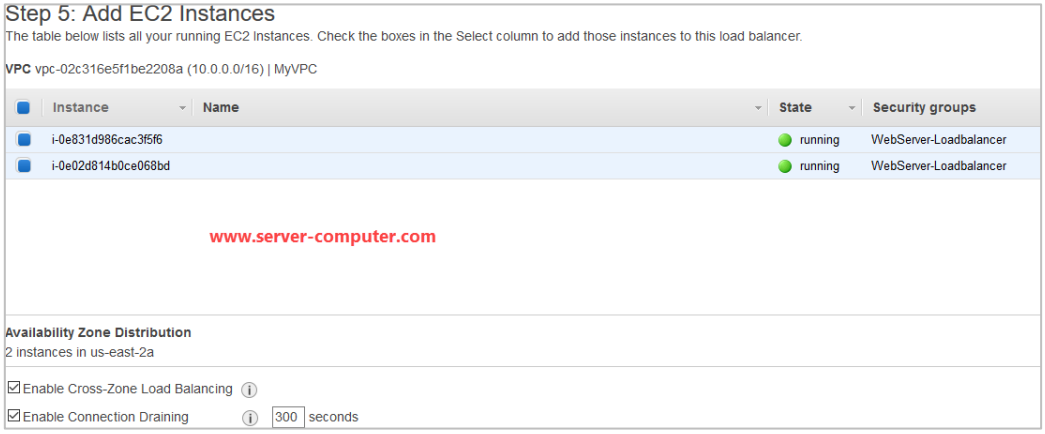
Click Next: Assign Security Groups



Click **Next: Security Settings**

Click Next: Configure Health Checks



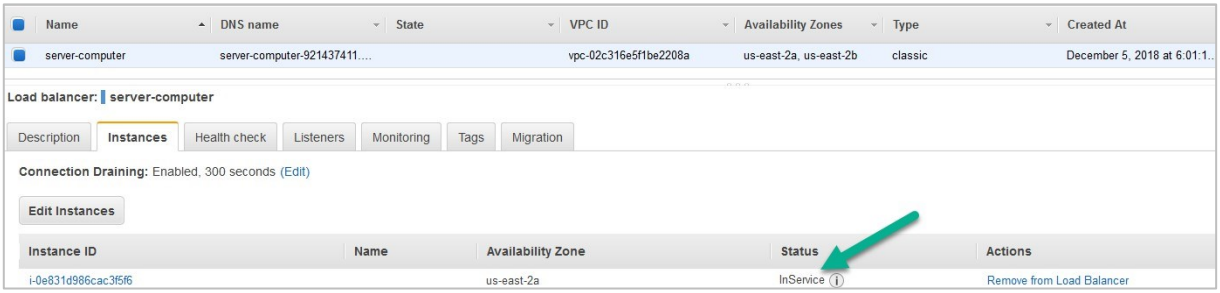


Specify your default web file in this example I am using /index.html Click **Next: Add EC2 Instances**

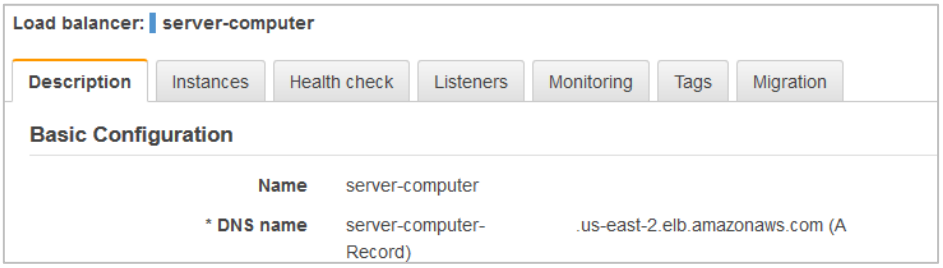
Click **Next: Add Tags**

Click **Review and Create**

Click **Create**



Check instances status should be InService



Load Balancer DNS Name copy it and paste in web browser now fresh twice you will see response is coming from Server1 and Server2



Which concludes load balancer is working fine.

Result

Two web servers were launched with user data scripts, a Classic Load Balancer was created with a health check on /index.html, both instances reached the InService status, and the DNS name of the load balancer returned responses alternately from Server1 and Server2.

Viva Questions

1. What is Elastic Load Balancing?

Ans. A managed service that distributes incoming application traffic across multiple targets and scales with the traffic demand.

2. What are the types of load balancers in AWS?

Ans. Application Load Balancer, Network Load Balancer, Gateway Load Balancer and Classic Load Balancer.

3. What is a health check?

Ans. A periodic request sent by the load balancer to each target. Targets that fail the check are marked unhealthy and taken out of rotation.

4. What is user data in EC2?

Ans. A script passed at launch time that runs on the first boot to configure the instance, for example to install a web server.

5. What is the difference between an internet-facing and an internal load balancer?

Ans. An internet-facing load balancer has a public DNS name that resolves to public IP addresses. An internal load balancer resolves to private IP addresses inside the VPC.

Assignment Questions

1. Replace the Classic Load Balancer with an Application Load Balancer using a target group and configure path based routing so that /app1 and /app2 are served by different instances.

2. Stop the web service on Server1 and observe the health check status and the traffic distribution. Record the time taken for the load balancer to mark the instance unhealthy.

Extra Experiment 2: AWS CloudTrail and Athena Analytics

Aim

To create a CloudTrail trail that records the account activity into an S3 bucket and to analyze the CloudTrail logs using Amazon Athena.

Description

AWS CloudTrail records the API calls and console actions made in the account as events. A trail delivers those events as log files to an S3 bucket and can cover all regions. The logs show who did what, when and from where, which supports governance, compliance and security auditing.

Amazon Athena is an interactive query service that runs standard SQL on data stored in S3. By creating an external table with the CloudTrail SerDe over the log location, the JSON log files can be queried like a database table without loading them anywhere else.

Solution

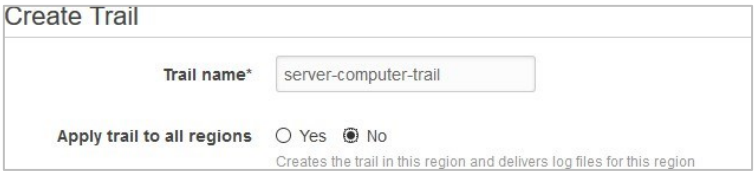
AWS CloudTrail – Enable Governance and Auditing

AWS CloudTrail is an AWS service that helps you enable governance, compliance, and operational and risk auditing of your AWS account. Actions taken by a user, role, or an AWS services are recorded as events in CloudTrail. Events include actions taken in the AWS Management Console, AWS Command Line Interface, and AWS SDKs and APIs.

CloudTrail is enabled on your AWS account when you create it. When activity occurs in your AWS account, that activity is recorded in a CloudTrail event. You can easily view recent events in the CloudTrail console by going to Event history.

Visibility into your AWS account activity is a key aspect of security and operational best practices. You can use CloudTrail to view, search, download, archive, analyze, and respond to account activity across your AWS infrastructure. You can identify whom or what took which action, what resources were acted upon, when the event occurred, and other details to help you analyze and respond to activity in your AWS account.

How to Create CloudTrail

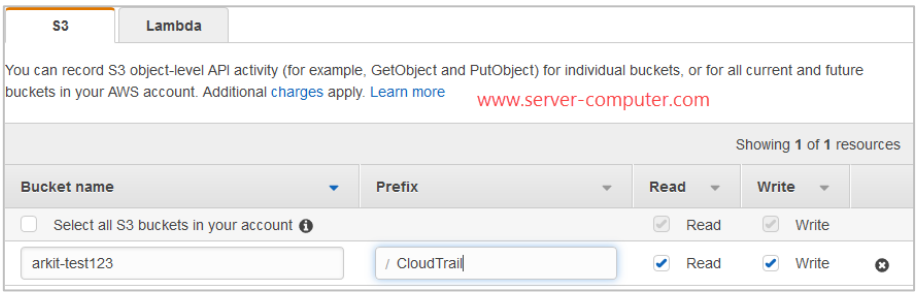


Login to AWS Console → Services → Management & Governance → CloudTrail Click on **Create Trail**

Provide trail name as your wish in this case **server-computer-trail**

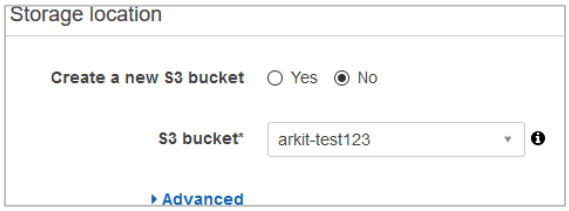
Note: If you want to audit all regions by default select "Yes" radio, button otherwise select "No"





Select S3 bucket where you want to store CloudTrail Logs. CloudTrail logs uses S3 bucket for storing audit logs.

If you did not have S3 bucket created, provide bucket name in storage location section by selecting "Yes" radio button, it will create it for you. Select no if you have existing S3 bucket.



Click **Create**



CloudTrail has been created successfully.

Athena Analytics

If you would like to create a table in hive using existing logs, you can create by clicking on **Athena table creation**.

CREATE EXTERNAL TABLE cloudtrail_logs_server-computer_test123 (eventVersion STRING,

userIdentity STRUCT< type: STRING, principalId: STRING, arn: STRING,

accountId: STRING, invokedBy: STRING, accessKeyId: STRING, userName: STRING, sessionContext: STRUCT<

attributes: STRUCT< mfaAuthenticated: STRING, creationDate: STRING>,

sessionIssuer: STRUCT< type: STRING, principalId: STRING, arn: STRING,

accountId: STRING, userName: STRING>>>,

eventTime STRING, eventSource STRING, eventName STRING, awsRegion STRING, sourceIpAddress STRING, userAgent STRING, errorCode STRING, errorMessage STRING, requestParameters STRING, responseElements STRING, additionalEventData STRING, requestId STRING,

eventId STRING, resources ARRAY<STRUCT<

arn: STRING,

accountId: STRING, type: STRING>>,

eventType STRING, apiVersion STRING, readOnly STRING, recipientAccountId STRING, serviceEventDetails STRING, sharedEventID STRING, vpcEndpointId STRING

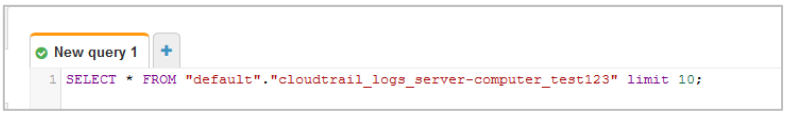
)

COMMENT 'CloudTrail table for server-computer-test123 bucket' ROW FORMAT SERDE 'com.amazon.emr.hive.serde.CloudTrailSerde'

STORED AS INPUTFORMAT 'com.amazon.emr.cloudtrail.CloudTrailInputFormat' OUTPUTFORMAT 'org.apache.hadoop.hive.ql.io.HiveIgnoreKeyTextOutputFormat' LOCATION 's3://server-computer-test123/AWSLogs/687993403879/CloudTrail/' TBLPROPERTIES ('classification'='cloudtrail');

Create table and query using athena interface

Analytics → Athena



You can see the data in tabular format

DROP TABLE cloudtrail_logs_server-computer_test123;

Delete Athena table using above like query (replace table name).

Otherwise, for RAW log go to your S3 bucket and click on bucket name → **AWSLogs** → **Account Number** → You can see all the CloudTrail logs over there.

Download the json.gz file and analyze the activities

Result

A trail was created and the CloudTrail log files were delivered to the S3 bucket. An Athena table was created over the log location and the account activity was queried in tabular form.

Viva Questions

1. What is AWS CloudTrail?

Ans. A service that records API activity and console actions in an AWS account as events for auditing and compliance.

2. Where does CloudTrail store its logs?

Ans. In an S3 bucket chosen when the trail is created. Events can also be sent to CloudWatch Logs.

3. What is Amazon Athena?

Ans. A serverless query service that runs SQL directly on data stored in S3 and charges for the amount of data scanned.

4. What is a SerDe?

Ans. A serializer-deserializer library that tells Athena how to read a file format, for example the CloudTrail SerDe for CloudTrail JSON logs.

5. What is the difference between CloudTrail and CloudWatch?

Ans. CloudTrail records who made which API call. CloudWatch monitors the performance metrics and logs of resources.

Assignment Questions

1. Write an Athena query that lists all ConsoleLogin events of the last seven days with the user name and the source IP address.

2. Configure the trail to send events to CloudWatch Logs and create a metric filter and an alarm that fire when an IAM user is deleted.

Extra Experiment 3: Auto Scaling

Aim

To create a launch configuration and an Auto Scaling group with scaling policies that add an instance when the CPU utilization is 60% or more and remove an instance when it is 20% or less, and to test the policies by generating CPU load.

Description

Amazon EC2 Auto Scaling keeps the right number of instances running for an application. The launch configuration or launch template defines the AMI, the instance type, the key pair and the security group used for new instances. The Auto Scaling group defines the minimum, maximum and desired capacity and the subnets in which instances are launched.

Scaling policies use CloudWatch alarms on metrics such as CPU utilization to add or remove instances. A load balancer can be attached so that new instances receive traffic automatically, and notifications can be sent through SNS when instances are launched or terminated.

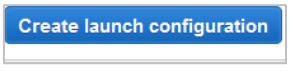
Solution

Auto Scaling

Amazon EC2 Auto Scaling helps you ensure that you have the correct number of Amazon EC2 instances available to handle the load for your application. You create collections of EC2 instances, called Auto Scaling groups. You can specify the minimum number of instances in each Auto Scaling group, and Amazon EC2 Auto Scaling ensures that your group never goes below this size. You can specify the maximum number of instances in each Auto Scaling group, and Amazon EC2 Auto Scaling ensures that your group never goes above this size. If you specify the desired capacity, either when you create the group or at any time thereafter, Amazon EC2 Auto Scaling ensures that your group has this many instances. If you specify scaling policies, then Amazon EC2 Auto Scaling can launch or terminate instances as demand on your application increases or decreases.

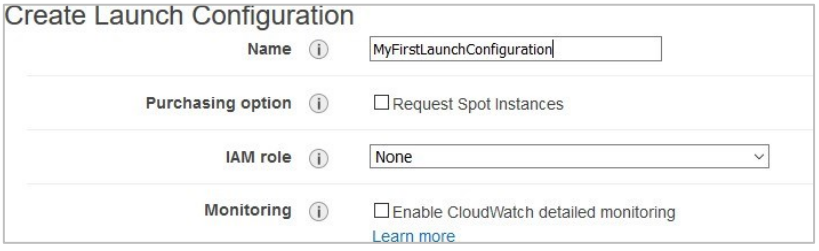
Launch configuration

Login to **AWS Console → EC2 →** (Under Auto Scaling) Click on **Launch Configurations**

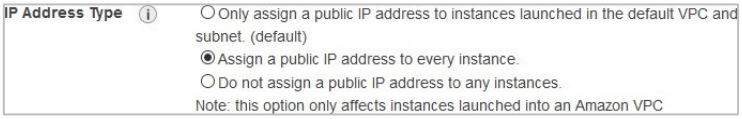


→ **Choose AMI** (I select Ubuntu 18.04 LTS)

→ **Choose Instance Type** (t2.micro) Click Next: Configure Details



>> Click **Advanced Details**

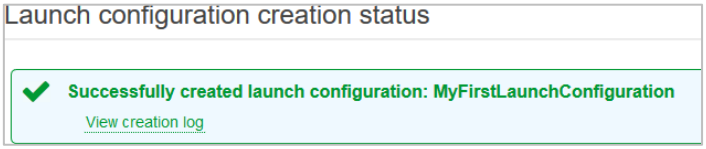


Note: In case there is no default VPC available in selected zone (In my case I deleted default VPC).

Click **Next: Add Storage**

Click Next: Configure Security Group

Select existing Security group or create new security group, as you are wish, (Selecting existing would be good) Click **Review**

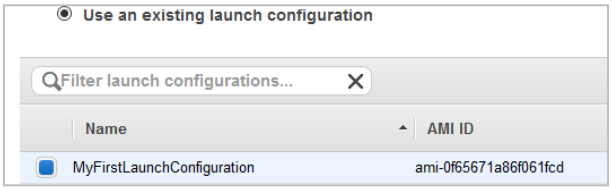


Click Create Launch Configuration Select the Key Pair or create key pair

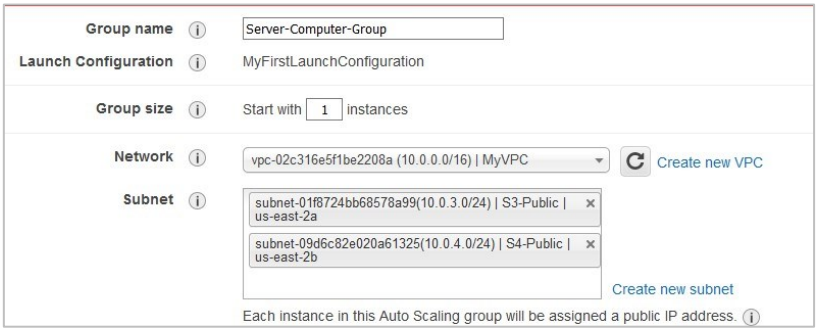
Launch configuration created successfully. Click **Close**

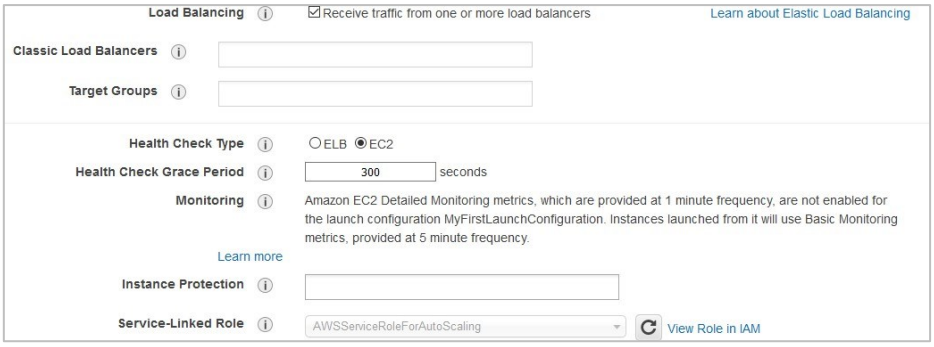
Auto Scaling Groups

Select Auto Scaling Groups → Create Auto Scaling Group → Select Launch Configuration



Click **Next Step**

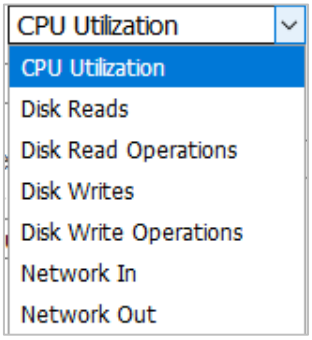


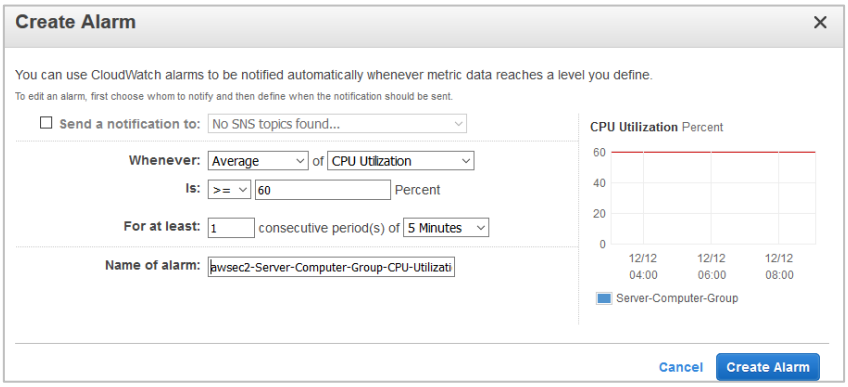


○ Keep this group at its initial size

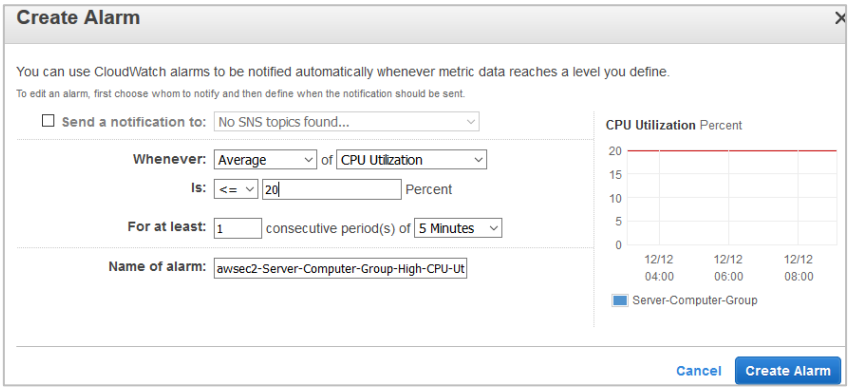
◉ Use scaling policies to adjust the capacity of this group If you are auto-scaling group, want load balancer you can add ELB to auto scaling group Click **Next: Configure Scaling Policies**

If you do not want to create scaling policy, select first radio button otherwise select use scaling policies button Below are the conditions you can use for auto scaling EC2 instances

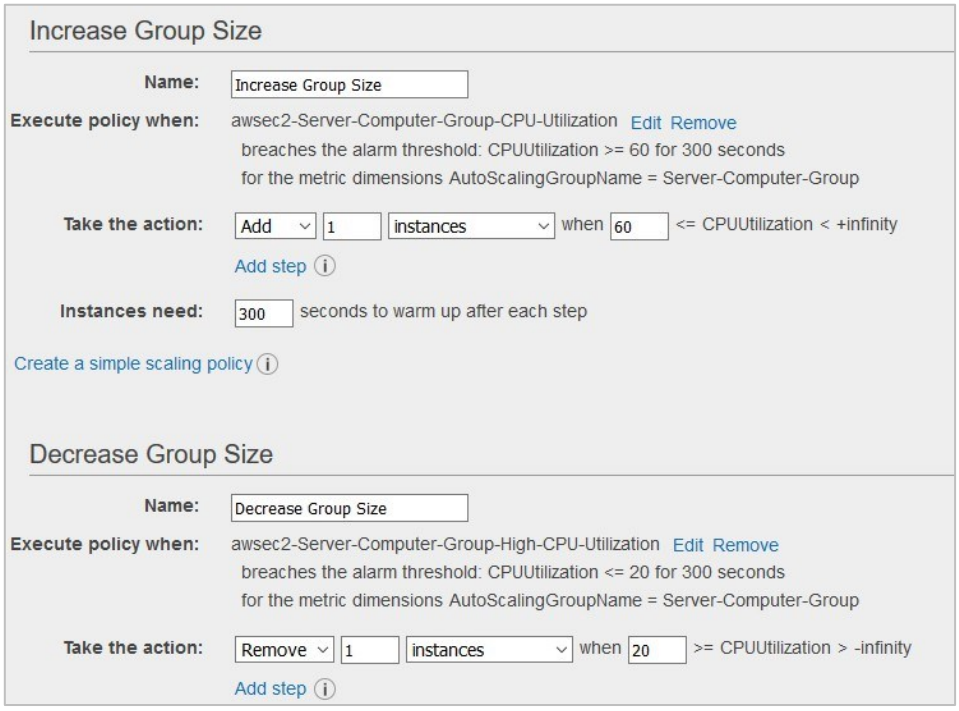




Created Auto increase group IF CPU Utilization is Greater than or equal to 60 for 5minutes add new EC2 instance to auto scaling group

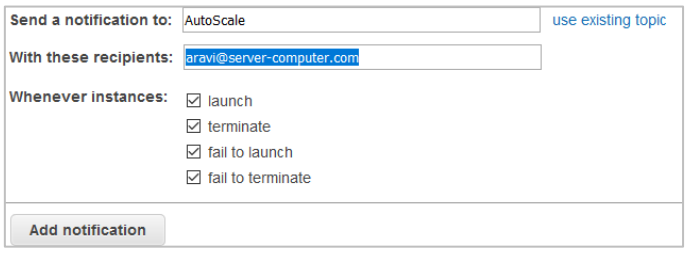


Create auto decrease group IF CPU Utilization is less than or equal to 20 for 5 minutes remove on EC2 instance from scaling group



Click <u>Next: Configure Notifications</u>

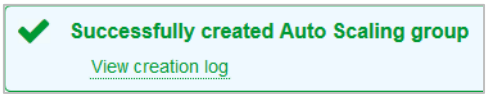
If you want notifications when auto scale triggers create notification



Click **<u>Next: Configure Tags</u>**

Add tags for recognizing auto scale instances Click **<u>review</u>**

Click <u>Create Auto Scaling Group</u>



Now go back to instances you would see EC2 instances launched by auto scaling group configuration.

In order to create a CPU load to test auto scaling use below scripts

while true; do true; done &

dd if=/dev/zero of=/dev/null &

Execute above scripts multiple times in your EC2 instances, to create CPU Load is more than 60 percent for 5 minutes it will automatically launch another EC2 instance.

Wait for 5 Minutes and see

To scale down identify the background running jobs and kill them

jobs

fg <Job Number> CTRL + C

OR

ps –aux |grep dd |awk '{print $2}' | xargs kill -9 ps –aux |grep bash |awk '{print $2}' | xargs kill -9 OR

kill -9 <PID>

Wait for 5 minutes EC2 instances will be terminated automatically which are launched using auto scale option.

Result

A launch configuration was created and an Auto Scaling group was created with scale-out and scale-in policies. Running the CPU load scripts caused a new instance to be launched after five minutes, and killing the load jobs caused the extra instance to be terminated.

Viva Questions

1. What is Auto Scaling?

Ans. A service that automatically adds or removes EC2 instances based on demand so that the application has the required capacity at the lowest cost.

2. What is a launch configuration?

Ans. A template that specifies the AMI, the instance type, the key pair, the security groups and the storage used by the Auto Scaling group to launch instances.

3. What are minimum, maximum and desired capacity?

Ans. The lower and upper limits on the number of instances in the group and the number of instances the group tries to maintain.

4. What is a cooldown period?

Ans. The time after a scaling activity during which Auto Scaling waits before starting another scaling activity.

5. Name the types of scaling policies.

Ans. Target tracking, step scaling, simple scaling and scheduled scaling.

Assignment Questions

1. Create an Auto Scaling group from a launch template with a target tracking policy that keeps the average CPU utilization at 50%, and attach it to the load balancer from the ELB experiment.

2. Configure a scheduled scaling action that sets the desired capacity to 3 at 9 AM and back to 1 at 6 PM, and verify it from the activity history.

Extra Experiment 4: AWS CloudFormation

Aim

To provision an EC2 web server stack automatically using an AWS CloudFormation template.

Description

AWS CloudFormation describes infrastructure in a text template written in JSON or YAML. The template declares resources such as instances, security groups and buckets with their properties. CloudFormation creates all the resources as a stack in the correct order, and deleting the stack removes every resource it created.

This is infrastructure as code. The template is the single source of truth for the environment, can be kept under version control and can be reused across regions and accounts. There is no charge for CloudFormation itself; only the resources created by the stack are billed.

Solution

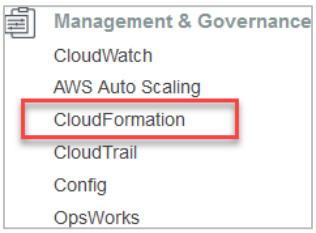
ClodFormation

AWS CloudFormation provides a common language for you to describe and provision all the infrastructure resources in your cloud environment. CloudFormation allows you to use a simple text file to model and provision, in an automated and secure manner, all the resources needed for your applications across all regions and accounts. This file serves as the single source of truth for your cloud environment.

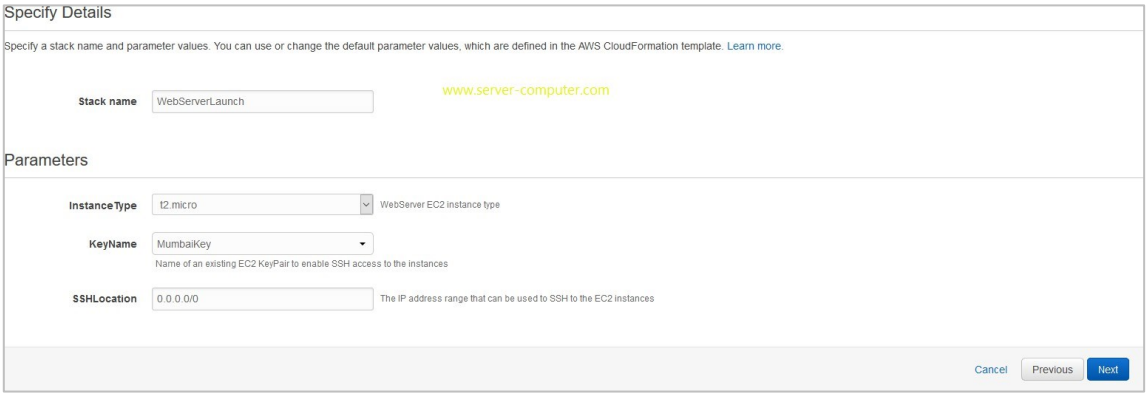
AWS CloudFormation is available at no additional charge, and you pay only for the AWS resources needed to run your applications.

"*Cloud Infrastructure as Code*"

Login to **AWS Web console →Services**

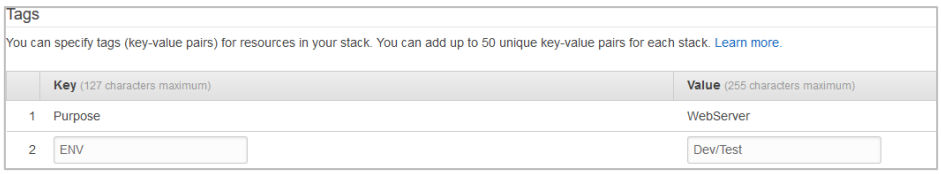


Create Stack



https://github.com/techtutorials/aws-lab-guide/blob/aws/LaunchEC2WebServer.template

Download and upload the template file Click **Next**



Add Tags



Click **Next**

Click on **Create**

It will create S3 bucket for CF template store and keeps your CloufFormation templates in it If you delete CloufFormation, it will automatically delete associate stack/resources

Result

The LaunchEC2WebServer template was uploaded, the stack was created with the required tags, and the EC2 web server resources defined in the template were provisioned automatically.

Viva Questions

1. What is AWS CloudFormation?

Ans. A service that provisions AWS resources from a template file in an automated and repeatable way.

2. What is a stack?

Ans. The collection of resources created from one template that is managed as a single unit.

3. Which formats can a CloudFormation template use?

Ans. JSON and YAML.

4. What is infrastructure as code?

Ans. Managing infrastructure through machine readable definition files instead of manual configuration in the console.

5. What happens when a stack is deleted?

Ans. CloudFormation deletes all the resources it created for the stack, unless a resource has a Retain deletion policy.

Assignment Questions

1. Write a YAML template that creates a VPC, a public subnet, an internet gateway and a security group, and create a stack from it. Then update the stack to add a second subnet.

2. Use a change set to preview the changes before updating the stack and explain what the change set shows.

Extra Experiment 5: Amazon FSx

Aim

To create an Amazon FSx for Windows File Server file system and configure its storage, network, authentication and backup settings.

Description

Amazon FSx provides fully managed third-party file systems. FSx for Windows File Server offers SMB file shares integrated with Active Directory, and FSx for Lustre offers a high performance file system for HPC and machine learning workloads. FSx takes care of hardware provisioning, patching and backups.

While creating a file system the storage capacity, the throughput capacity, the VPC, the availability zone, the subnet, the security group, the Active Directory and the backup and maintenance windows are configured. The file system is accessed from Windows instances by mapping the share through its DNS name.

Solution

Amazon FSx

Amazon FSx provides fully managed third-party file systems. Amazon FSx provides you with the native compatibility of third-party file systems with feature sets for workloads such as Windows-based storage, high-performance computing

(HPC), machine learning, and electronic design automation (EDA). You don't have to worry about managing file servers and storage, as Amazon FSx automates the time-consuming administration tasks such as hardware provisioning, software configuration, patching, and backups. Amazon FSx integrates the file systems with cloud-native AWS services, making them even more useful for a broader set of workloads.

Services → Storage → FSx → Create File system



Note: If you're looking for HPC High Performance computer then select FSX for Lustre

Click **Next**

File System name

Minimum 300GB and Max 65536GB

Default throughput 8MB/s you can also select different values of throughput

Select Network & Security

- VPC
- AZ
- Subnet
- Security Group

Windows Authentication

Note: Must be active directory or create new active directory in AWS

Encryption

Maintenance preferences Select backup window time Click **Next**

Result

The FSx file system settings for storage capacity, throughput, network and security, Windows authentication, encryption and maintenance were configured and the file system creation steps were completed.

Viva Questions

1. What is Amazon FSx?

Ans. A managed service that provides third-party file systems such as Windows File Server and Lustre.

2. When is FSx for Lustre used?

Ans. For high performance computing, machine learning and other workloads that need fast parallel file access.

3. What is required for Windows authentication in FSx?

Ans. An Active Directory, either AWS Managed Microsoft AD or a self-managed directory.

4. What is the storage capacity range of an FSx for Windows file system given in the manual?

Ans. A minimum of 300 GB and a maximum of 65536 GB.

5. What is the difference between EFS and FSx for Windows?

Ans. EFS provides NFS file systems for Linux. FSx for Windows provides SMB file shares with Windows features such as ACLs and Active Directory integration.

Assignment Questions

1. Create an Amazon EFS file system and mount it on two Linux instances at the same time. Write a file from one instance and read it from the other.

2. Prepare a comparison table of EBS, EFS, FSx and S3 covering the access method, the protocol, sharing between instances and typical use cases.

Extra Experiment 6: SQS – Simple Queue Service

Aim

To create an Amazon SQS queue, send a message to it and poll the queue to view the message.

Description

Amazon SQS is a fully managed message queue that decouples the components of an application. Producers send messages to the queue and consumers poll the queue to receive and delete them. Standard queues provide at-least-once delivery with best-effort ordering and nearly unlimited throughput. FIFO queues provide exactly-once processing and strict ordering with limited throughput.

Messages are kept in the queue for a retention period of up to 14 days. The visibility timeout hides a received message from other consumers while it is being processed, and a dead-letter queue collects messages that could not be processed.

Solution

SQS – Simple Queue Service

Amazon SQS provides several advantages over building your own software for managing message queues or using commercial or open-source message queuing systems that require significant up-front time for development and configuration.

These alternatives require ongoing hardware maintenance and system administration resources. The complexity of configuring and managing these systems is compounded by the need for redundant storage of messages that ensures messages are not lost if hardware fails.

What can be used to communicate between components?

- Amazon Simple Queue Service (SQS)

Standard Queue

- At-Least-Once Delivery
- Best-Effort-Ordering

FIFO Queue

- Exactly-Once Processing

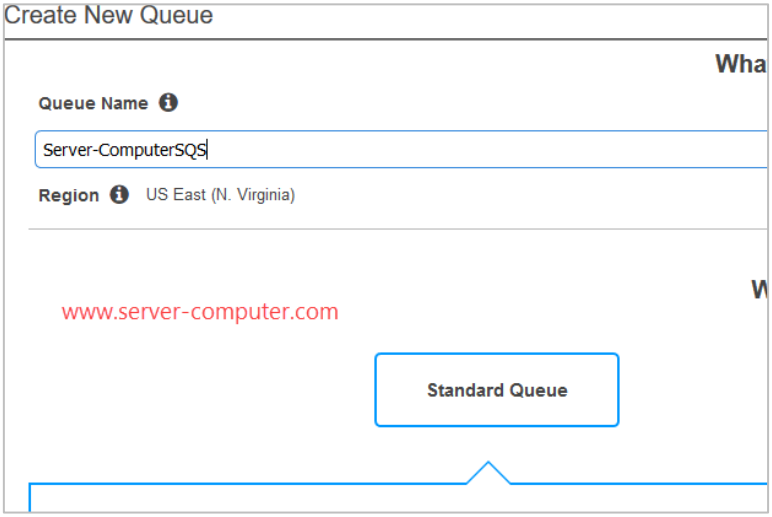
Duplicates are not introduces

- Limited Throughput

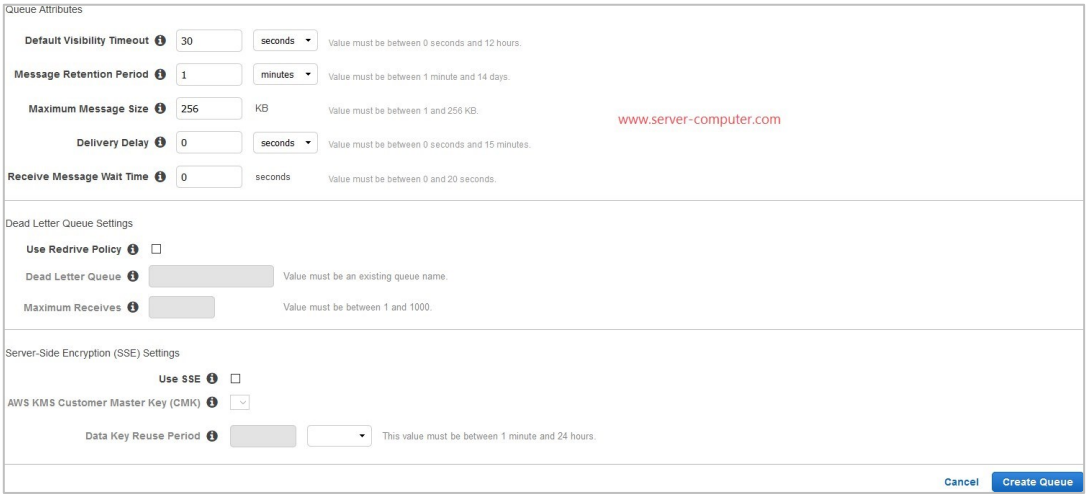
Up to 300 send, receive, delete per second

Services → Application Integration → Simple Queue Service





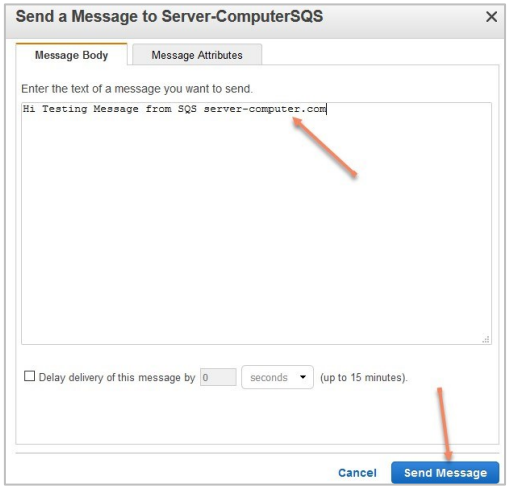
Provide queue name and Click **Configure Queue**



Click **Create Queue**

New queue created successfully. Now send message and poll to see the message queue Select newly created queue name and Actions → send message

Write the message in message box example is shown in below screenshot Click **Send Message**



Close the popup window, select queue name **Actions → View/Delete messages → start polling for messages**



This scenario is only for testing SQS or practicing SQS. If you know use case or project, where you can integrate SQS try

Result

A queue was created, a message was sent to it from the console, and the message was received by polling the queue.

Viva Questions

1. What is Amazon SQS?

Ans. A managed message queuing service used to send, store and receive messages between software components.

2. What is the difference between a standard queue and a FIFO queue?

Ans. Standard queues give at-least-once delivery and best-effort ordering. FIFO queues give exactly-once processing and preserve the order of the messages.

3. What is the visibility timeout?

Ans. The period during which a message received by one consumer is hidden from the other consumers.

4. What is long polling?

Ans. A receive request that waits up to 20 seconds for a message to arrive, which reduces empty responses and cost.

5. What is a dead-letter queue?

Ans. A queue that receives messages that could not be processed successfully after a set number of attempts.

Assignment Questions

1. Create a FIFO queue, send five messages with the same message group ID and show that they are received in the same order.

2. Write a Python script using boto3 that sends ten messages to the queue and another script that receives and deletes them.

Extra Experiment 7: SNS – Simple Notification Service

Aim

To create an SNS topic, subscribe an email address and an SQS queue to it, publish a message and verify the delivery through SNS, SQS and a Lambda function.

Description

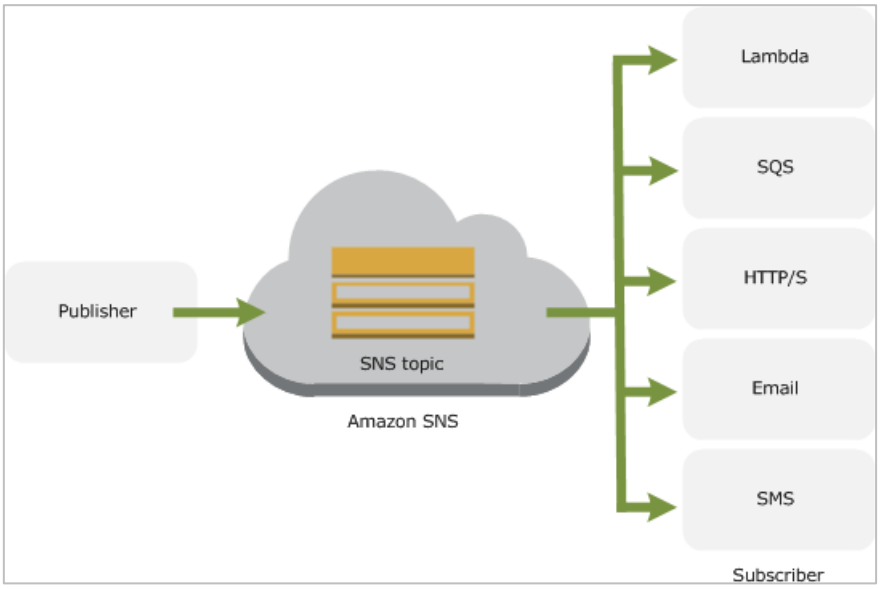
Amazon SNS is a publish and subscribe messaging service. Publishers send messages to a topic and SNS delivers a copy to every subscriber of that topic over the chosen protocol: email, SMS, HTTP or HTTPS, SQS or Lambda. An SQS queue subscribed to a topic must have an access policy that allows SNS to send messages to it.

This fan-out pattern lets one event reach several consumers at the same time. Combining SNS, SQS and Lambda gives a chain in which a published message is queued and then processed by a function.

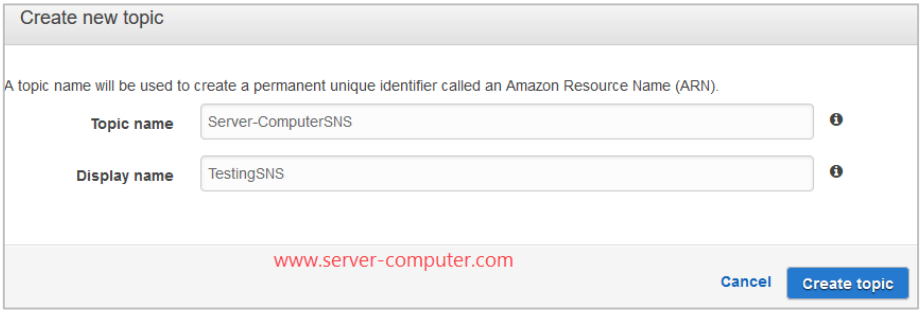
Solution

SNS – Simple Notification Service

Amazon Simple Notification Service (Amazon SNS) is a web service that coordinates and manages the delivery or sending of messages to subscribing endpoints or clients. In Amazon SNS, there are two types of clients—publishers and subscribers—also referred to as producers and consumers. Publishers communicate asynchronously with subscribers by producing and sending a message to a topic, which is a logical access point and communication channel. Subscribers (i.e., web servers, email addresses, Amazon SQS queues, AWS Lambda functions) consume or receive the message or notification over one of the supported protocols (i.e., Amazon SQS, HTTP/S, email, SMS, Lambda) when they are subscribed to the topic.



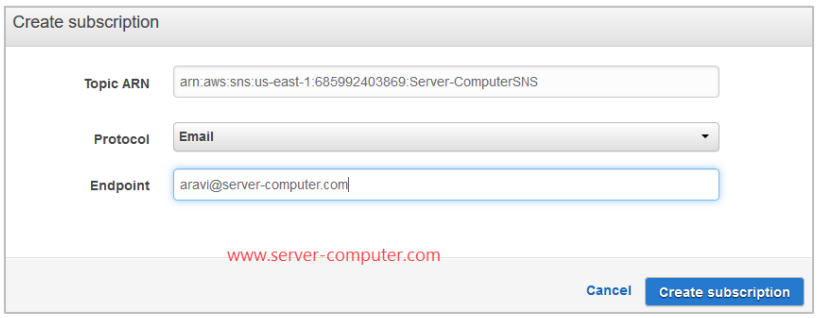
Services → Simple Notification Service → Create topic



Click **Create Topic**

Topic created successfully. Click on topic

Create subscription

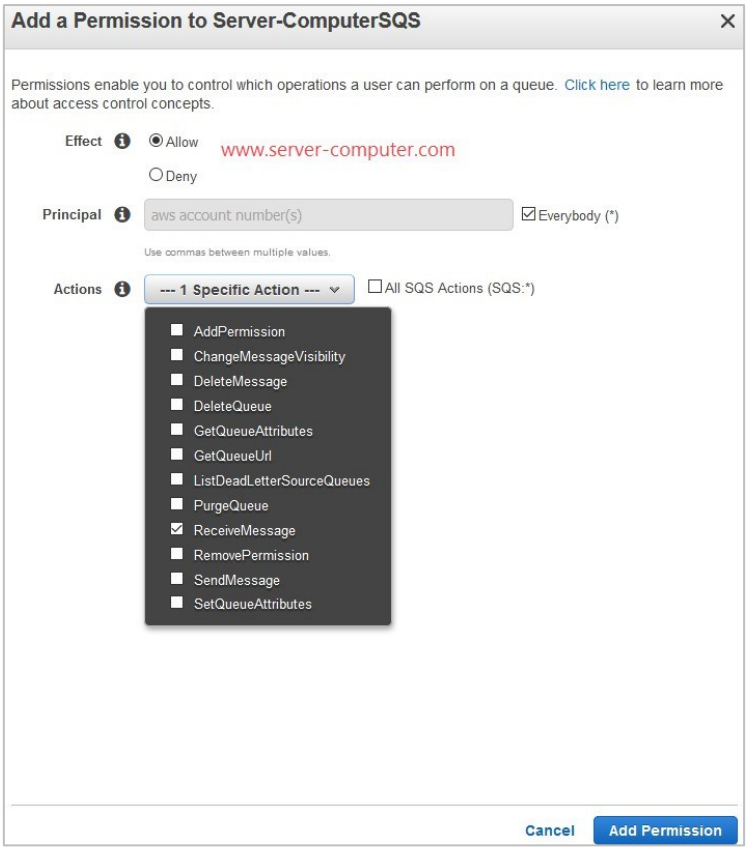


Subscription will sent an email for verification after verification you will see subscription ID Click **Publish to Topic**

Write Subject and Message body click publish

All the subscribers will receive email immediately Will do Flow as **SNS → SQS → Lambda function**

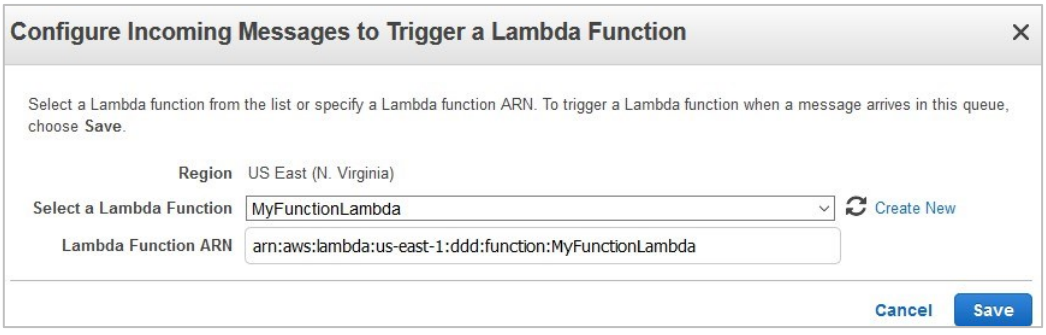
Go to SQS and provide permissions to SNS to send notifications using ARN value Select SQS Queue and add permissions



Copy the ARN value from Details tab on SQS arn:aws:sqs:us-east-1:585692493869:Server-ComputerSQS Change back to SNS and create Subscription under topic

- Topic ARN : Autofil
- Protocol: Amazon SQS
- EndPoint: ARN Value copied from SQS Send topic

Go back to SQS and View/Delete Messages → Start poling messages you can see the message from SNS Similar to this create Lambda function, get ARN value from Lambda, and add to SQS for further triggers Queue Actions → configure Trigger for Lambda Function



Go back to SNS and publish to topic

As soon as SNS trigs SQS will send message after words lambda will execute the defined function.

Result

An SNS topic was created, an email subscription was confirmed and a published message was received by email. The SQS queue was subscribed to the topic with the required permissions, the message was seen in the queue, and the Lambda trigger on the queue executed the function.

Viva Questions

1. What is Amazon SNS?

Ans. A managed publish and subscribe service that delivers messages from publishers to the subscribing endpoints.

2. Which protocols can an SNS subscription use?

Ans. Email, Email-JSON, SMS, HTTP, HTTPS, Amazon SQS, AWS Lambda and mobile push.

3. What is the fan-out pattern?

Ans. Publishing one message to a topic that is delivered to several subscribers, such as multiple SQS queues, for parallel processing.

4. Why must the SQS queue policy be changed before subscribing the queue to a topic?

Ans. SNS needs permission to call SendMessage on the queue. The queue access policy must allow the topic ARN to send messages.

5. What is the difference between SNS and SQS?

Ans. SNS pushes messages to many subscribers immediately. SQS stores messages in a queue until a consumer polls and processes them.

Assignment Questions

1. Create a topic with two SQS queue subscriptions and show that a single published message is delivered to both queues.

2. Use a subscription filter policy so that only messages with the attribute priority = high are delivered to one of the queues.

Reference Material

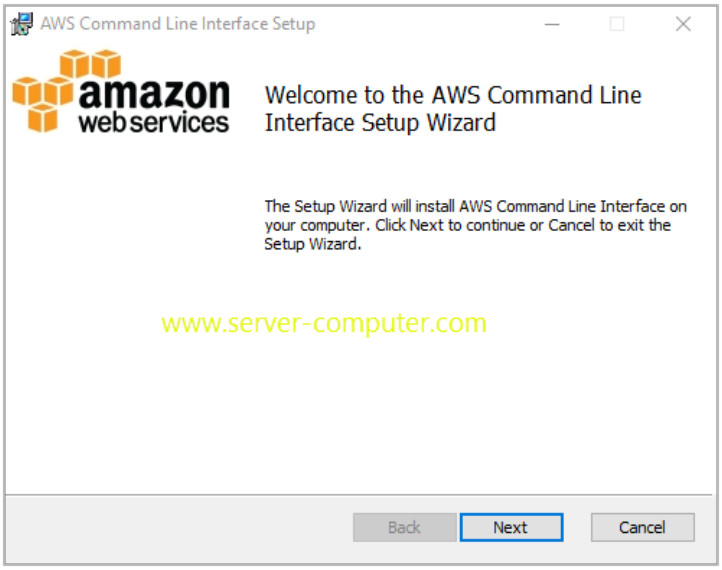
5. AWS CLI

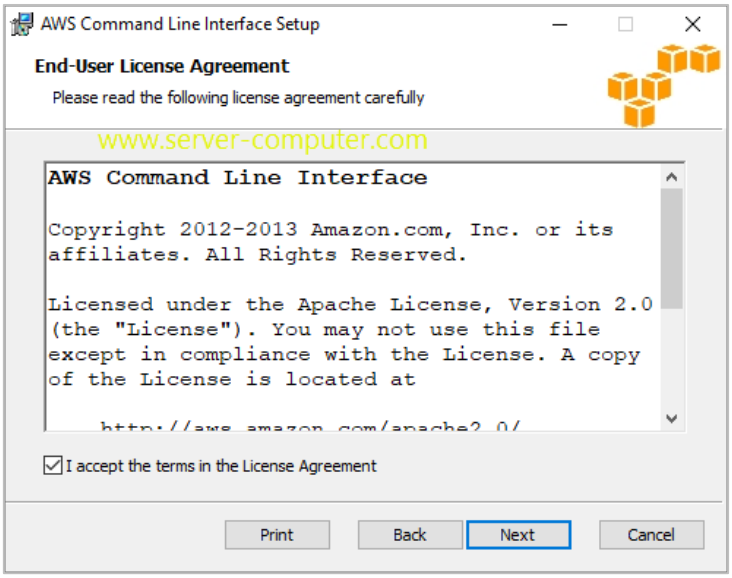
The AWS Command Line Interface (CLI) is a unified tool to manage your AWS services. With just one tool to download and configure, you can control multiple AWS services from the command line and automate them through scripts. The AWS CLI introduces a new set of simple file commands for efficient file transfers to and from Amazon S3.

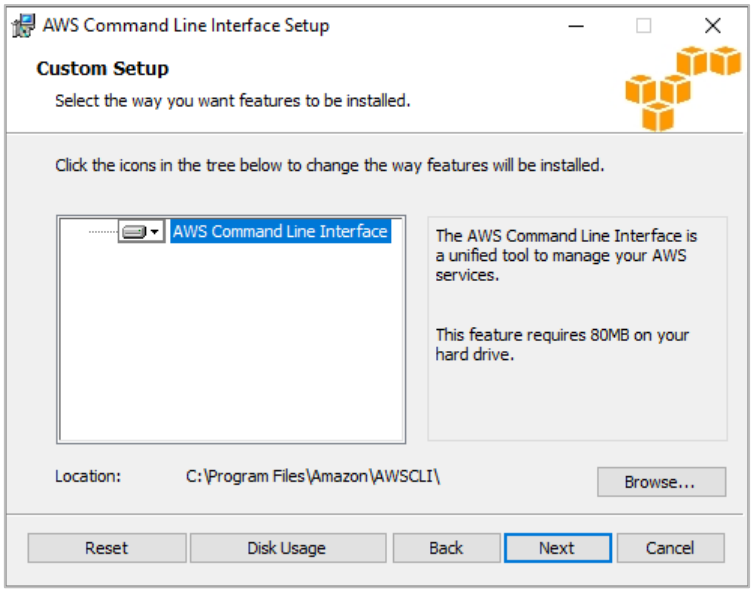
Aws cli configuration for Linux

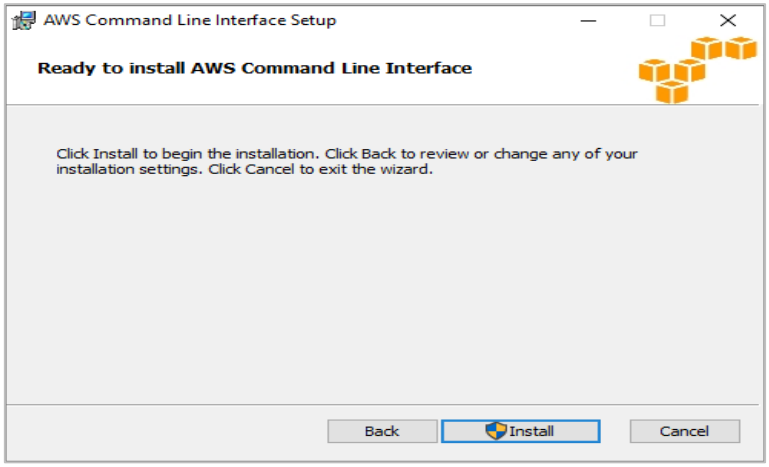
Download AWS CLI for Windows 64 Bit

You should require administrator privileges to install this package in windows machine Double click on .msi file









Click **Install**

Click **Finish**

Login back to AWS Management console and create user with programmatic access **Refer Topic 5 download** ACCESS Key and secret key

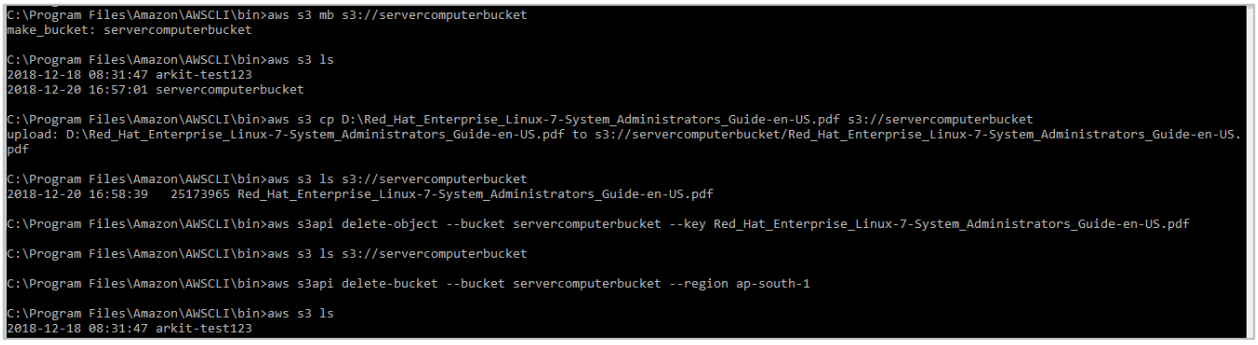
Start Menu → Run → cmd

cd C:\Program Files\Amazon\AWSCLI\bin Change your directory path to above mentioned

>aws configure

```
C:\Program Files\Amazon\AWSCLI\bin>aws configure
AWS Access Key ID [****************ULVQ]:
AWS Secret Access Key [****************XhN2]:
Default region name [ap-south-1]:
Default output format [None]:
```

Now successfully installed and configure aws cli, run few aws cli commands to manage AWS infrastructure



Create S3 Bucket

Bin> aws s3 mb s3://servercomputerbucket

make_bucket: servercomputerbucket

List S3 buckets

Bin> aws s3 ls

2018-12-18 08:31:47 arkit-test123

2018-12-20 16:57:01 servercomputerbucket

Upload Object to S3 Bucket

Bin> aws s3 cp D:\Red_Hat_Enterprise_Linux-7-System_Administrators_Guide-en-US.pdf s3://servercomputerbucket

upload: D:\Red_Hat_Enterprise_Linux-7-System_Administrators_Guide-en-US.pdf to s3://servercomputerbucket/Red_Hat_Enterprise_Linux-7-System_Administrators_Guide-en-US.pdf

List Objects in S3 Bucket

Bin> aws s3 ls s3://servercomputerbucket

2018-12-20 16:58:39 25173965 Red_Hat_Enterprise_Linux-7-System_Administrators_Guide-en-US.pdf

Delete Object from S3 bucket

Bin> aws s3api delete-object --bucket servercomputerbucket --key Red_Hat_Enterprise_Linux-7-System_Administrators_Guide-en-US.pdf

Bin> aws s3 ls s3://servercomputerbucket

Delete S3 bucket

Bin> aws s3api delete-bucket --bucket servercomputerbucket --region ap-south-1

6. Creating EC2 Instance using AWS CLI

Before creating an EC2 instance using AWS CLI collect few details

- AMI ID
- Instance Type
- Key Name (If there is no Key Pair create one)
- Security Group ID
- Subnet ID

Bin> aws ec2 run-instances --image-id ami-06bcd1131b2f55803 --count 1 --instance-type t2.micro --key-name KEYNAME --security-group-ids sg-857f92e9 --subnet-id subnet-51e5592c

7. Few AWS Articles

→ Mount S3 Bucket in Linux using S3FS

→ Use S3 Bucket as Windows Local Drive

→ AWS Basic Interview Questions and Answers

→ AWS Certification course Content

→ List all AWS Instances from All Regions

→ How To create your First Free Tier AWS Account

→ AWS Add New User Accounts with SSH Access Linux Instance

→

8. AWS Services and abbreviations

- S3 – Simple Storage
- EC2 – Elastic Compute Cloud
- EBS – Elastic Block Storage
- EFS – Elastic File System
- ECS – Elastic Container Service
- EKS – Elastic Container Service for Kubernetes
- RDS – Amazon Relational Database Service
- IAM – Identity, Access Management
- VPC – Virtual Private Cloud (isolated Network)
- ELB – Elastic Load Balancer
- EMR – Elastic MapReduce
- MSK – Managed Streaming for Kafka
- SQS – Amazon Simple Queue Service
- SNS – Amazon Simple Notification Service
- SES – Amazon Simple Email Service

- ECR –Amazon Elastic Container Registry
- SWF – Amazon Simple Workflow Service

Foundations of Cloud Computing

- AWS Certified Cloud Practitioner
- The Cloud
- 6 Advantages of Cloud
- Cloud Deployment Models
- AWS Global Infrastructure
- AWS Free Tier Account

Technology

- AWS Cloud Compute
- AWS Lambda Function-as-a-service
- AWS Fargate
- What is CDN
- AWS S3
- AWS Storage Services
- AWS CloudFront
- AWS VPC
- AWS Networking Services
- How to Launch First EC2 instance
- AWS Database Services Deep Dive
- AWS Migration Services
- AWS Analytics Services

Capstone Project Problem Statements

The following problem statements are intended for the capstone project of this course. Each project is to be completed in teams of two or three students using the AWS services covered in the experiments. Every team submits an architecture diagram, the configuration steps with screenshots, the templates or scripts used, a test report and a cost estimate from the AWS Pricing Calculator.

1. Scalable Three-Tier Web Application on AWS

Problem Statement: A college wants to host its student portal on AWS so that it stays available during result announcements when the traffic increases many times. Design and deploy a three-tier architecture with a web tier, an application tier and a database tier inside a custom VPC spread across two availability zones. The web and application servers must sit in an Auto Scaling group behind an Application Load Balancer, and the database must be an RDS MySQL instance with Multi-AZ in the private subnets. Demonstrate scaling by generating load and show that the application survives the failure of one availability zone.

AWS Services: VPC, EC2, Auto Scaling, Elastic Load Balancing, RDS, CloudWatch, IAM

Expected Deliverables: Architecture diagram, VPC and security group design, CloudFormation template for the stack, load test results showing scale-out and scale-in, and a monthly cost estimate.

2. Serverless Image Processing Pipeline

Problem Statement: A photo sharing start-up needs thumbnails generated automatically for every image uploaded by its users. Build an event driven pipeline in which an image uploaded to an S3 bucket triggers a Lambda function that creates a thumbnail, stores it in a second bucket and records the file name, size, upload time and thumbnail location in a DynamoDB table. Failed invocations must be sent to a dead-letter queue and an SNS email must be sent to the administrator. A CloudWatch dashboard must show the invocation count, the duration and the errors.

AWS Services: S3, Lambda, DynamoDB, SQS, SNS, CloudWatch, IAM

Expected Deliverables: Lambda function code, IAM roles and policies, bucket event configuration, DynamoDB table design, test report with sample uploads and the dashboard.

3. Cloud Cost Governance and Audit Framework for a Department

Problem Statement: A university department gives AWS accounts to several project teams and wants to keep the spending and the security under control. Create an IAM structure with groups and least privilege policies for students, faculty and administrators, enable CloudTrail for all regions with the logs stored in S3, and use Athena to produce an activity report of who created or deleted resources. Set up billing alarms, AWS Budgets and SNS notifications, and write a Lambda function scheduled through CloudWatch Events that stops EC2 instances left running after working hours.

AWS Services: IAM, CloudTrail, S3, Athena, CloudWatch, SNS, Lambda, AWS Budgets

Expected Deliverables: IAM policy documents, CloudTrail and Athena configuration, Athena queries and sample reports, the scheduling Lambda script and an audit report of one week of activity.

4. Secure Document Sharing Portal

Problem Statement: A law firm wants to share confidential documents with its clients without emailing attachments. Build a portal on EC2 that stores documents in a private S3 bucket with versioning and server-side encryption, generates time limited pre-signed URLs for downloads, keeps old versions for 90 days and then moves them to S3 Glacier through a lifecycle rule. Access to the portal must require MFA for staff users, and every download must be logged in CloudTrail and DynamoDB. Prepare a recovery procedure for an accidentally deleted document.

AWS Services: S3, EC2, IAM, DynamoDB, CloudTrail, CloudWatch, Lambda

Expected Deliverables: Portal code, bucket policies and lifecycle rules, IAM and MFA configuration, download audit table design and a demonstration of version restore.

5. IoT Sensor Data Collection and Alerting System

Problem Statement: A smart campus project collects temperature, humidity and power readings from sensors installed in laboratories. Simulate the sensors with a Python script that sends readings to an SQS queue every few seconds. A Lambda function must consume the queue, validate the readings and store them in DynamoDB. Publish custom CloudWatch metrics for each laboratory, build a dashboard for the readings and create alarms with SNS notifications when the temperature or the power consumption crosses the defined limits.

AWS Services: SQS, Lambda, DynamoDB, CloudWatch, SNS, IAM

Expected Deliverables: Sensor simulator script, Lambda consumer code, DynamoDB table design, dashboard and alarm configuration, and a test report with triggered alerts.

6. Disaster Recovery Setup for a Database Driven Application

Problem Statement: An e-commerce company running a web application with an RDS database in one region wants to survive a regional outage. Design and implement a disaster recovery plan with cross-region AMI copies of the application servers, scheduled EBS and RDS snapshots copied to the secondary region, S3 cross-region replication for the uploaded product images and a CloudFormation template that rebuilds the whole environment in the secondary region. Measure the recovery time objective and the recovery point objective achieved and document the failover and failback steps.

AWS Services: EC2, AMI, EBS, RDS, S3, CloudFormation, CloudWatch, Lambda

Expected Deliverables: Backup and replication configuration, the rebuild CloudFormation template, a failover drill report with the measured RTO and RPO, and a runbook for the operations team.

7. Event Driven Order Processing System

Problem Statement: An online store needs its order processing to be decoupled so that the inventory, billing and customer notification steps do not block one another. Build a system in

which a new order is published to an SNS topic that fans out to three SQS queues. Three Lambda functions process the queues, update the inventory and billing tables in DynamoDB and send an order confirmation email through SNS. Each queue must have a dead-letter queue, and a CloudWatch dashboard must show the queue depth, the function errors and the processing time.

AWS Services: SNS, SQS, Lambda, DynamoDB, CloudWatch, IAM

Expected Deliverables: Topic and queue configuration with access policies, Lambda code for the three consumers, DynamoDB table design, dashboard, and a test report showing fan-out delivery and dead-letter handling.

8. Infrastructure as Code for a College Laboratory Environment

Problem Statement: The laboratory in-charge spends several hours before every lab session creating instances and users for the students. Automate the setup with CloudFormation templates that create a VPC with public and private subnets, a fixed number of EC2 instances with a pre-installed tool set from a custom AMI, IAM users and groups for the students with limited permissions, and an SNS topic for announcements. Add a Lambda function on a schedule that stops all laboratory instances at the end of the day and starts them before the next session, and produce a daily cost report by email.

AWS Services: CloudFormation, VPC, EC2, AMI, IAM, Lambda, CloudWatch Events, SNS, Cost Explorer

Expected Deliverables: CloudFormation templates with parameters, the custom AMI build steps, the scheduling Lambda code, a demonstration of creating and deleting the full environment, and the cost report format.

9. Hybrid Network Connectivity Between a Corporate Office and AWS

Problem Statement: A company wants its on-premises data centre to communicate securely with the applications running in AWS. Simulate the on-premises network with a second VPC in another region running a software VPN appliance on an EC2 instance. Connect the two networks with a site-to-site VPN using a virtual private gateway and a customer gateway, place the application servers in private subnets that reach the internet only through a NAT gateway, and configure the route tables and security groups so that only the required traffic is allowed. Monitor the VPN tunnel status and the data transfer in CloudWatch.

AWS Services: VPC, Virtual Private Gateway, Customer Gateway, Site-to-Site VPN, NAT Gateway, EC2, CloudWatch, SNS

Expected Deliverables: Network design with CIDR plan, VPN configuration on both sides, route table and security group design, connectivity test results, and alarms on tunnel status.

10. Centralized Log Analytics and Monitoring Platform

Problem Statement: A software company runs several web servers and wants one place to search their logs and watch their health. Install the CloudWatch agent on a fleet of EC2 instances to ship the web server access logs and system metrics to CloudWatch Logs. Export the logs to S3 on a schedule and query them with Athena to find the most requested pages, the error rates and the top

client IP addresses. Build dashboards for the fleet, create alarms for high error rates and disk usage, and send the alerts to an SNS topic subscribed by the operations team.

AWS Services: EC2, CloudWatch Logs, CloudWatch agent, S3, Athena, SNS, IAM, Lambda

Expected Deliverables: Agent configuration files, log group and export setup, Athena table definitions and queries, dashboards, alarm configuration and a sample weekly analytics report.